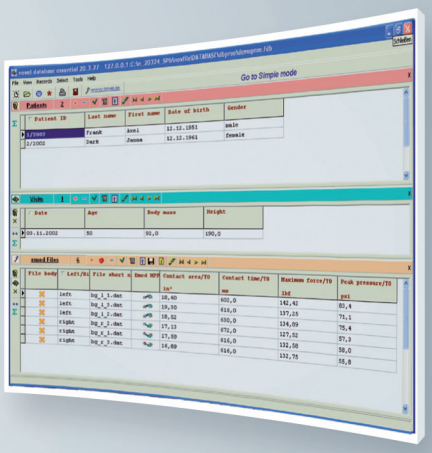
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database light

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WELCOME TO novel database light

This manual provides instructions for the installation and operation of novel database light. The program can be used to organize novel data files (emed, pedar and pliance) with additional clinical information.

The names of soft- and hardware components as well as proprietary names of companies are in general protected by laws of trademarks and brands or patent law.

All information in this manual was carefully compiled and written; however, errors may be present. novel gmbh cannot be held responsible for consequences of incorrect information in the manual.

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1 novel database light



1.1 General Information

novel database light is a basic database which contains a specific set of tables and fields to organize patient and visit information with novel data files.

The database consists of the following tables:

- Patients: Patient demographic data
- Visits: Basic data related to visit and information about amputations
- emed files: Data files of emed measurements
- pedar files: Data files of pedar measurements
- pliance files: Data files of pliance measurements
- user files: Additional files of any type; desktop files, links to video files, and reports are stored here. Any other files such as photo files, ASCII files, etc can also be stored here.

The database is especially designed for the basic demands and applications in the field of medicine. Some database features are listed below.

- The database contains the basic information (database fields) that has been widely used for the data analysis.
- The database can be run in a computer network.
- The database allows direct access to all installed novel programs and report creation.
- Data exchange with other clinical databases is available.

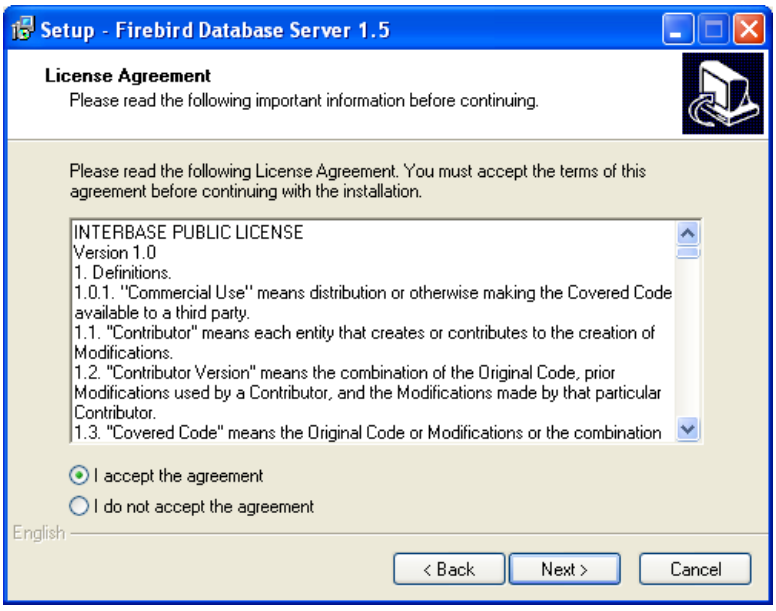
1.2 Installation of novel database light Software

1.2.1 Installation of the Firebird Database Server

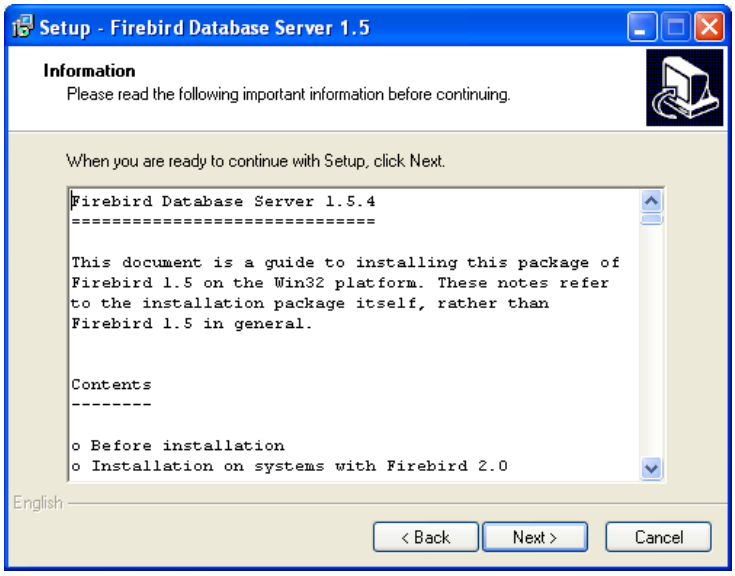
1. Insert the CD with the Firebird Database Server. The Firebird Database Server setup should be on the same CD as all novel software.
2. An automated window should appear. Click **Yes** to install the Firebird Database Server.
3. The following window will appear:



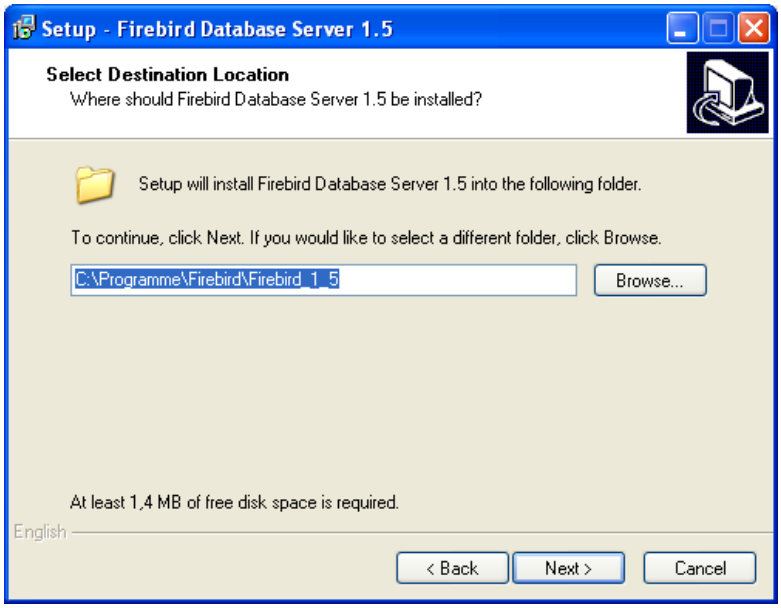
Click the **Next** button.



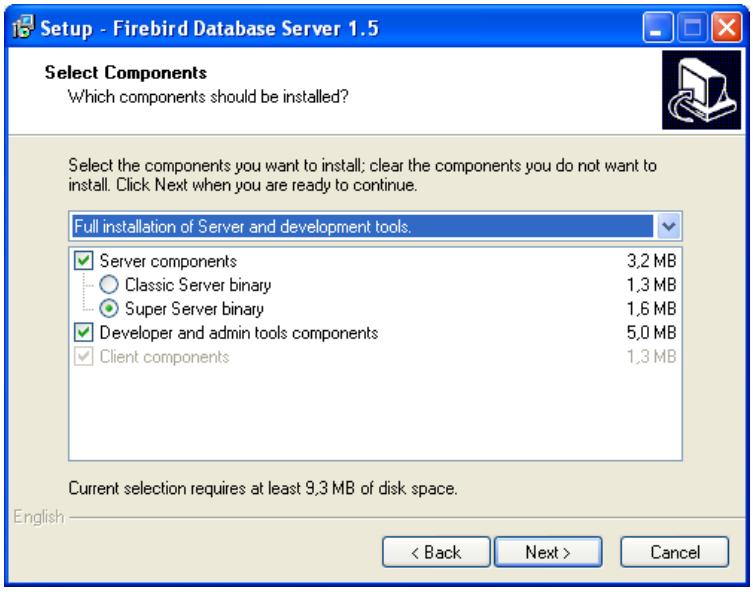
Select the option **I accept the agreement** and click the **Next** button.



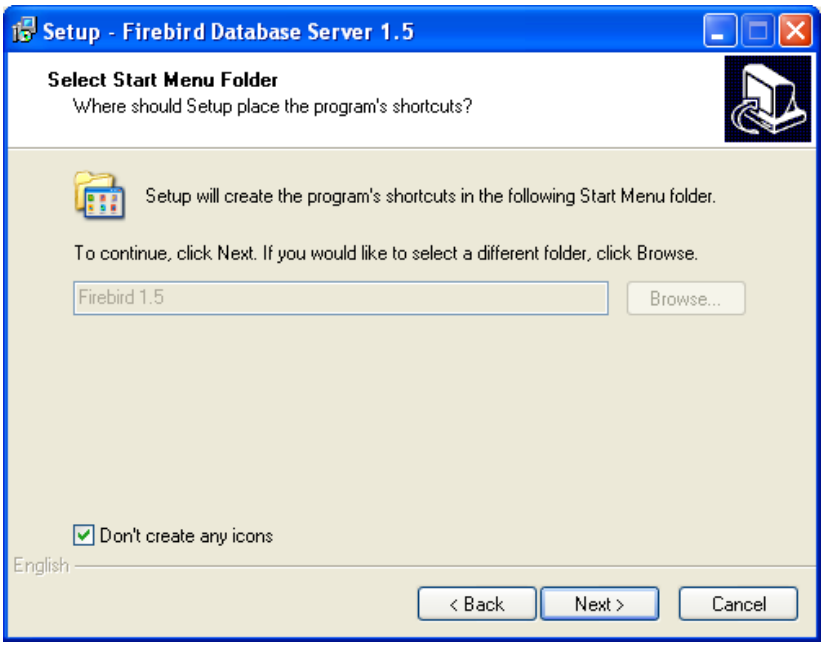
Click the **Next** button.



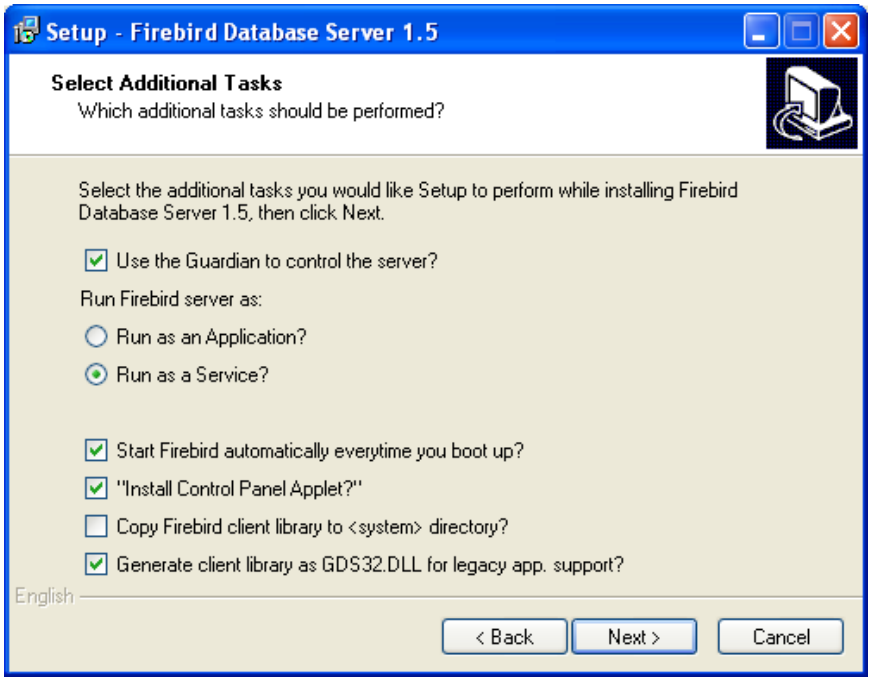
Click the **Next** button.



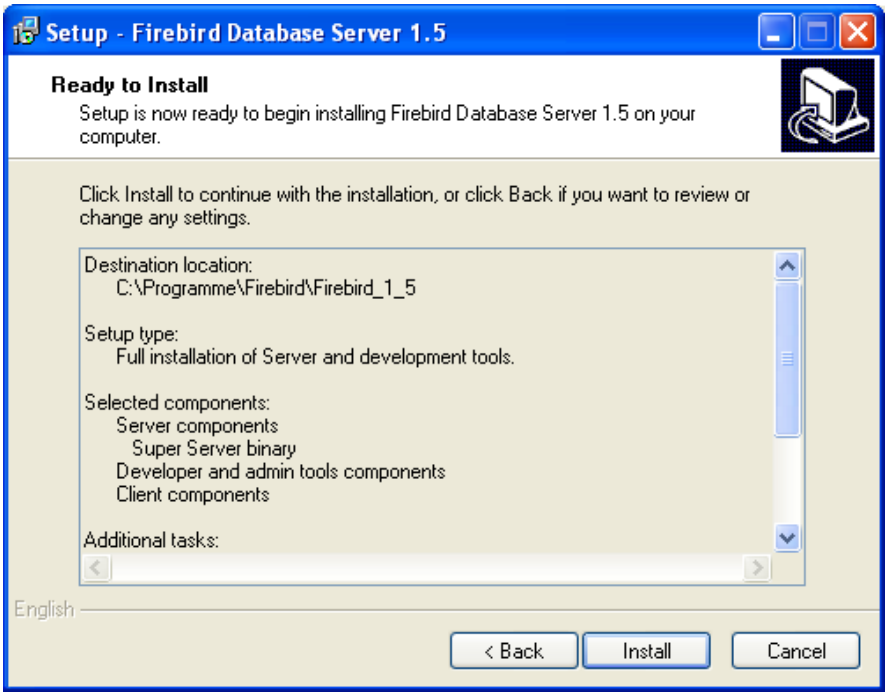
Select the default option **Full installation of server and development tools** and click the **Next** button.



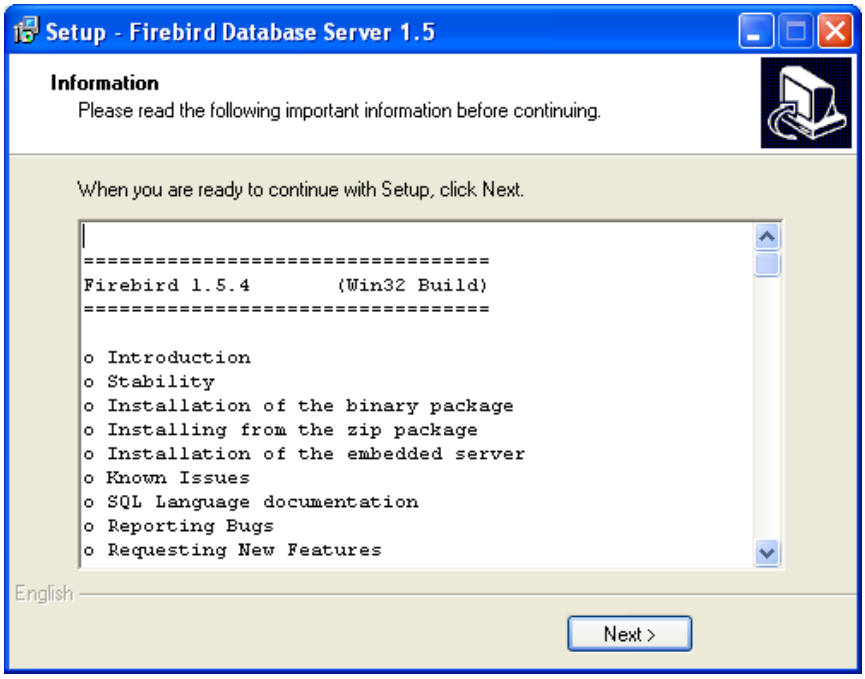
Select the **Don`t create any icons** check box and click the **Next** button.



Select the default options **Use the Guardian to control the server**, **Run as A Service?**, **Start Firebird automatically everytime you boot up?**, **Install Control Panel Applet**, and **Generate client library to <system> directory?**.
Click on the **Next** button.



Click the **Install** button.



Click the **Next** button.



Click the **Finish** button to finish with the installation.

The installation of Firebird Database Server 1.5 is now complete.

1.2.2 Installation of novel Software

1. Insert the disk or CD containing the software.
2. An automated menu should appear. Click on the software names to start the setup wizards. Skip to step 6.
3. If the automated menu does not appear, open My Computer or Windows Explorer.
4. Select the drive where the disk/CD is located.

5. Open (double click on) the setup.exe file.
6. Then follow the instructions that appear on the screen to install the software.

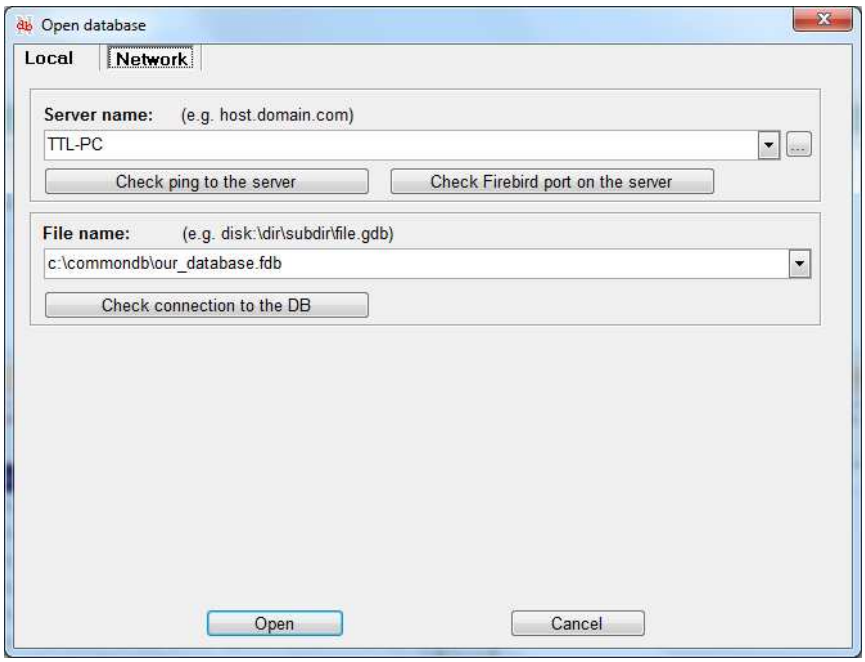
IMPORTANT

The directory to which the program will be installed may be changed from the default directory. However, once the directory is selected and the software is installed, the files can NOT be moved. The programs can NOT be opened from the *.exe files. They can only be opened from the shortcuts. Please consult your novel representative before moving any files or shortcuts.

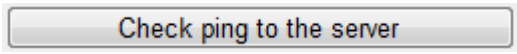
After the installation, the program novel database light can be found in the **Start** menu, under **Programs**.

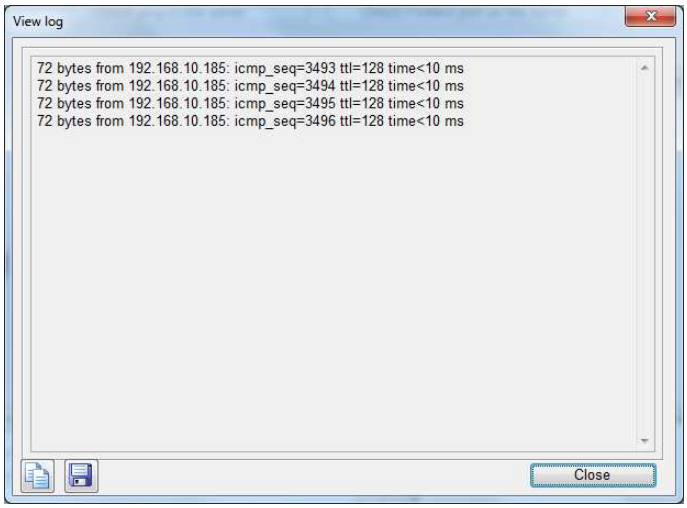
1.2.3 Installation of the novel database light in a network

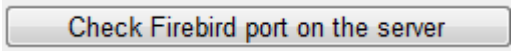
1. Install "Firbird SQL server version 1.5" on the server.
2. Install the database software on each client station.
3. Create a new database (*. fdb file) on one of the stations with menu option **File|New DB** (see section 4.1.1).
4. Copy the created database file from client station on the server hard drive.
   Note: It is not necessary to share this folder on the network!
5. Copy the file dbproudf.dll from the folder "…\Program\Firebird\Firebird_1_5\UDF" on local station to folder with same name on the server.
6. Open the database from the server with menu option **File|Open**, click on the **Network** tab. Specify the server name or the network address and the path to the server where the database is stored.

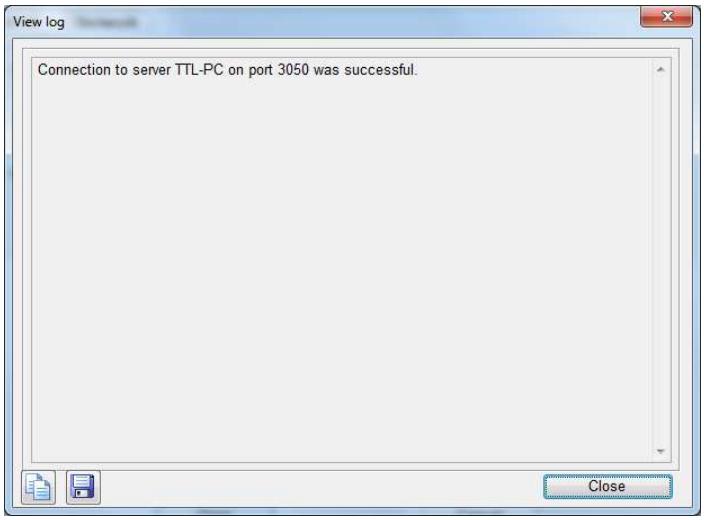


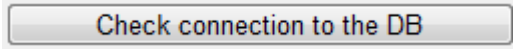
The following buttons test the connection to the network database. The log window will display the results.



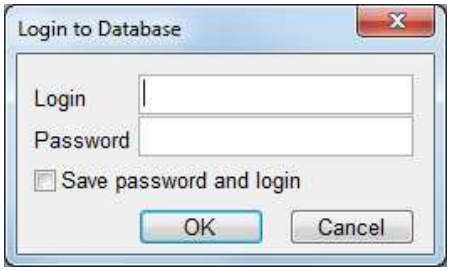


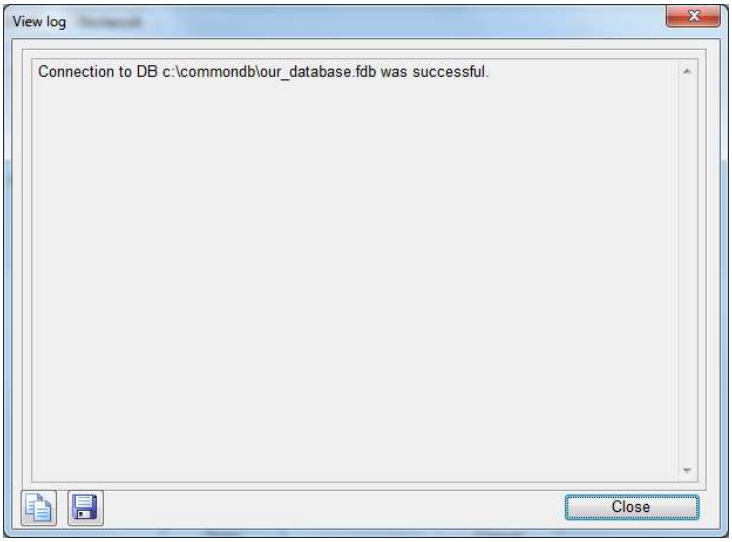






Please type the login name and the password in the following window to check the connection. Log view window will display the result. See below for information on the default login name and password.





Use **Copy** button to copy the text.

Use the **Save as** button to save the text as *.txt file.

7. After clicking **Open** the following login dialog appears. Please type the login name and the password:



The default login name is “SYSDBA” and the default password is “masterkey” in any Interbase database. The PC will now open the database located on the network.

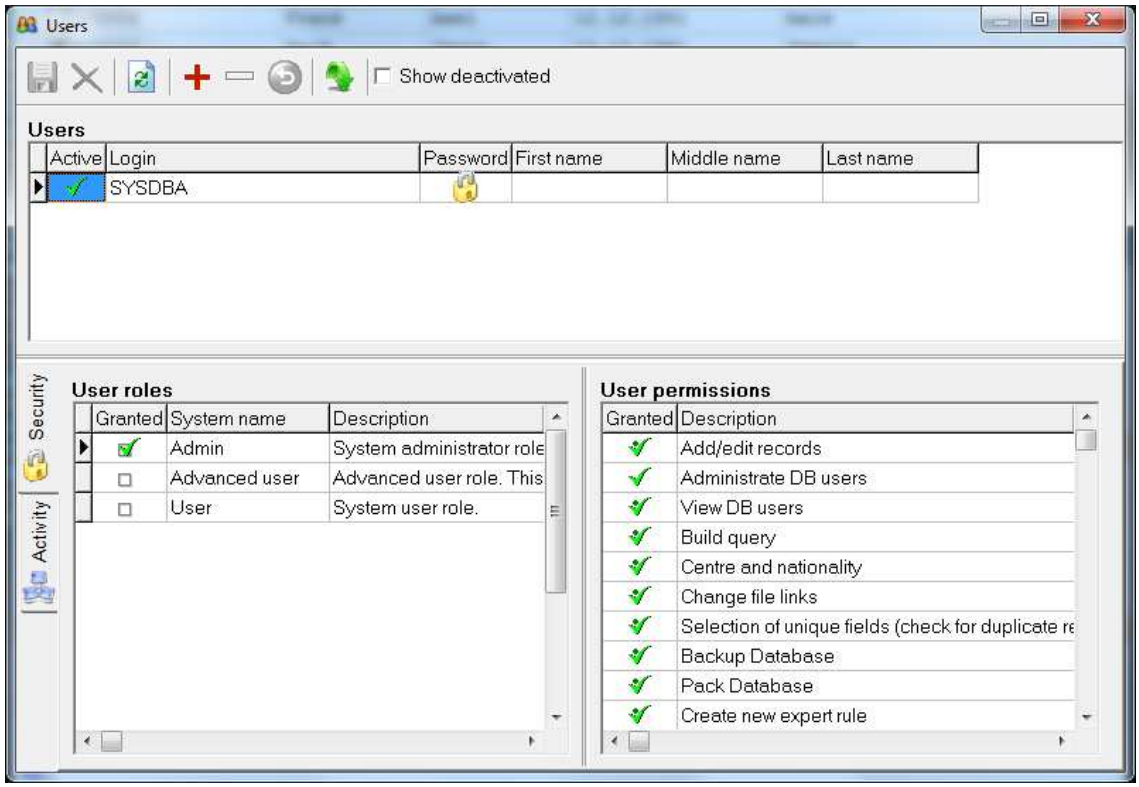
IMPORTANT

These user name and password are known for all users. If you do not change the password for the SYSDBA user, any person can gain access to the data and can change or destroy the database structure or delete the important data.

IMPORTANT

Please, change this password for your data security and safety.
Please, store this password in a safe place.

8. Create one or more users with individual passwords and individual access rights (if it is necessary) using menu item **File|Administration**.



Click on the button  to add a new user. Click on the  button to change the password for this user (including the SYSDBA user). By clicking on the different user roles listed, it is possible to give different permissions for each user (the permissions are listed on the right side). After you finished creating the appropriate users click the **Save** button.

IMPORTANT

The network Firebird server uses TCP/IP port 3050 by default. If there is Firewall enabled on the computer where the network Firebird configuration is installed and setup, port 3050 needs to be enabled for incoming connections (the details how to enable port 3050 are given in

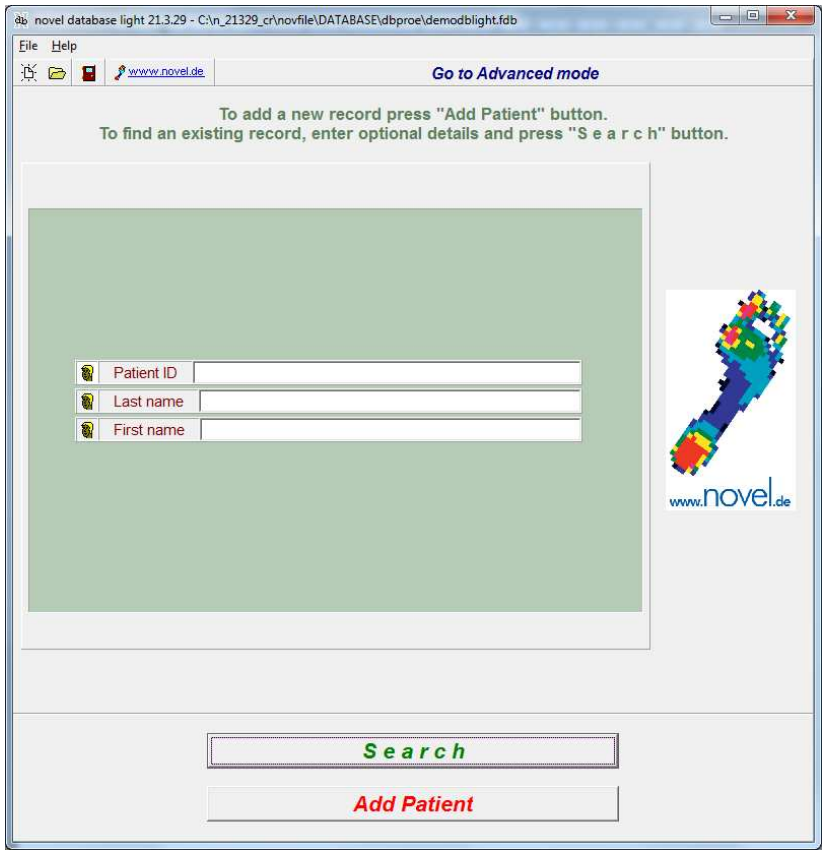
APPENDIX K: Example 5: **Opening port 3050 for the incoming connections to Firebird server via TCP/IP protocol (Windows 7, Firewall)**).

##########

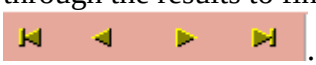
1.2.4 novel database light SIMPLE MODE

This section describes the use of the simple mode in novel database light. The simple mode was created as a means for easy data entry. It is used to simply add new patients and edit information to existing patients. It is also possible to collect data for each patient using the simple mode. To complete data analysis it is necessary to switch to advanced mode.

When opening the database, it will open to the last mode which was used. To change between modes, simply click on the **Go to Advanced/Simple Mode** toggle button located in the top toolbar. The main page for the simple mode will look similar to the following window.



To search for an existing patient:

1. Simply type Last Name or/and First Name or/and Patient ID data.
2. Click on the **Search** button. If more than one Patient matches the search criteria, a message will appear stating multiple records were found. You will need to scroll through the results to find the correct Patient using the scroll buttons:
    .
3. To edit existing information, click the **Edit** button in the top right corner of the Editor Window and simply add or change the appropriate information. Descriptions of the fields and screens can be found in **Section 5**.
4. Click **Done** when finished.

To add a new patient:

1. Simply click on the **Add Patient** button.
2. The screens as described in **Section 5** will appear.

3. Type in all appropriate information. First add the patient information then click the **Add Visit** button and add the visit related information.
4. Click **Done** when finished.

To collect data for a patient:

1. First find an existing patient or add a new patient.
2. Click on the appropriate Data file tab (emed, pedar, or pliance). Existing data files can be added using the **add record** button  . New files can be collected using the 

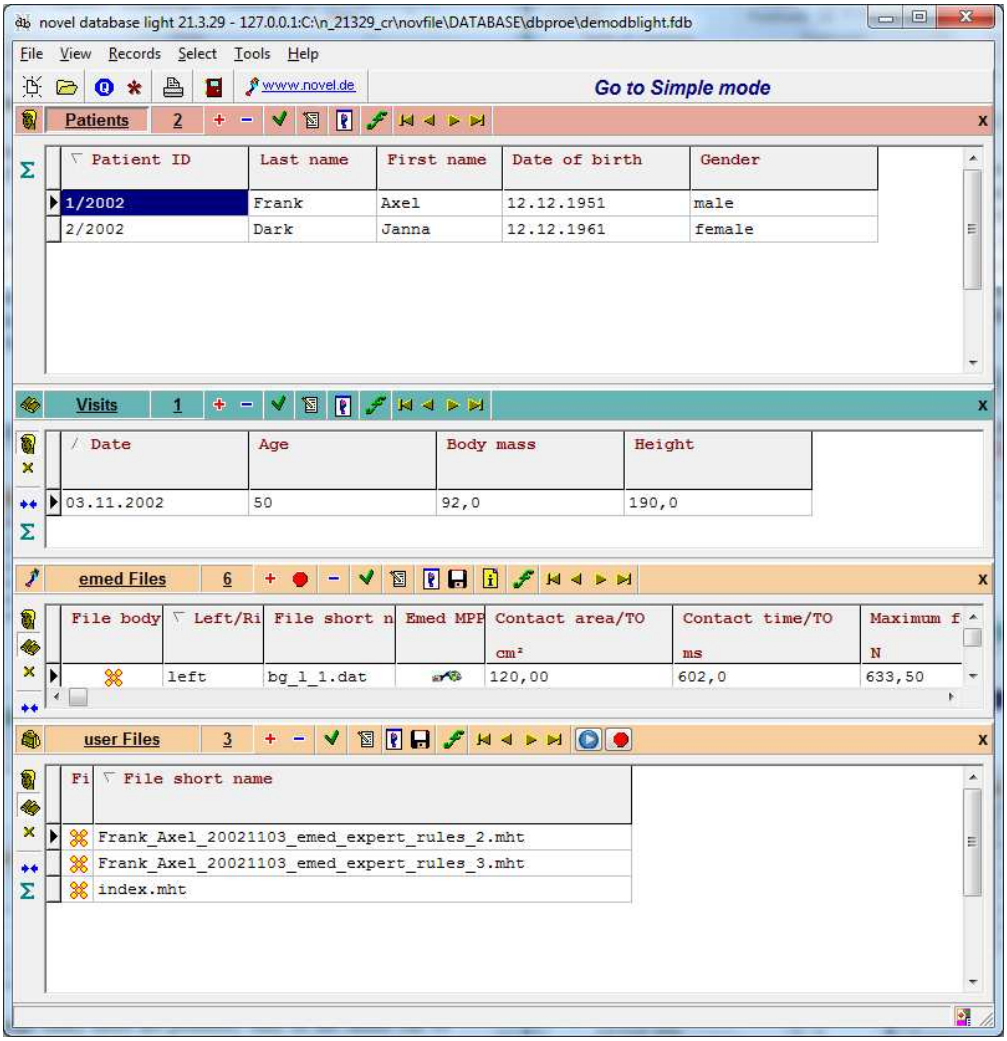
button. See the details in **APPENDIX G: Example 1: Daily procedure of collecting data**
3. Click **Done** when finished.

##########

1.2.5 QUICK START WITH novel database light

This Section gives an overview of the most important functions of the database advanced mode to enable immediate use of novel database light.

1. Open novel database light. An example database should open and a screen similar to the following will appear. The program will default to the simple mode. Click **Go to Advanced mode** button. If a database does not open automatically, use the open icon and select an existing database from the list. It will help to practice with the existing demo database, demoproe.fdb, before trying to create a completely new database.



Four tables are represented here (although many more are possible). Any of the tables can be individually closed using the **X** icon in the upper right corner of the table. Any of the tables can be opened again by selecting the **View** menu and then any of the tables listed.

A list of possible tables is shown in **Section 1.1**.
The following tables are seen above.

- Patients: Patient demographic data
- Visits: Clinical parameters
- emed files: Data files of emed measurements

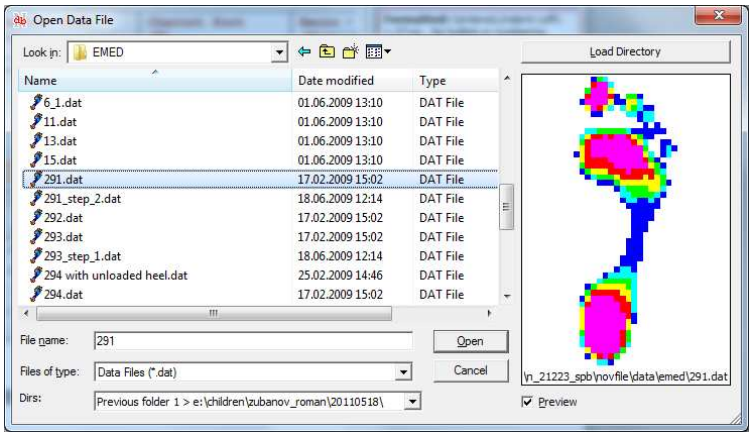
- user files: Files of any data format e.g. text documents, pictures, videos, MHT and RTF reports, etc.

2. Select a patient to view the corresponding data.
   All of the data files shown in the emed, pedar, and pliance **files** tables relate to the selected visit (marked with a black arrow) in the **Visits** table. All of the visits shown in the **Visits** table relate to the selected patient in the **Patients** table.
3. Edit data for the selected patient.
   a. Select the **Edit** icon in the **Patients** toolbar. A window will appear with data pertaining to that patient. There are several tabs down the right side of the window with different information for each Table. Click through the pages across the top to view the types of information available in the default setup.

IMPORTANT

To revert back to the original information click the **Cancel** button.

4. Add a new patient.
   a. Select the **Add** icon in the **Patients** toolbar. A **Database editor** window similar to the **Edit record** window will appear. Add information pertaining to the new patient. Make sure to click through all the tabs to make sure that data entry is complete. Data can be added or changed at any time. To change data once you have completed adding a new patient click the **Edit** button.
5. Add a visit to a Patient
   a. Select the **Add visit** icon. Add information pertaining to the new visit. Make sure to click through all the tabs to make sure that data entry is complete. Data can be added or changed at any time.
6. Add a (or measure a new) data file to a Patient.
   a. Select the Patient in the Patient Table.
   b. Select the Visit in the Visit Table.
   c. To add an emed data file, in the emed file table click the add file icon.
   d. An **Open Data File** window will open.



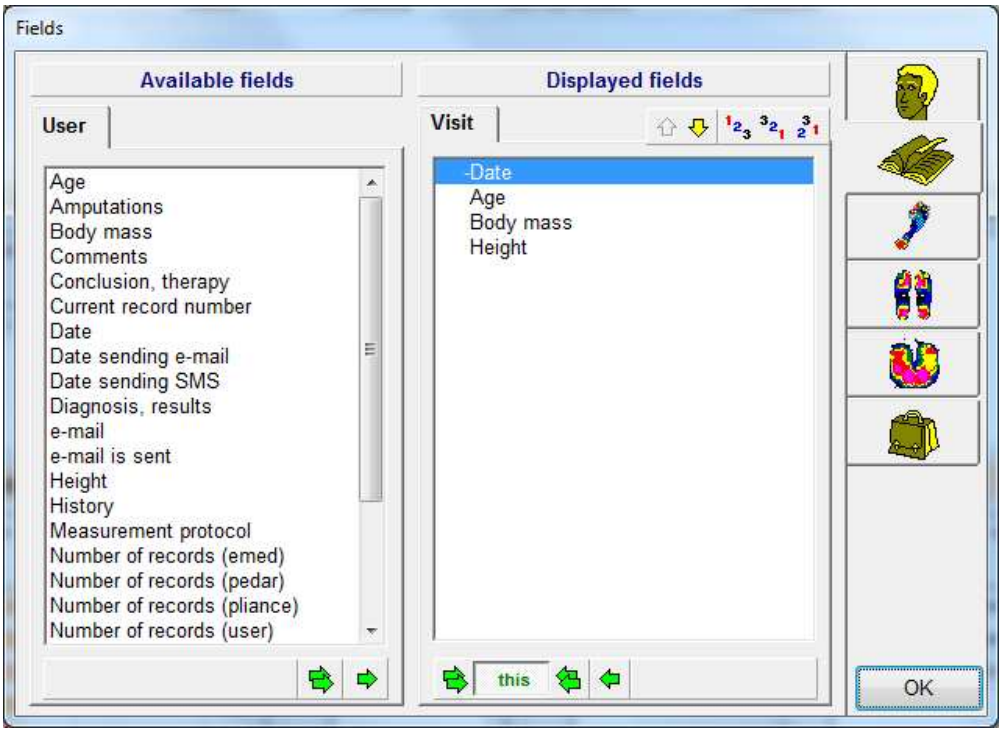
   e. Select the *.dat or *.kor file to add and click **Open**.

f. The new file will be listed in the table.

g. Choose  to measure a new data file. The software will open the installed data collection program for the respective system (`emed, pedar, pliance`). Follow the appropriate data collection steps for the selected system.

7. Customize fields displayed in advanced mode.

a. Select the **Choose fields to display** icon  . A window similar to the following will appear.



b. All possible fields are listed under the **Available fields** section. All fields to be displayed are listed under the **Displayed fields** section. Simply select the fields you would like to add from the **Available fields** to the **Displayed fields** section and click the single arrow (to move just those selected). Fields can be removed from the **Displayed fields** section by selecting the fields and clicking on the single arrow icon. This can be done for each table represented by the tabs along the right side of the screen.

c. Use the Up and Down arrow icons  to put the fields in the order desired.

d. Use the  icons to sort the data by increasing, decreasing or without sorting.

e. Click **OK** when finished to return to the main database view. See **Section 3.4.4** for additional details.

8. Use the Run application feature to analyze data files or open them in any other installed programs. If it is the access from emed Files table, then do the following.

a. Select a patient, and visit, and at least one data file. To select multiple data files or a group of patients use the Ctrl key +left click.

b. Select the **Run application** icon  in the data files table. A window will appear with multiple choices for data analysis. The programs which are available to use are dependent upon what software was installed (purchased). When any of the icons are selected, the data files that have been selected within the database will automatically open in the program.

1.3 The Buttons of the Main Toolbar



	**New DB**	Create a new database
	**Open DB**	Open an existing database
	Build query	Query for a specific record
	**Select all**	Select all records of the database
	**Print**	Print the tables of the database
	**Exit**	Exit the program
	**Path to novel Home page**	http://www.novel.de

1.4 The Buttons of the Table Toolbars

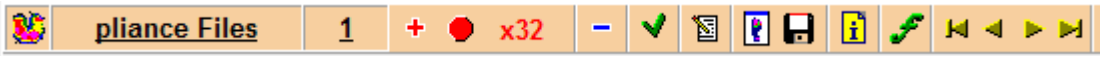
All table toolbars include similar sets of buttons and buttons specific to the table.











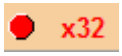


The buttons similar in each table are listed below.

	**Table name**	Shows the name of the table
	**Number of records**	Shows the number of records in the table
	**Add**	Add a record to the database

	**Delete**	Delete a record from the database
	**Mark all**	Mark all records of the database and/or table
	**Edit**	Edit the selected record
	**Run application**	Opens the window with the available programs to run further novel programs
	**Choose fields to display**	Choose fields or columns of the table to display
	**First/ Previous/ Next/ Last**	Change to the first / previous / next / last record

emed/pedar/pliance Files tables contain the measurement button.

 **Measure**	Starts emed measurement system
 **Measure…**	Starts pedar-x measurement system
 **Measure…**	Starts pliance-x32 measurement system.

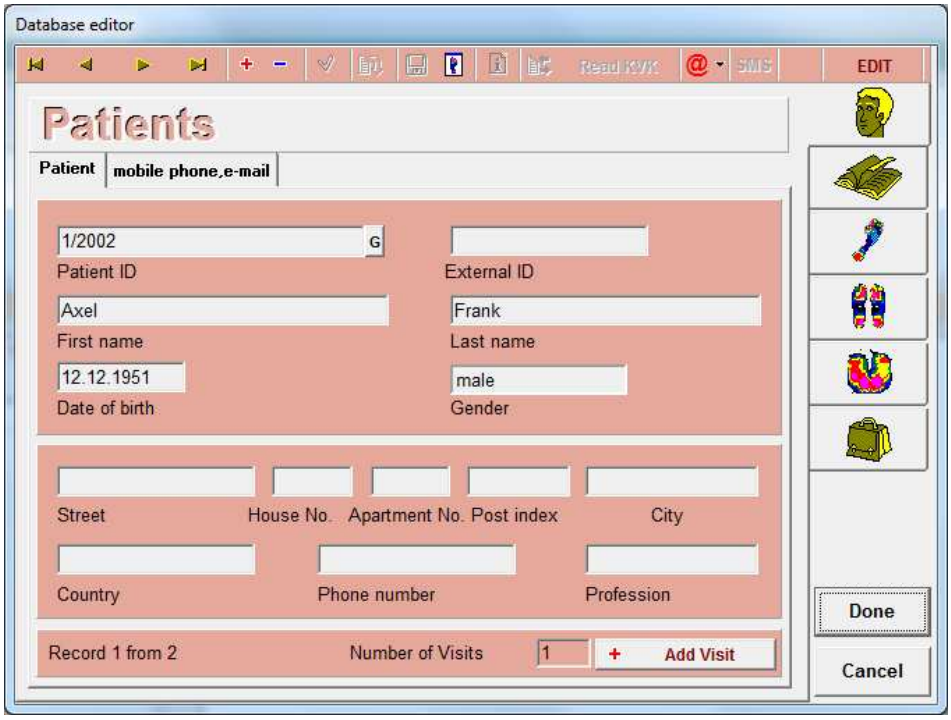
User Files table contains the buttons starting the user defined applications

 **Scan files**	Waiting of the scan files in the selected folder
 **Run user defined applications**	Starts Import 3D scans, Scan with NexEngine etc.

##########

2 Database editor window

Different buttons are enabled in different tables.



Buttons functions are listed below.

 - use these buttons to navigate between first, previous, next, and last records.

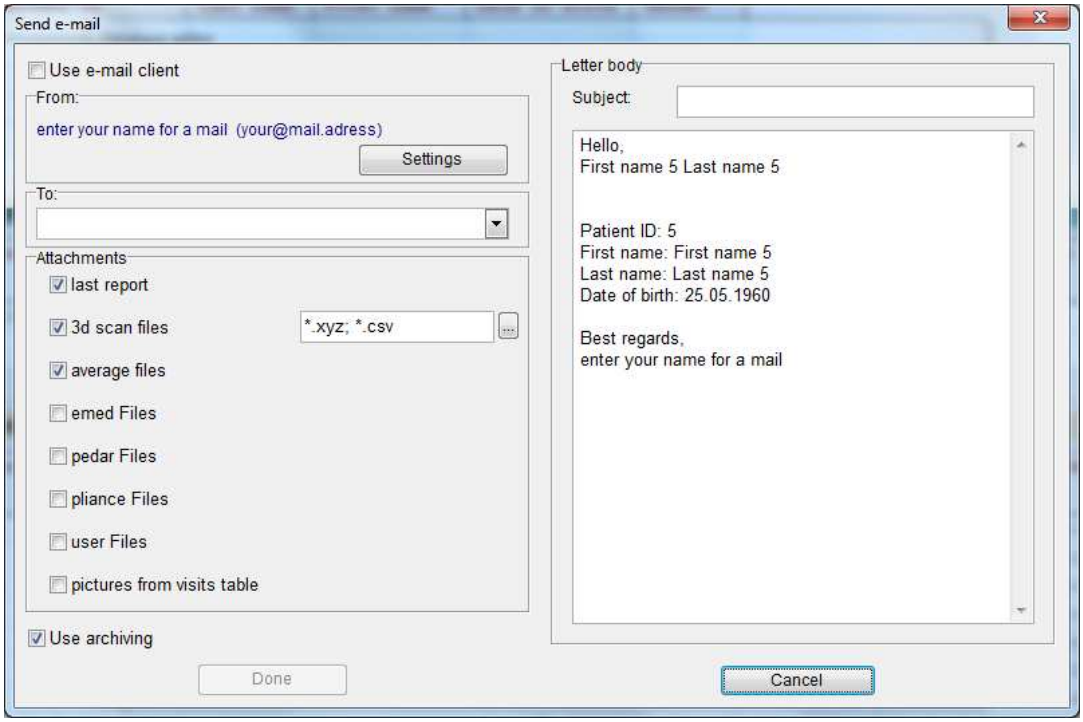
 - use these buttons to add or delete the record.

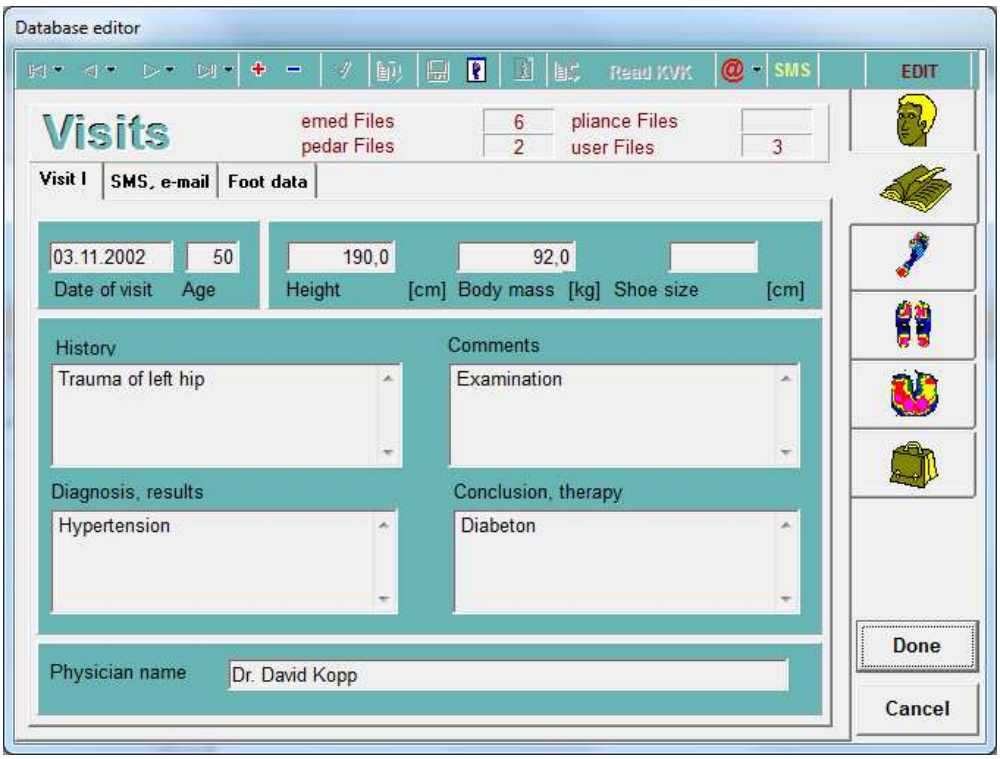
 - use this button to start an application via the **Run application** window.

 - use this button to send the e-mail using the external post client.

The report, 3d scan files, average files, and measured data files may all be attached to the e-mail.

Mark the **Use archiving** checkbox to archive the files.



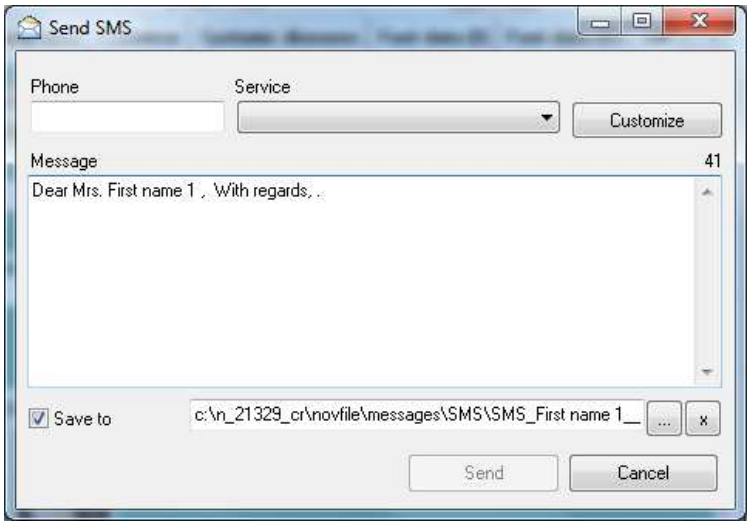




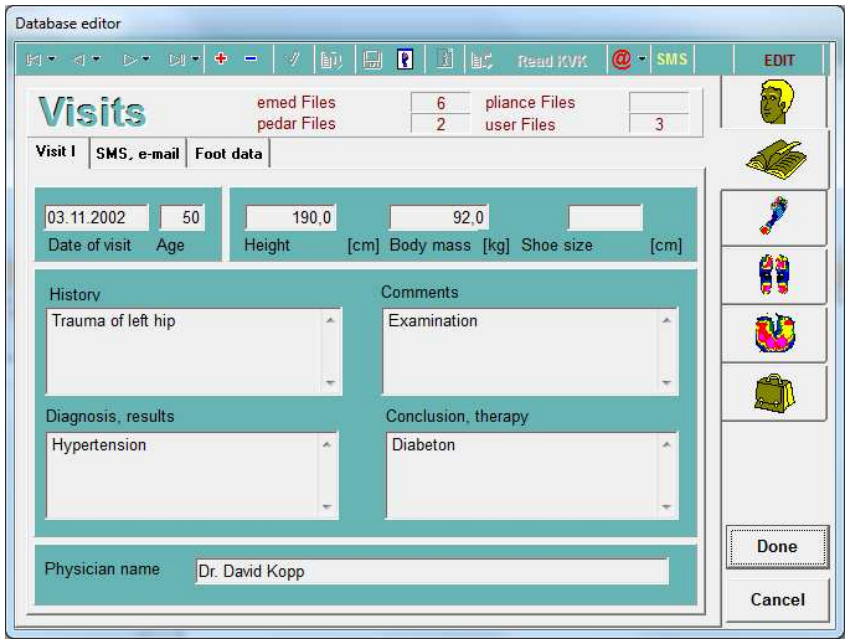
- use these buttons to navigate between first, previous, next, and last records on this (**Visits**) or upper level (**Patients**).

 - use this button to copy data from previous visit.

SMS - use this button to send SMS.
When this button is clicked, the following window will appear.



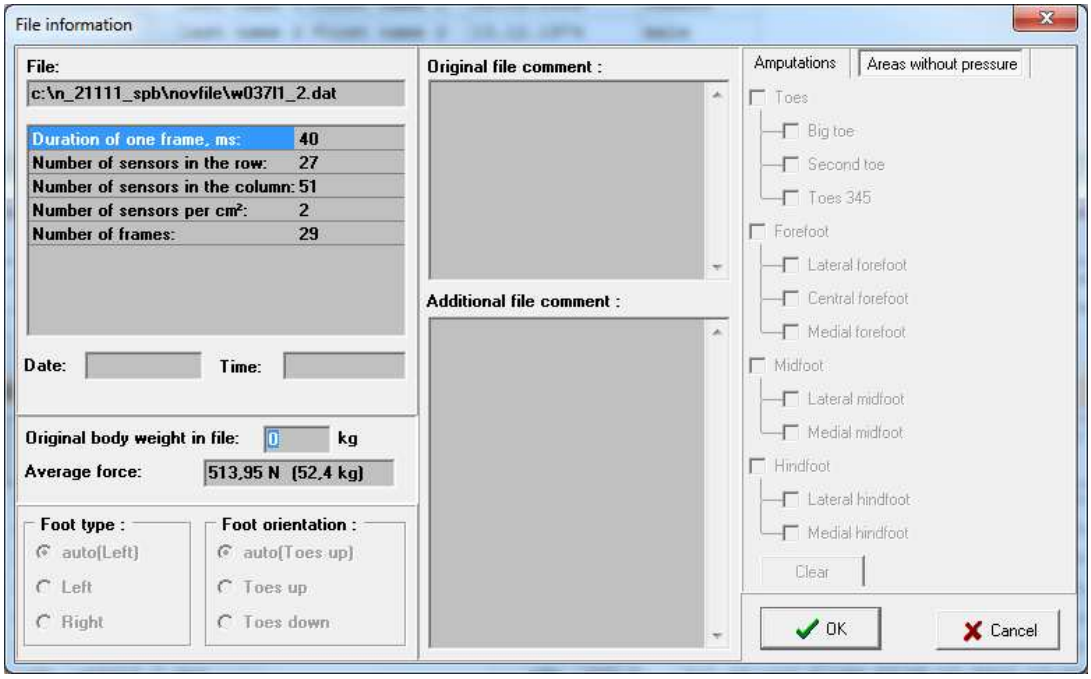
The number of emed, pedar, pliance, and user Files are displayed across the top of the database editor window.



- use this button to mark all records.

- use this button to save the file (files).

- use this button to view/edit file information.
When the button is clicked, the following window will open.



Foot type, **Foot orientation**, and **Areas without pressure** can be edited in the **Edit** mode.

 - use the **Paste-exchange data** button to insert data from an external database. The button is enabled when the corresponding *.exe is installed.

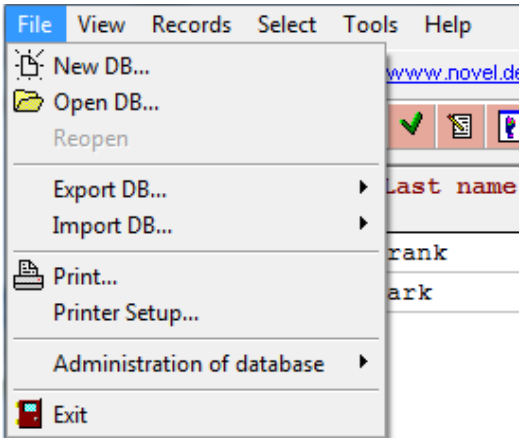


- use this button to read a KVK. This button is enabled when a card is inserted.

##########

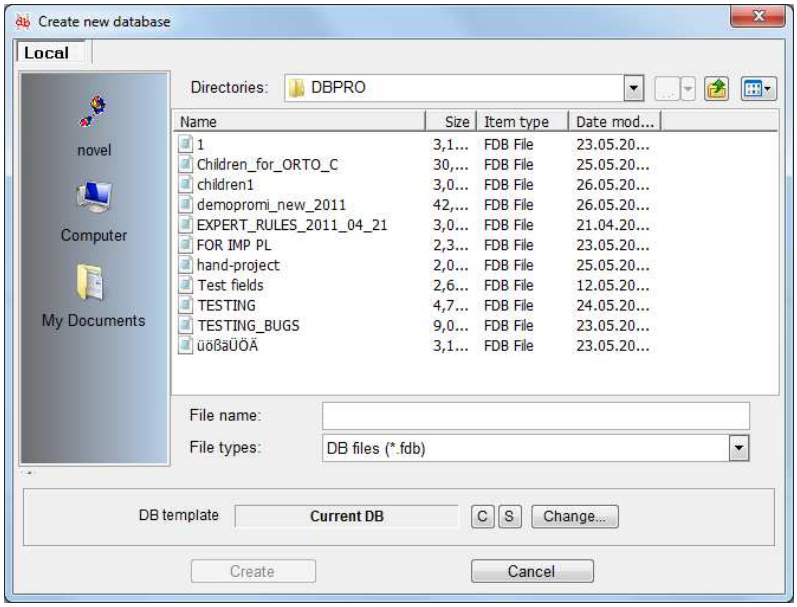
3 THE FUNCTIONS OF THE DATABASE novel database light

3.1 File Menu



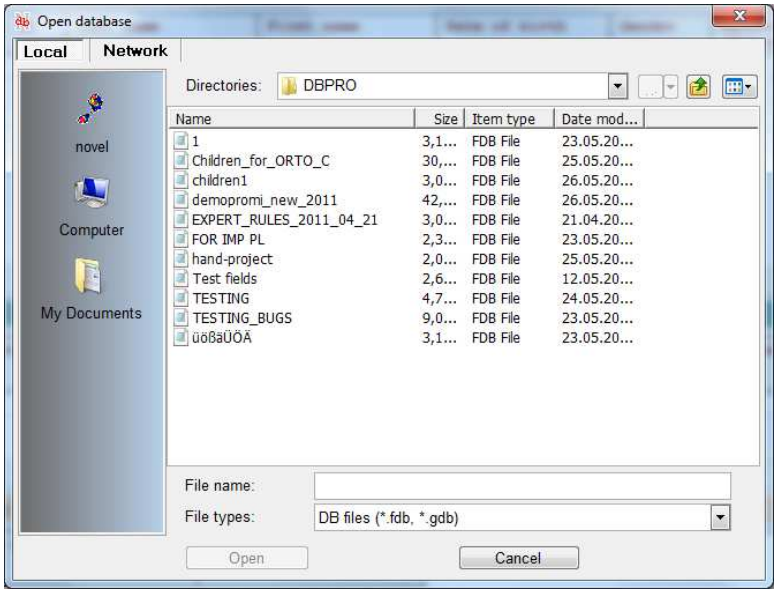
3.1.1 New DB

Use the **New DB** command to create a new database. In the **Create new database** window, the file name and directory can be chosen. The file extension of a novel database light is **.fdb (*.gdb** is an old file extension and is also available**).**



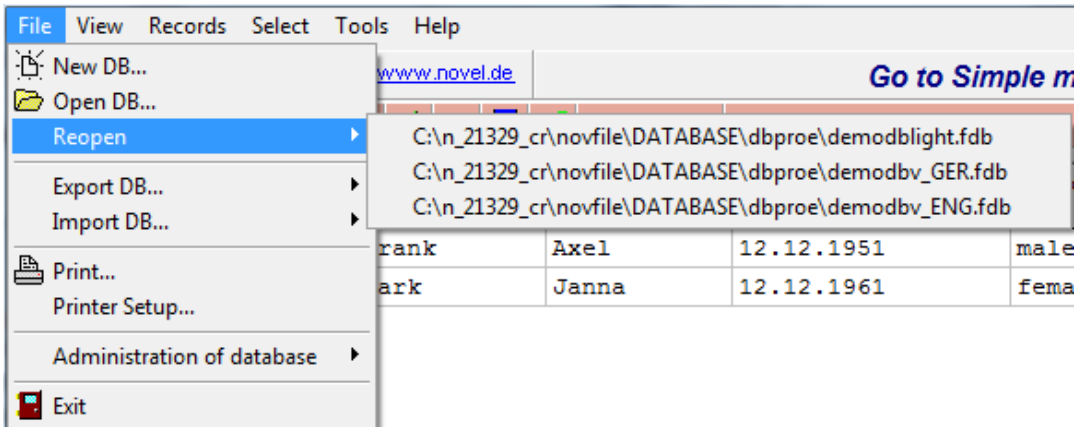
3.1.2 Open DB

Use the command **Open DB** to open an existing database. The file extension of a novel database light is ***.fdb (*.gdb** is also available**).**

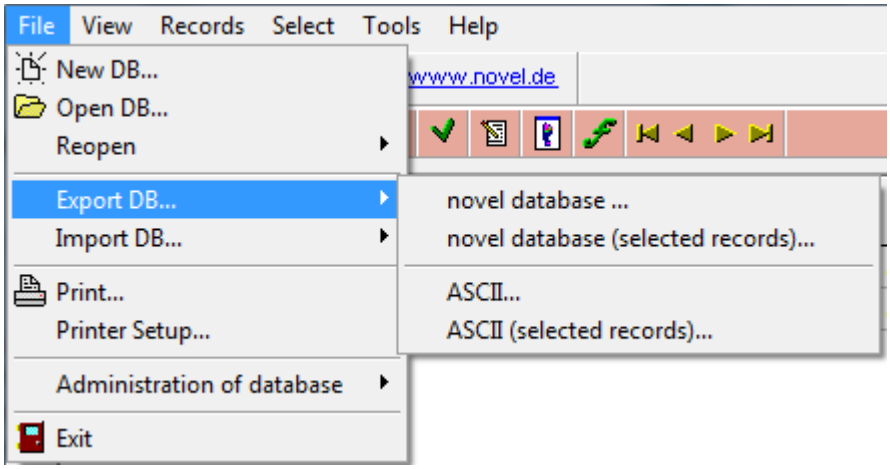


3.1.3 Reopen

Use this option to open one of the databases that were opened previously (databases opened previously are in the list).

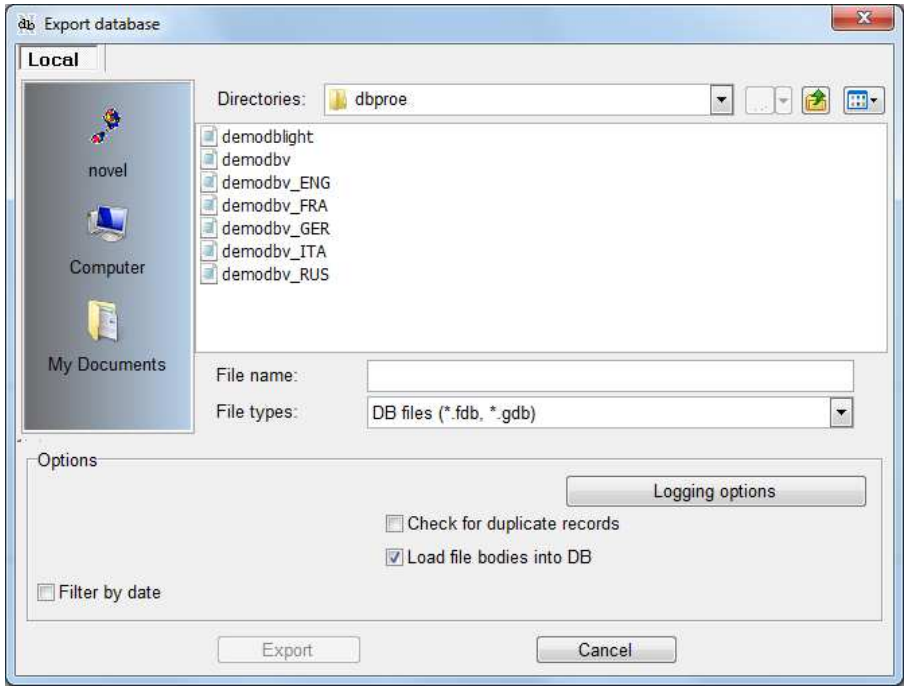


3.1.4 Export DB



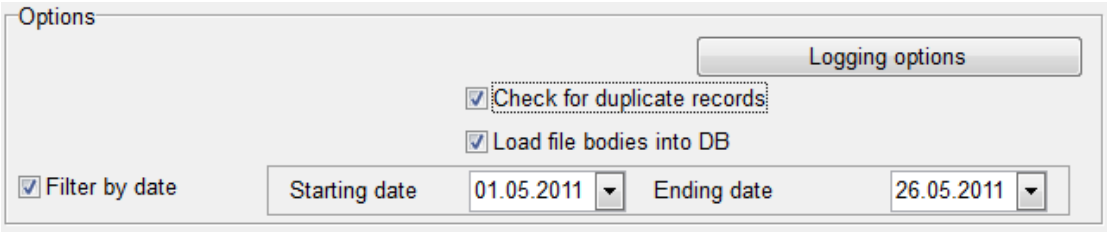
The database can be exported in two different formats: database and ASCII.

- **novel database**
  Exports the database in the format of the novel databases.



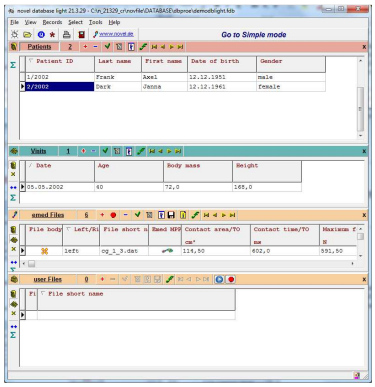
The database will be saved as a file in the directory "c:\novel\novfile\database\dbproe" with the file extension ***.fdb (*.gdb).** There are three additional options for the export of a database:

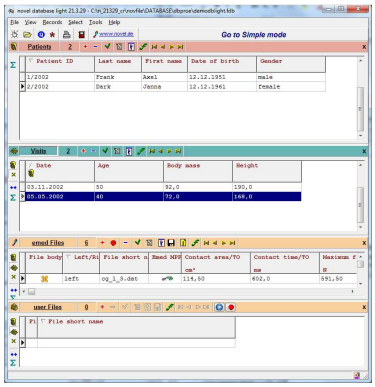
1. **Filter by date –** select this option to export the records in which the date of visit is in the range of typed starting and ending dates (date format is specific to computer settings).



2. **Check for duplicate records –** select this option to identify any repeated records.
3. **Load file bodies into DB –** select this option to include data files inside of the database.

- **novel database (selected records)**
  Exports the selected records in the format of the novel databases.
  Select the records to be exported either in Patients or Visits table and select the Export database by selection option.





NOTE: If records are selected in `emed`, `pedar` and `pliance` Files table, the Warning message will appear.



- **ASCII**
  Exports the database in ASCII format.

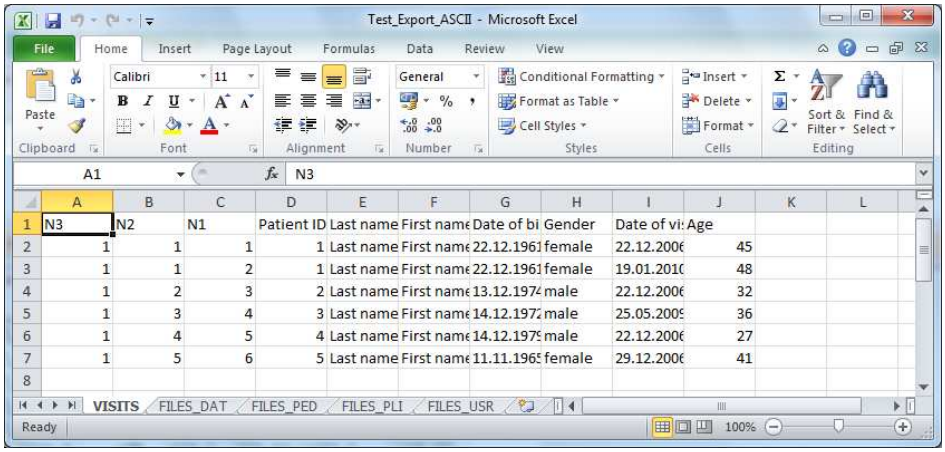


There is only one option for to export the database to ASCII:
Filter by date – select this option to export the records in which the date of visit is in the range of entered starting and ending dates.

The database may be exported to ASCII in two file types: ASCII DB files, and Excel Files:

1. The exported ASCII DB files are saved in the directory "c:\novel\novfile\ascii\asciiout" and are saved in several different files:
   - ***.avs**: Includes the data from the Patients and Visits tables
   - ***.adt**: Includes the data from the emed files table
   - ***.apd**: Includes the data from the pedar files table
   - ***.apl**: Includes the data from the pliance files table
   - ***.aur**: Includes the data from the User files table

2. When the Excel files type is selected, the following window will be opened in background (Windows 7) or foreground (Windows XP).



A page will be created for each of the tables from the database

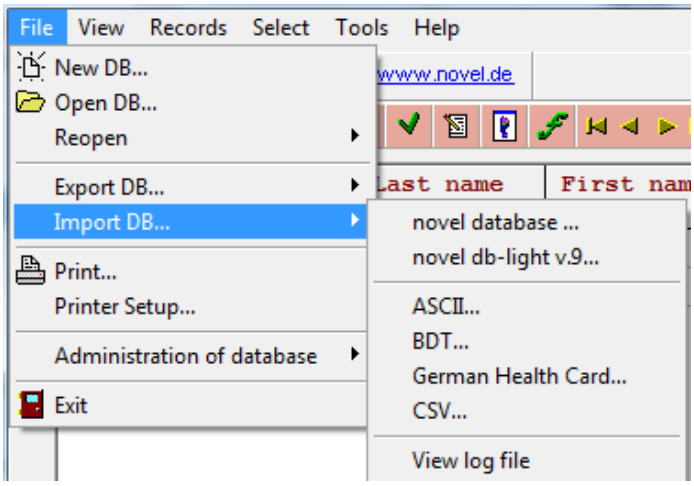
- **VISITS**: Includes the data from the Patients and Visits tables
- **FILES_DAT**: Includes the data from the emed files table
- **FILES_PED**: Includes the data from the pedar files table
- **FILES_PLI**: Includes the data from the pliance files table
- **FILES_USR**: Includes the data from the User files table

IMPORTANT

MS Excel 2003 or later should be installed.

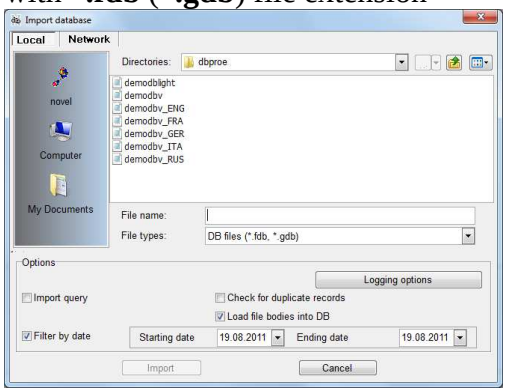
- **Export ASCII (selected records)**
  Select the records to be exported either in Patients or Visits table and select the Export ASCII (selected records) option.

3.1.5 Import DB



Different types of files can be imported into an opened database:

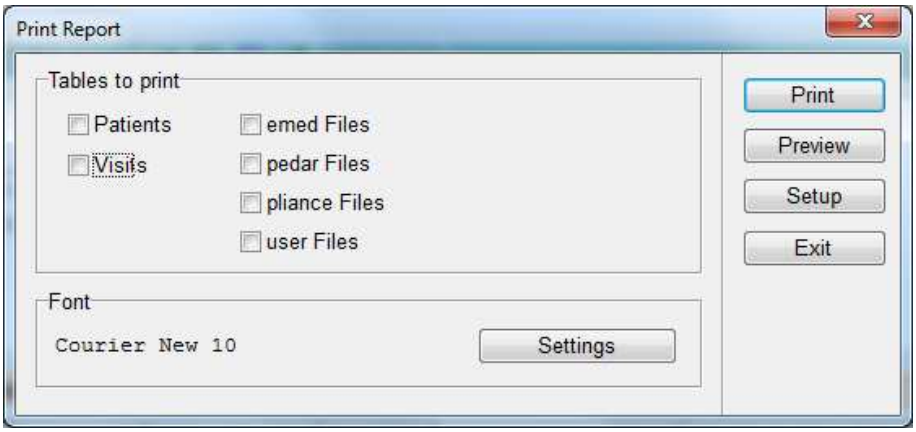
- **novel database** — Database with ***.fdb** (***.gdb**) file extension



  Four options are available.
  - **Filter by date –** select this option to import the records in which the date of visit is in the range of typed starting and ending dates.
  - **Check for duplicate records –** select this option to identify any repeated records.
  - **Load file bodies into DB –** select this option to include data files inside of the database.
  - **Import query –** select this option to import queries of source database
- **novel db-light v.9** — Database with ***.db** file extension.
  Format of the `novel db-light` (version 9 only!)
  NOTE: Do not use this option for the `novel database light` import.
- **ASCII** — Standard import of text files with extension *.avs and *.asc.
- **BDT** — Standard import of text files with extension *.bdt.
- **German Health Card** — Standard import of text files with extension *.kvk.
- **CSV** — Standard import of text files with extension *.csv.
- **View log file** — View of a log file is available.

3.1.6 Print

Use this command to print the database.



With **Tables to print** it is possible to print tables individually or in specified combinations. Select the tables to be printed.
Using the **Settings** button, the display and printing font can also be changed here.
Using the **Setup** button, the printer setup can be specified.

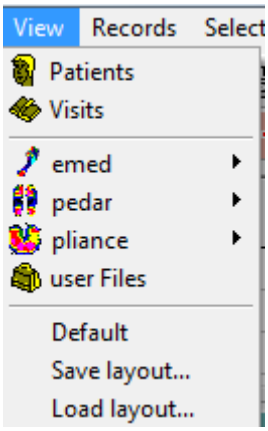
3.1.7 Printer Setup

Use this option to change the printer setup. This is a standard Windows feature.

3.1.8 Exit

Exits the program.

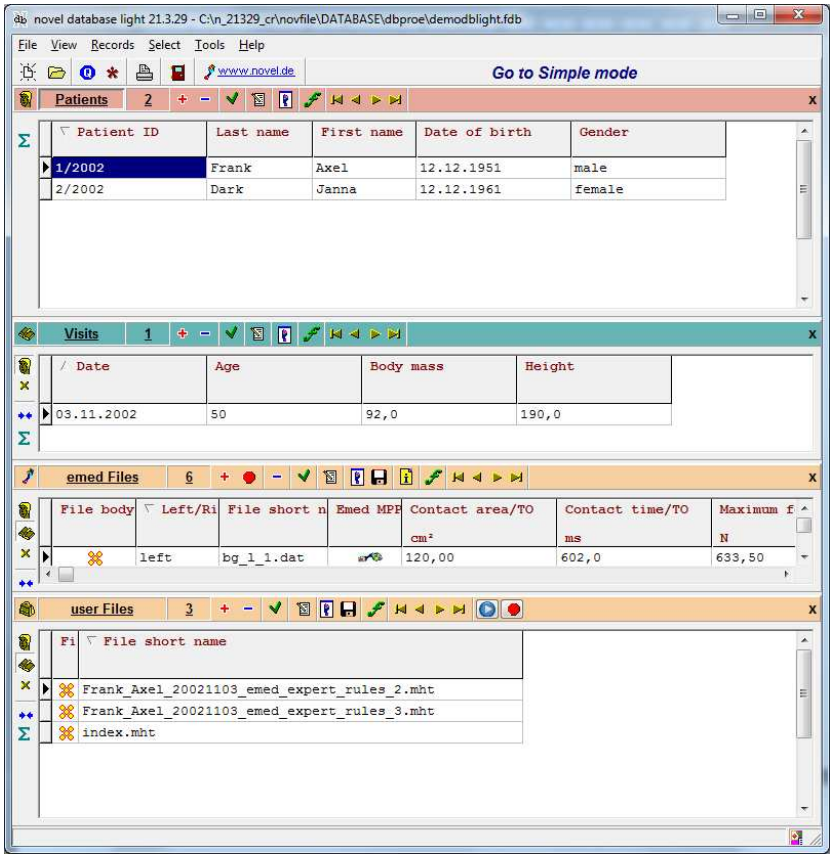
3.2 View Menu



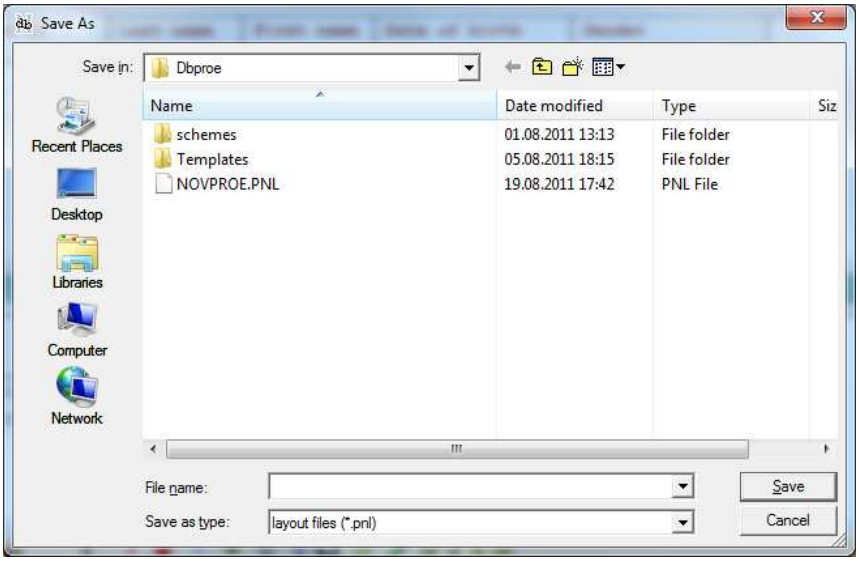
This menu is used to activate (view) and deactivate (hide) each of the tables available in the database. A check mark indicates the table is displayed on the screen. See **Section 1.1** for a list of tables available.

3.2.1 Database Layout

1. The **Default** option returns the database to the default layout of the tables.

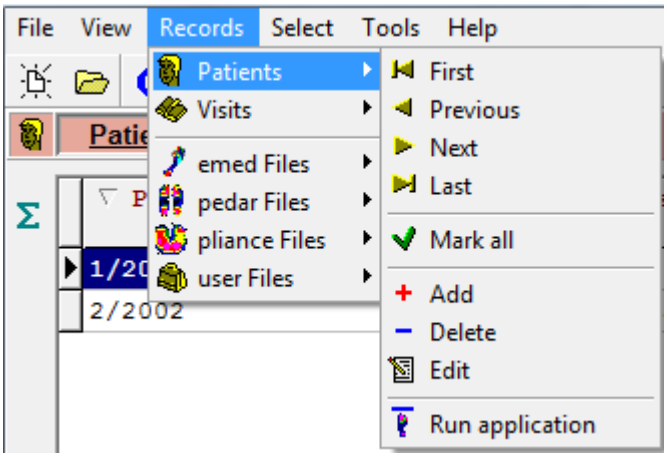


2. The **Save Layout** option saves the current layout (file with extension *.pnl) for future use or use with other databases.



3. The **Load Layout** option loads previously saved layouts (file with extension *.pnl).

3.3 Records Menu

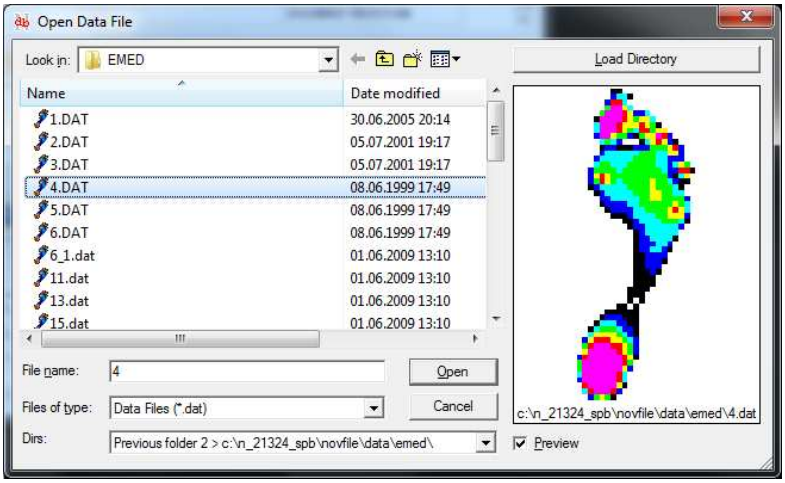


For each of the six tables listed in this menu, similar features are available. All of the features listed in this menu are available in the toolbars for each table. A list of all features will follow.

a. First, Previous, Next, Last 
Within the table you can change to **First, Previous, Next** or **Last** record.

b. Mark All 
Mark all records of the table.

c. Add 
New records can be added to the Patients or Visits tables. See **Section 5.**
The command **Add** opens a selection window to add data files in the emed, pedar, and pliance files tables.



Select the data files to add and click **Open**. The data files will be included in the corresponding table. The file can be saved in the database or a link to the file can be saved. (see **Section 3.5.3, 3.5.4)**. To select all files from this folder

click the **Load Directory** button. If the **Preview** checkbox is selected, it is possible to view the file before the adding procedure.

d. Delete

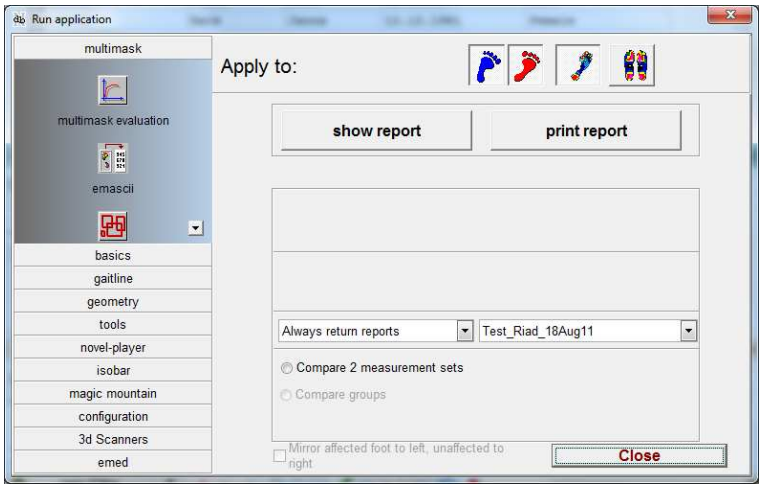
Delete the marked record (records) or data file (files) from the table.

e. Edit

Edit records (see section 5) in tables.

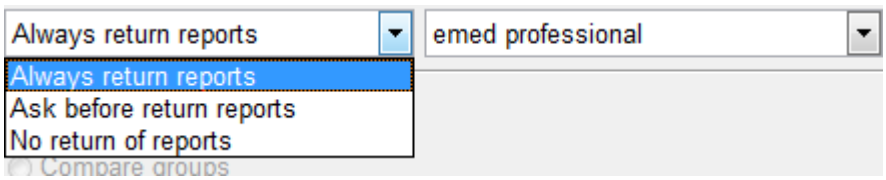
f. Run Application

All installed novel programs can be applied to the measured data files. The records must be selected before the application program is run. A window similar to the following will appear.



All tabs and buttons represent all novel programs. Click on a button to open the corresponding program.

1. The red and blue feet at the top of the screen are toggle buttons used to select right and/or left feet to apply analysis to.
2. The multi colored foot at the top of the screen indicates emed files are being analyzed. If pedar or pliance files are being analyzed, this icon will change accordingly.
3. A report which is created may be returned to the database and stored within the **User files** table. There are three options available: Always return reports, ask before returning reports (a pop up window will occur after the report is created and closed) or never return reports.



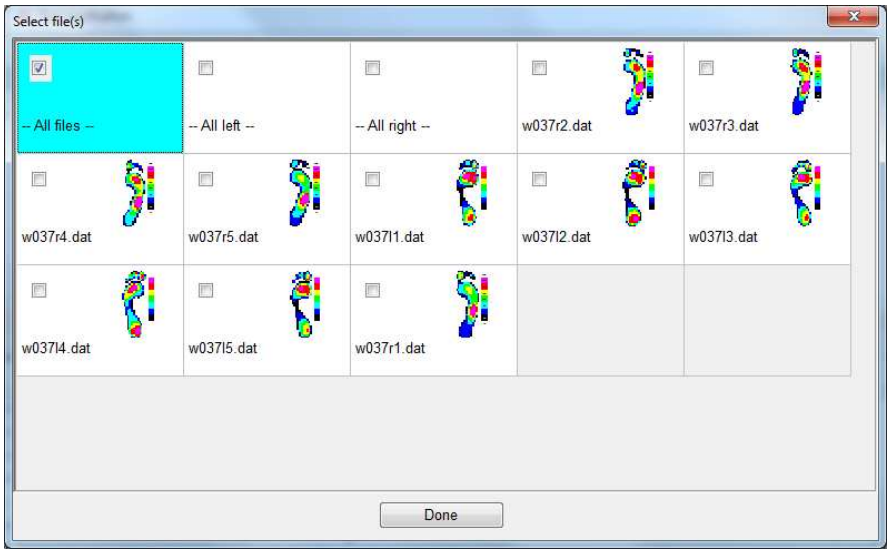
4. **Compare 2 measurement sets** radio button:

5. When the radio button is marked, it is possible to select different visits from the same patient (if more than one visit of patient is stored in the database), different sets of files, or two visits of different patients (if these visits were selected at the Visit level in database prior to clicking **Run application**).

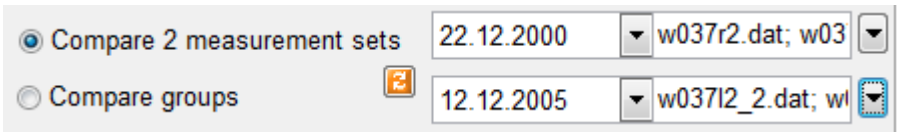


Use the **Swap selections** button to change the order of visits.

Click the button to select a specific set of files. The following window will appear:



Select the appropriate file(s). The names of the selected files will be displayed.



6. **Compare groups** radio button
Before using this option groups should be created with the **Build query** option. See **section 3.4.1**.



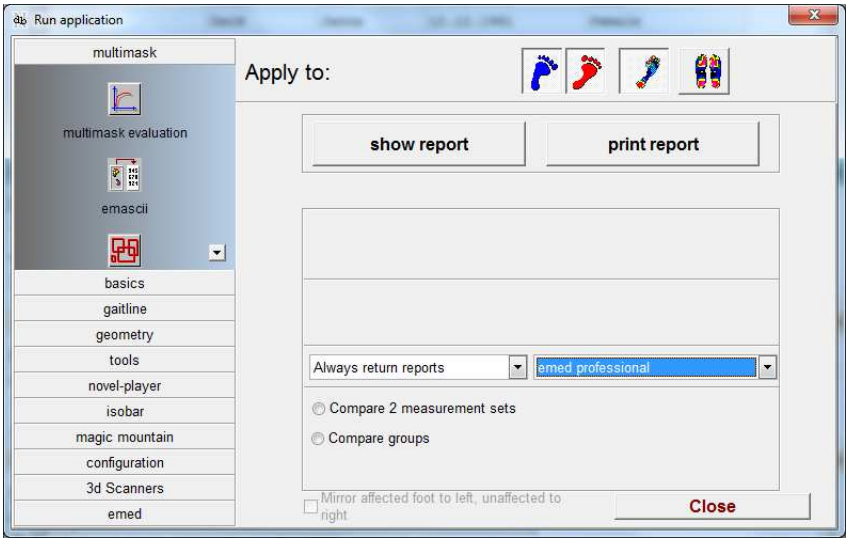
Use **Swap selections** buttons to change the order of the groups.

Click the button to select the group.

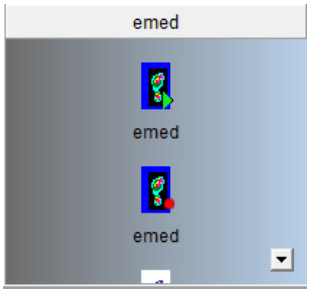
Click the **Fast query select** button to view all existing queries (groups).

NOTE: Groups must first be created using the query function. See **Section 3.4.1**.

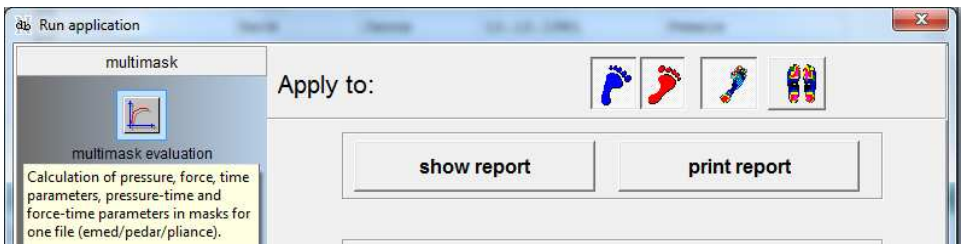
7. All remaining buttons will open the corresponding novel program.



The **Run application** window supports the installed programs dynamically, i.e. a dynamic set of programs is available instead of a predetermined set. The dynamic set of programs is taken from the nvlsoft.ini file (see in **APPENDIX B: novel.ini description**).

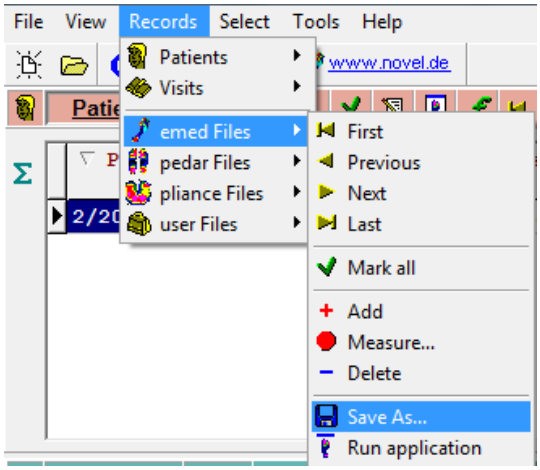


Programs icons are supplied with an extended hint function. To see the hint, move the mouse over the program icon.



g. Save As

Saves the selected data files outside the database (i.e. extracts the file from the database) .

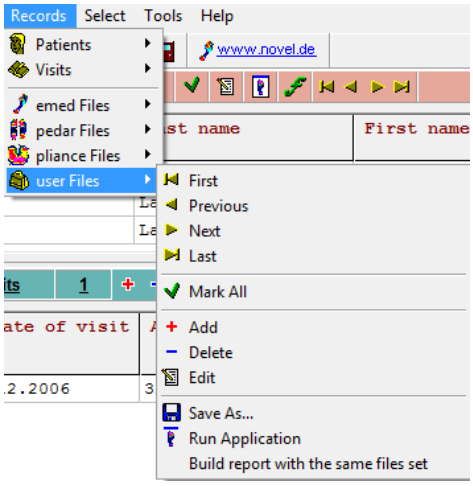


emed Files, pedar Files, and pliance Files tables contain the **Measure** button.

h. **Measure**

Select this icon to collect data using the corresponding system. For example, if the measure icon is selected in the emed files table, the emed data collection software will open. The data files are automatically stored in the database for the selected patient.

User Files table contains the **Edit**, **Run application**, **Build report with the same files set** buttons.



i. Edit

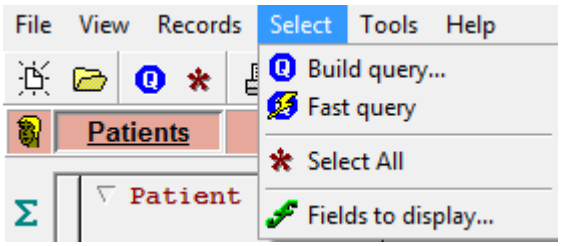
Use this option to edit a report. This option is enabled only for the files with extensions *.mht, *.html, and *.rtf. When a report is selected and the button is clicked (or the option is selected) the report editor form will be opened.

j. Run Application

If file extension is registered in the operating system, then file will be opened in the corresponding program, e.g. *.doc file will be opened in MS Word, *.mht will be opened in Internet Explorer etc.

3.4 Select Menu

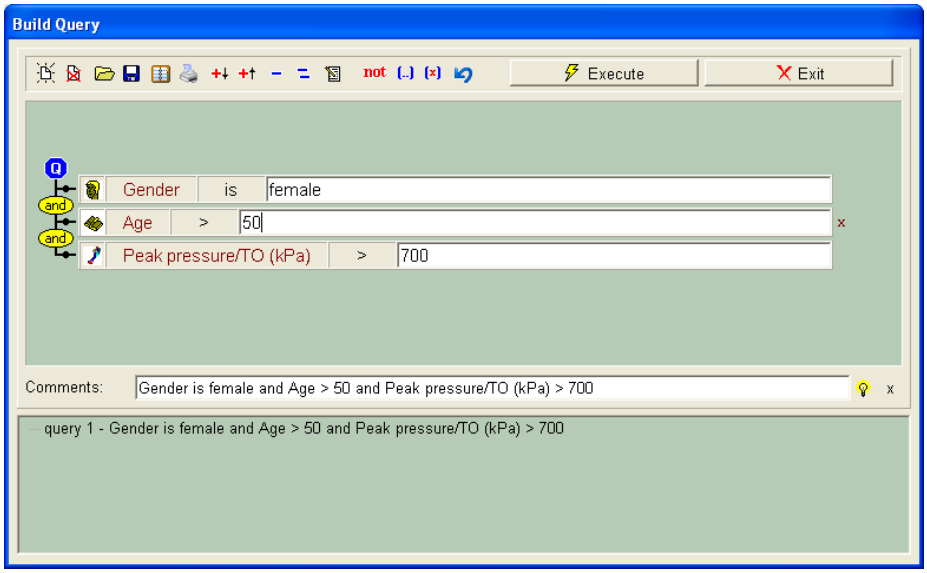
There are four options available in this menu.



3.4.1 Build Query

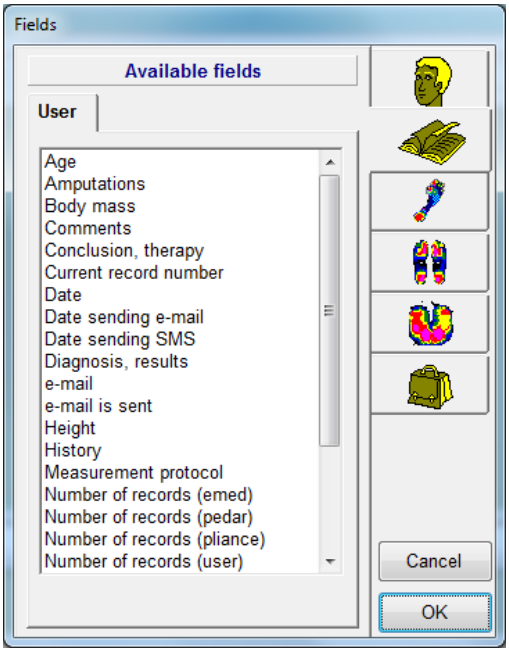


This function is used to build a complex query to look up selective patient data and define subgroups. For the query, all parameters of the database fields can be used. The following example query will select all female patients over 50 with a peak pressure exceeding the 700 kPa. In the comments window a name is given for that selection or patient group. Several different patient groups can be defined and analyzed in the corresponding reports.



To create a new query:

1. Select the query icon  in the database. A blank query window will appear.
2. Click on the **Append condition** button and choose a parameter from the appearing **Available fields** window.

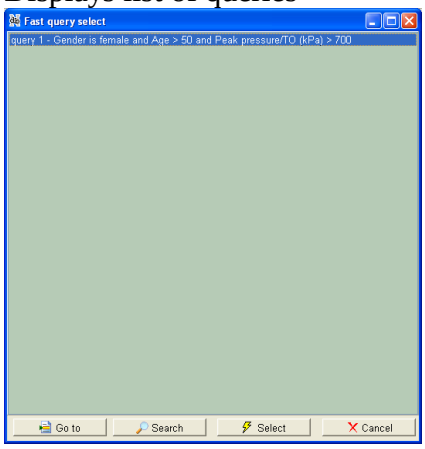
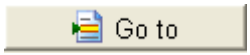
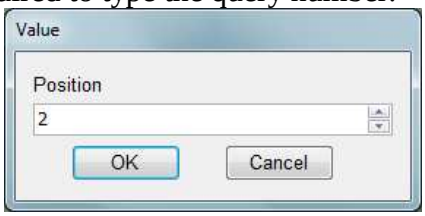
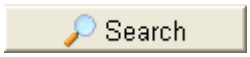
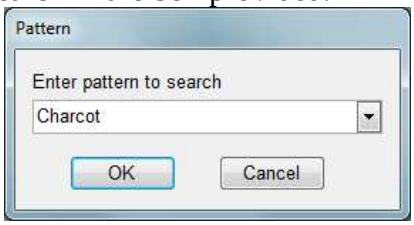
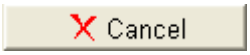


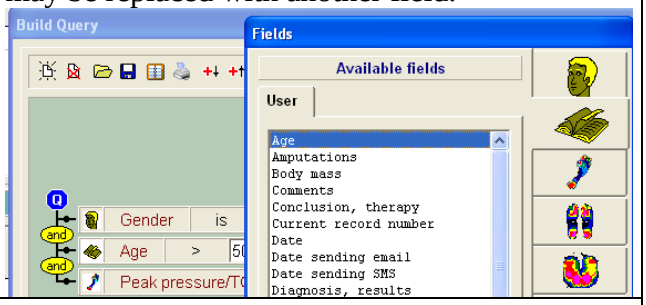
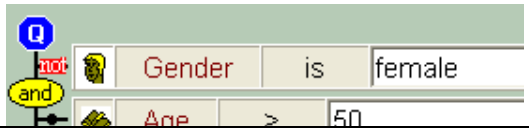
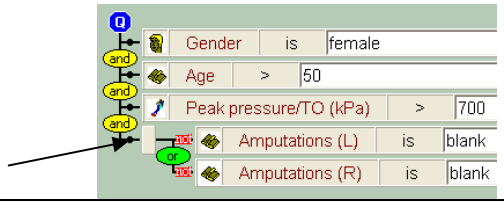
3. Mark the parameter you want to select and click **OK**.
4. Choose the corresponding operator (=, <, >, etc.) and fill in the field on the right side.
   - To use the brackets:
     Click the **Insert brackets** button and mark both positions with a mouse click where the brackets should be inserted. Brackets must be used when an **OR** condition is to be carried out before an **AND** condition.
5. Repeat steps 1-3 until all conditions are defined.
6. Run the query by clicking **Execute** button.

The query can be changed any time by adding a new, editing an existing, or deleting an old condition. Use the **Save query** button to save the settings of this query and make it usable some other time.

The following functions are found in the toolbar:

	**New query**	Creates a new query
	**Delete query**	Deletes the query
	**Load query**	Loads already existing and saved query
	**Save query**	Saves a defined query

	**Fast query select**	Displays list of queries   - click this button to select a specific query number. When button is clicked, it is required to type the query number.   - click this button to find the queries, containing the defined text. Enter the text to search in the box provided.   - click this button to open the selected query.  - click this button to close the window.
	**Print list of queries**	Prints a list of queries.
	**Append condition**	Appends a condition to the query
	**Insert condition**	Inserts a condition in the query. First select a condition (with left mouse button click) within the query definition above which the condition will be added and click this button.

–	**Delete condition**	Deletes a condition of the query. First select the condition (with left mouse button click) and then click this button.
=	**Delete all conditions**	Deletes all conditions of the query
	Edit condition	Edits a condition of the query. When this button is clicked, the selected Field may be replaced with another field. 
not	**Invert condition**	Inverts a condition of the query.  When button is clicked, or with a double click, the condition is inverted. 
(..)	**Insert brackets**	Inserts brackets in the query. When the button is clicked a left bracket mouse cursor will appear (instead of the arrow) when cursor is over the first condition. Click to place the bracket. After this click a right bracket cursor will appear. Click to place the bracket on the last condition. Insert brackets when a logical **OR** should be done before a logical **AND**.
(x)	**Remove brackets**	Removes brackets in the query. Click this button to delete the brackets. 
	Undo	Cancels the last action (returns to the previous state).
	Redo	Repeats the last action
Execute	**Execute query**	Starts the execution of the query
Exit	**Exit**	Exits a query

- For additional **Comments** a field on the bottom of the window can be used.



The comments may be automatically filled (using the defined variables) or by typing a description manually.

- Fill comments automatically

- Clear comments

Note: If comments were included previously, but now the user would like to utilize the automatic fill, first clear the comments and then fill it automatically.

- Every condition consists of a field with the name of the parameter, an operator (=, <, >, etc.), and a field containing alphanumeric values, the date, or a predefined text.
- Every condition is connected with a logical **AND** by default. Double click on the **AND** to change it into a logical **OR**.

NOTE: First the **AND** conditions are completed and then the **OR** conditions are completed. If you want to change the sequence brackets must be inserted.

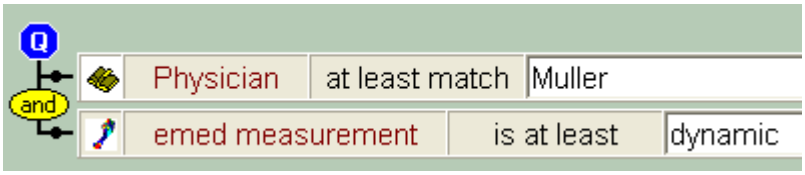
Use the button to delete the alphanumeric values, the date, text etc.

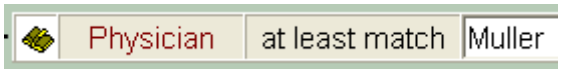
The available operators are listed below

Operator	Explanation
=	Field value is equal to filter value
<	Field value is less than filter value
<=	Field value is equal or less than filter value
>	Field value is greater than filter value
>=	Field value is equal or greater than filter value
is blank	Field value is blank
exact	Field value fully corresponds to the filter value
match	Field value matches to the filter value
at least exact	Field value fully corresponds to the filter value in one of the records of **Visits** or **emed Files** or **pedar Files** or **pliance Files** tables, all visits of this patient or all data files of this visit are displayed.
at least match	Field value matches to the filter value in one of the records of **Visits** or **emed Files** or **pedar Files** or **pliance Files** tables, all visits of this patient or all data files of this visit are displayed. Example. Query: ***Comments at least match Charcot foot*** All patients with all visits and all measured data files will be included in the query result if the words “Charcot foot” are in the Comments field in one of the visits of each found patient.
at least =	Field value is equal to the filter value in one of the records of **Visits** or **emed Files** or **pedar Files** or **pliance Files** tables, all visits of this patient or all data files of this visit are displayed. Example. Query: ***Date of visit at least = 22.12.2000***. All patients with all visits and all measured data files will be included

	in the query result if the Date of visit was 22.12.2000 in one of the visits of each found patient.
at least <	Field value is less than the filter value in one of the records of **Visits** or **emed Files** or **pedar Files** or **pliance Files** tables, all visits of this patient or all data files of this visit are displayed.
at least <=	Field value is equal or less than the filter value in one of the records of **Visits** or **emed Files** or **pedar Files** or **pliance Files** tables, all visits of this patient or all data files of this visit are displayed.
at least >	Field value is greater than the filter value in one of the records of **Visits** or **emed Files** or **pedar Files** or **pliance Files** tables, all visits of this patient or all data files of this visit are displayed.
at least >=	Field value is equal or greater than the filter value in one of the records of **Visits** or **emed Files** or **pedar Files** or **pliance Files** tables, all visits of this patient or all data files of this visit are displayed.
is	Field value coincides with filter value
is at least	Field value fully corresponds to the filter value in one of the records of **Visits** or **emed Files** or **pedar Files** or **pliance Files** tables, all visits of this patient or all data files of this visit are displayed. Example. Query: ***Amputations (L) is at least big toe OR Amputations (R) is at least big toe*** All patients with all visits and all measured data files will be included in the query result if the information about the amputated big toe is in the Amputations field in one of the visits of each found patient.
is one of	Field value at least coincides with selected filter values. Example. Query: ***Amputations (L) is one of big toe, 2nd toe, OR Amputations (R) is one of big toe, 2nd toe*** All patients with visits and corresponding measured data files will be included in the query result if the information about the amputations including big toe or 2nd toe or both is in the Amputations field in the visit of each found patient.
is only	Field value fully corresponds to the filter value (values) Query: ***Amputations (L) is only big toe OR Amputations (R) is one of big toe*** All patients with visits and corresponding measured data files will be included in the query result if the information about the amputations including big toe only is in the Amputations field in the visit of each found patient.
is first	Record with the earliest field date is selected
is last	Record with the latest field date is selected
is current date	Record with the current field date is selected
date =	Date in field value is equal to filter value
time =	Time in field value is equal to filter value

NOTE: It is not reasonable to use an operator containing **at least** with conditions joining with logical **AND**.
For example:



The 

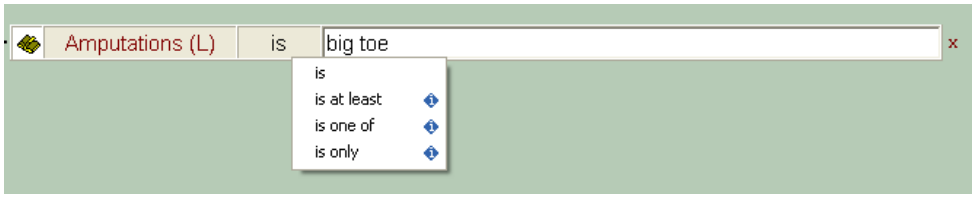
condition should display all visits of patients if at least in one visit the physician's name is Muller or Mullermann etc.

The 

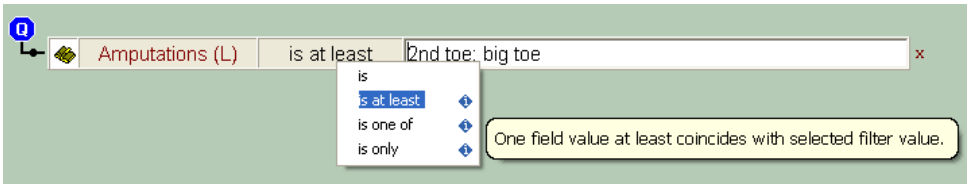
condition should display all files of visit in which at least one emed measurement is dynamic.

It is impossible to display simultaneously one visit to meet the requirements of the second condition and to display several visits to meet the requirements of the first condition. If this query is built then only one visit containing the files with dynamic emed measurement and physician name Muller will be displayed.

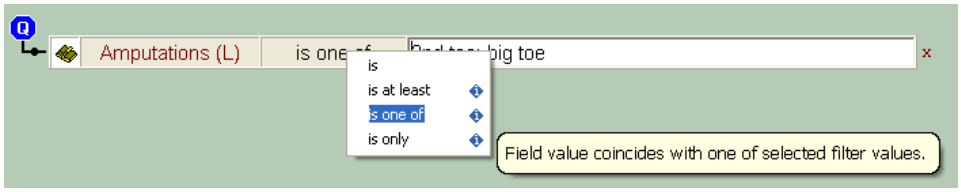
Hints to the following operators are available:
To see the hint, locate the mouse cursor and move it over the icon .



One value only can be selected.
(All records in which big toe is amputated or in concert with another minor amputation will be displayed.)

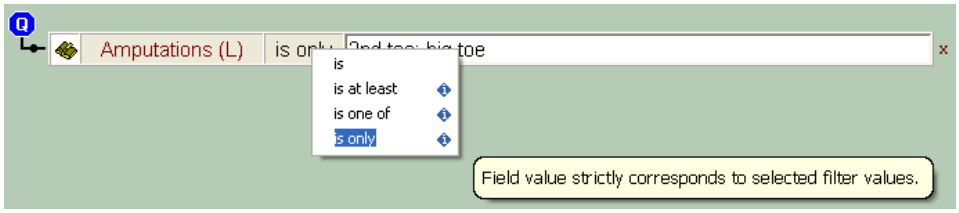


Several values can be selected.
(All patients with amputations of big and second toes only or in addition to other amputations at least in one visit will be found, i.e. all visits of these patients with and without these amputations will be displayed.)



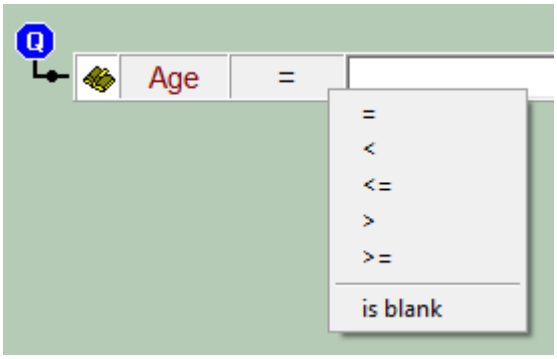
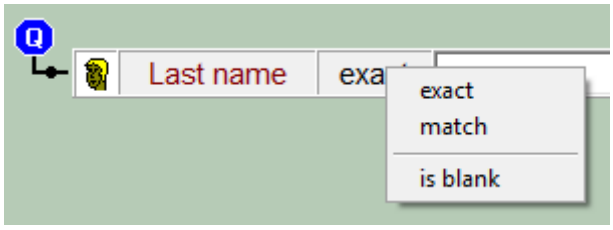
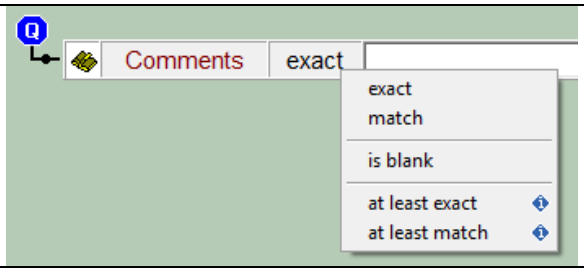
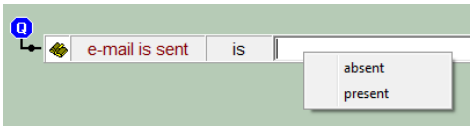
Several values can be selected.
All records from this table level will be found in which at least one field value coincides with the selected filter values.

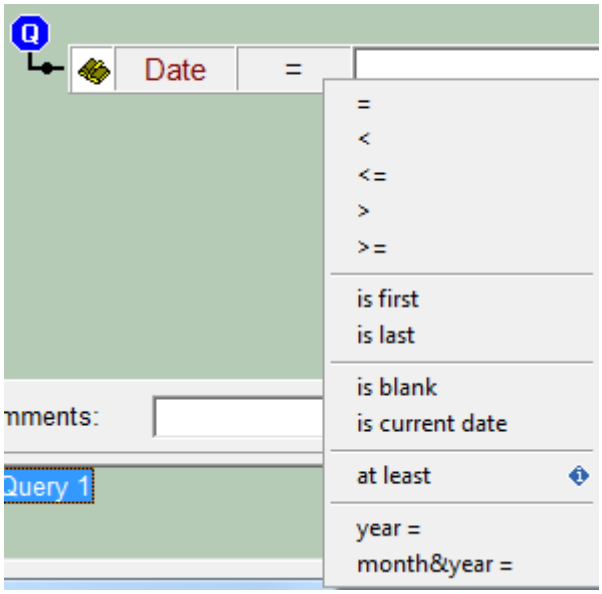
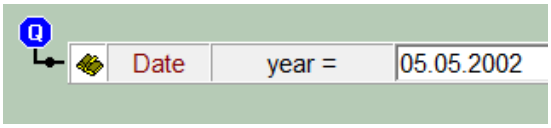
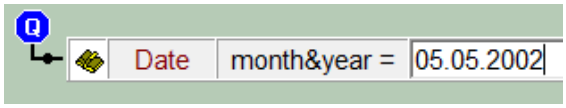
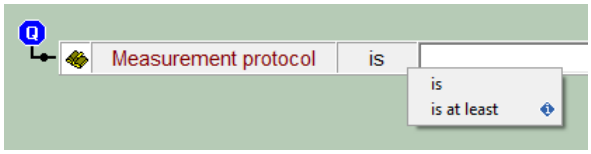
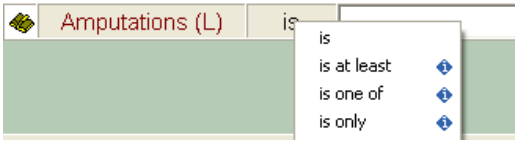
(All visits with amputations of the big or/and second toes in conjunction with other amputations will be found.)

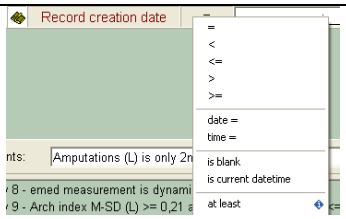


Several values can be selected.
All records from this table level will be found in which only one field value coincides with selected filter values.
(All visits with only amputations of big and second toes will be found.)

Available operators depend on the type of the field.

Type of the field	Operators
Numeric	
String	
memo	
boolean	

date	 **NOTE:** To query for a defined you any date and any month, and the appropriate year must be entered.  Any date may be entered and the appropriate month and year to find a required month and year. 
gradations	
multi gradations	
Date, time	



NOTE: To find a defined date, any time may be entered.



NOTE: To find a defined time, any date may be entered.

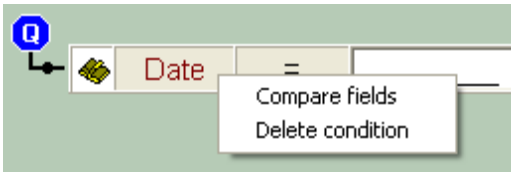


NOTE: When is current datetime operator is selected, the following query is built.

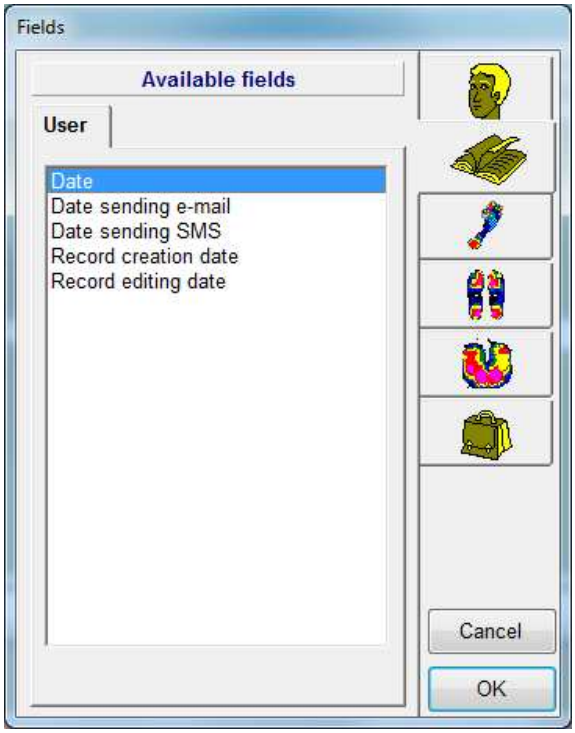


Use **is current datetime** the **date=** operators will find all records which were edited today.

To compare the date type fields select the date type field, right mouse click and select **Compare fields**.



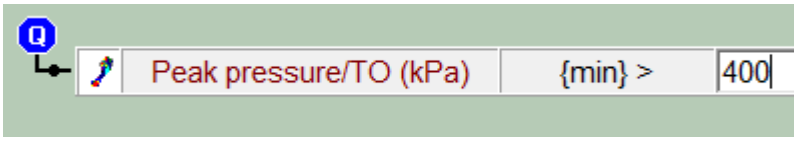
The **Fields** window will be opened.



Any date type field can be selected for comparison, e.g.



Query using **{min}** can be built**.**



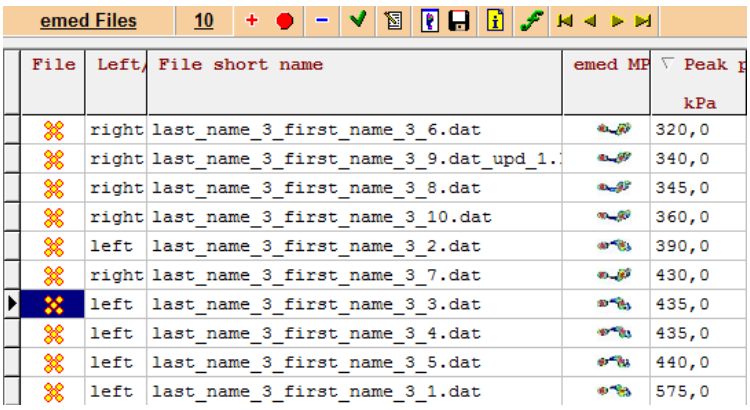
All visits with data files in which minimum peak pressure is greater than 400 kPa will be displayed.

Example:
First patient visit data

emed Files 10

File	Left/	File short name	emed MP	Peak p kPa
	right	Last_name_2_First_name_2_9.dat		490,0
	right	Last_name_2_First_name_2_1.dat		495,0
	right	Last_name_2_First_name_2_2.dat		505,0
	right	Last_name_2_First_name_2_3.dat		580,0
	right	Last_name_2_First_name_2_8.dat		625,0
	left	Last_name_2_First_name_2_6.dat		680,0
	left	Last_name_2_First_name_2_4.dat		710,0
	left	Last_name_2_First_name_2_10.dat		745,0
	left	Last_name_2_First_name_2_5.dat		845,0
	left	Last_name_2_First_name_2_7.dat		910,0

Second patient visit data

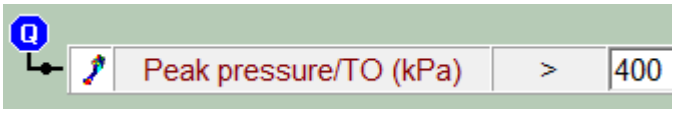


emed Files 10

File	Left/	File short name	emed MP	Peak p kPa
	right	last_name_3_first_name_3_6.dat		320,0
	right	last_name_3_first_name_3_9.dat_upd_1.:		340,0
	right	last_name_3_first_name_3_8.dat		345,0
	right	last_name_3_first_name_3_10.dat		360,0
	left	last_name_3_first_name_3_2.dat		390,0
	right	last_name_3_first_name_3_7.dat		430,0
	left	last_name_3_first_name_3_3.dat		435,0
	left	last_name_3_first_name_3_4.dat		435,0
	left	last_name_3_first_name_3_5.dat		440,0
	left	last_name_3_first_name_3_1.dat		575,0

Minimum peak pressure is defined in each visit.
It is equal to 490 kPa and 320 kPa respectively.
Only the ten files from the first patient will be included in the search result.

For a comparison, a query without **{min}** is built.

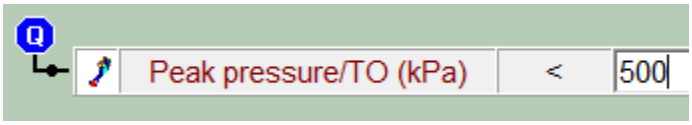


The query results in 15 records.

The data files in which peak pressure is greater than 400 kPa regardless of the minimum peak pressure in the data set referring to the visit will be displayed, i.e. 10 files (> 400 kPa) will be included from the first patient and 5 files (> 400 kPa) will be included from the second patient.

NOTE: The results of the next 2 queries will be identical.





NOTE: Only one aggregate function ({min},{max}) can be selected at a time in the query otherwise the following window will appear.



3.4.2 Fast query

Select the data needed to be found, e.g. Date of visit equal to 22.12.2006 and click the Fast search button.
All records with this date of visit will be found.

3.4.3 Select all



Use this command to display all records of the database after executing a query (i.e. clear the query).

3.4.4 Fields to Display



The fields to be displayed in each of the database tables (e.g. Patients, Visits, emed/pedar/pliance/user Files) can be selected from all available database fields. The sequence for the fields in the database tables can also be edited.

- **Available fields** displays all possibilities in the different groups of fields.
- **Displayed fields** lists all selected fields to be viewed in the tables.

Use these buttons for field (fields) selection

 - Add this list of parameters

 - Add selected field



- Add all available fields / Clear all displayed fields (across all tabs/tables – Patients, Visit, emed/pedar/pliance/User Files)

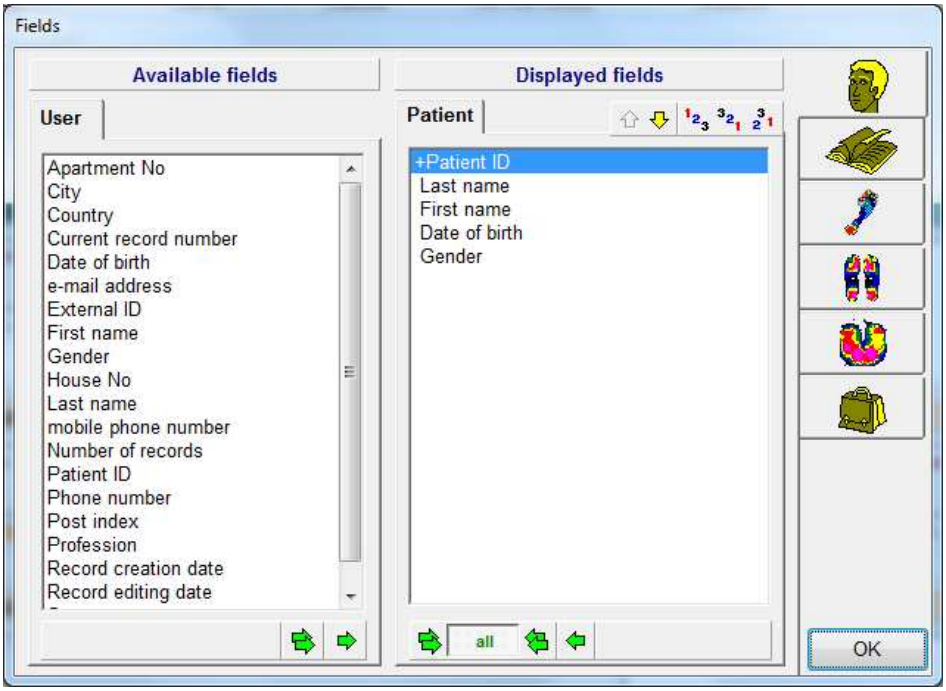


- Add all available fields from this tab / Clear displayed fields from this tab.

 - Clear selected field

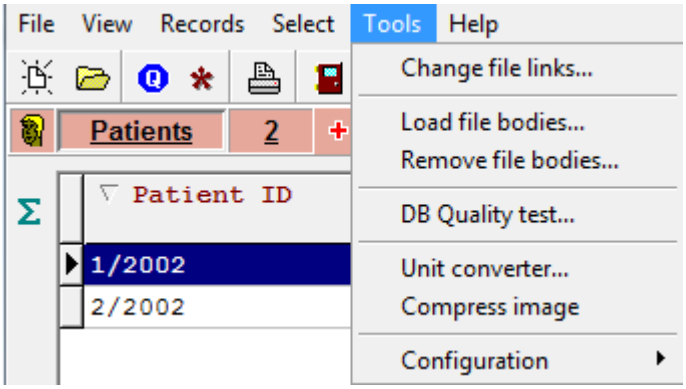
- Select the tabs along the right side of the window to see parameters from the different tables.
- To shift a field from one window to another, select the field and click on a button with a single green arrow pointing in the direction of movement or double click on the field.

- Using the twin arrow all parameters (**all**) or the parameters of the selected index card (**this**) can be moved with one click.



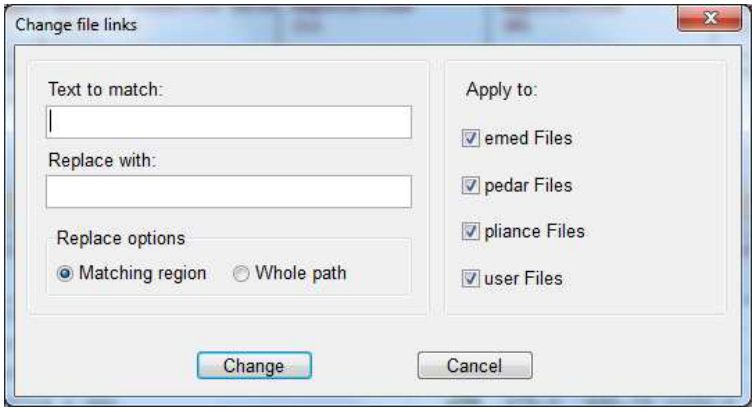
- The yellow arrows  change the sequence of the fields.
- To arrange a field in ascending or descending order, mark the field and select the corresponding number button  .

3.5 Tools Menu



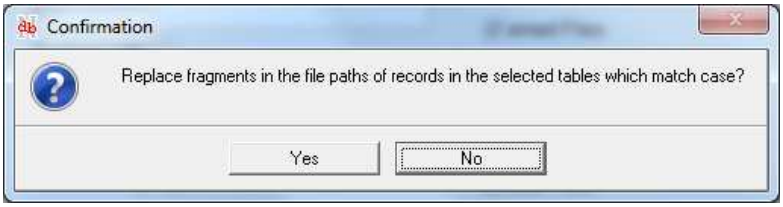
3.5.1 Change File Links

This option is used to change the path of the files which are linked to the database. This can be useful after exporting the database to another computer or if the directory structure has changed and not all data is contained within the database.



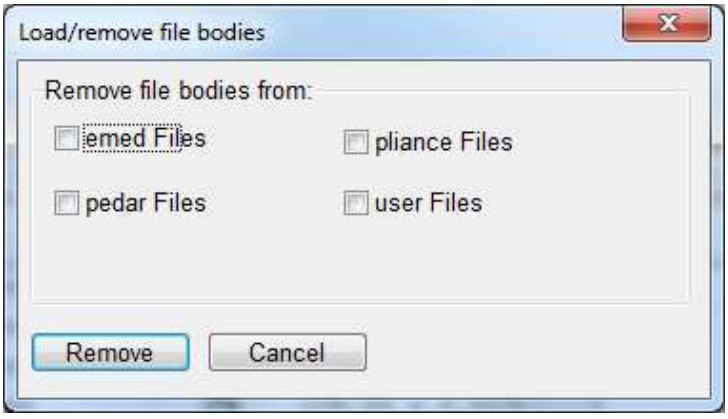
- Write the text you are looking for in the **Text to match** edit box**.**
- In the **Replace with** edit box write the new text you want to replace the original with.
- In the **Replace Options** section you can select if only the **Matching region** of the text or the **Whole path** should be replaced. **Matching region** should be used if the directory has changed for all of the data files, and **Whole path** should be chosen for editing the path of only one file.
- In the section **Apply to**, chose the types of files to apply the changes to. One or all file types can be chosen.

When all options within the window have been chosen, and the **Change** button is clicked, the following message window will appear.

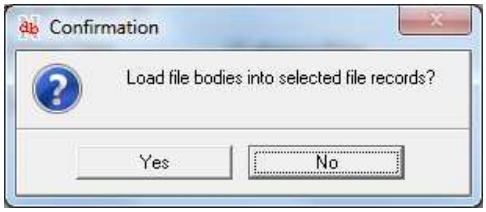


NOTE: **Text to match** and **Replace with** should start from the first character, e.g.:
Text to match: C:\novel
Replace with: D:\novel1.

3.5.2 Load File Bodies



This option is used to fully load data file bodies to the database where just the data file links are included. Selection of files is required before the **Load** field is clicked.
When the **Load** button is clicked, the following window will appear



Once the **Yes** button is clicked, the process will start.
If file is not found on the hard disk, the following window will appear.

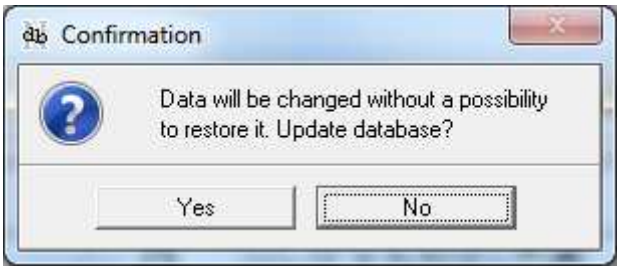


It is possible to skip the file (**OK**), to skip all absent files (**All**) or to cancel the procedure (**Cancel**).

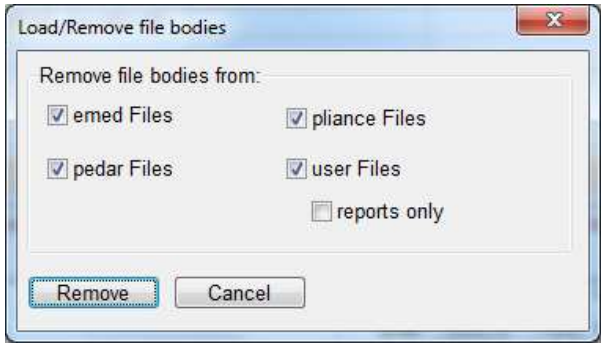
NOTE: By default the file bodies are loaded into the database. This can be changed under the **Tools | Configuration | Local** menu item.

3.5.3 Remove File Bodies

This option may be used to remove the data file bodies (copies of the data files contained WITHIN the database) from the database and leave behind just the link to the files. Selection of files is required before the **Remove** button is clicked.
When the **Remove File Bodies** option is clicked (required number is entered), the following window will appear.



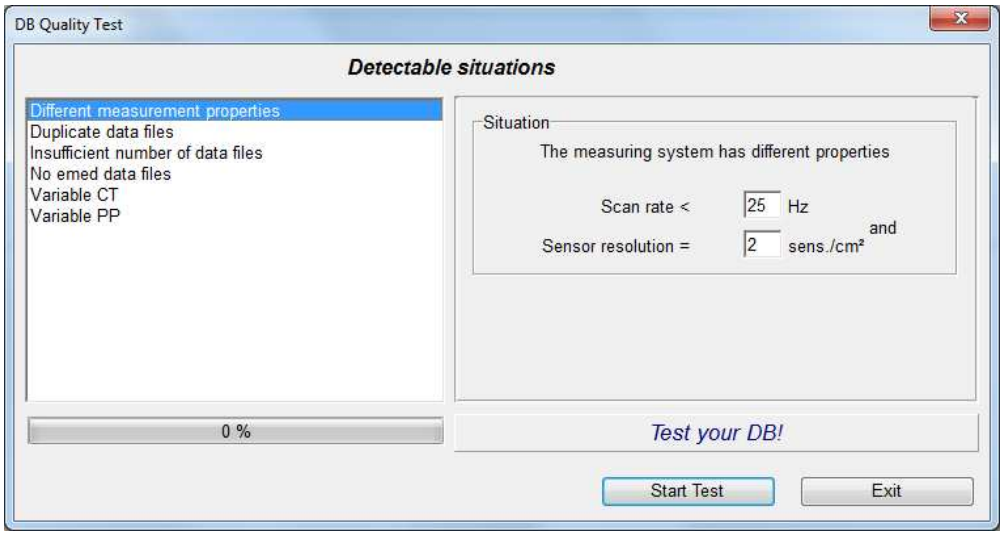
When the **Yes** button is clicked the window to select the files appears.



When the **Remove** button is clicked the procedure of removing the file bodies will begin.

3.5.4 DB Quality Test

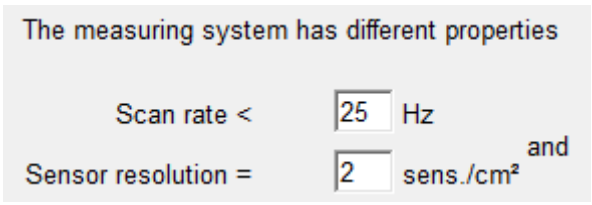
Use this test to find inconsistent or questionable records or data in the database.



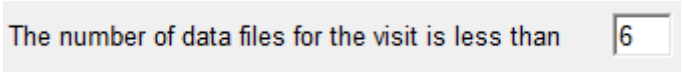
NOTE: This test is only available for emed data.

The following situations can be tested:

- **Different measurement properties**

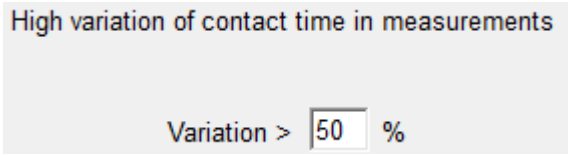


  Checks if equipment with different frequencies or sensor resolutions were used for the measurements.
- **Duplicate data files**
  Checks if identical data with different data names exists within the database.
- **Insufficient number of data files**
  Checks for a minimum number of data files which must be present for each visit.

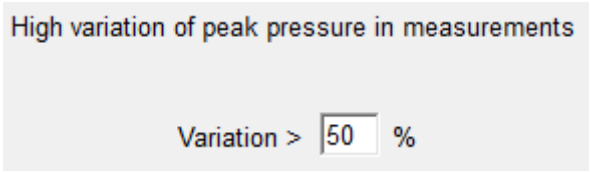


The query will find the records in the **Visits** table in which the number of measured emed files is less than value specified.

- **No emed data files**
  Finds the records within the **Visits** table in which no data files are present.
- **Variable CT**
  Tests if the variation of contact time is higher than a predefined limit.



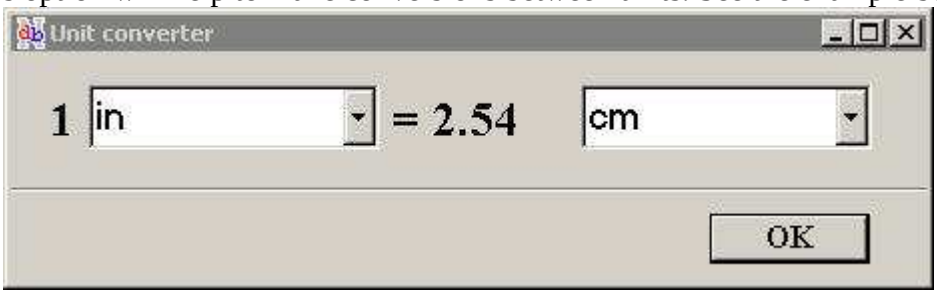
- **Variable PP**
  Tests if the variation of peak pressure is higher than a predefined limit.



After editing the parameters, click the **Start Test** button to run the function. The number of the detected errors will then be indicated. The result is shown in the columns of **Quality Factor** of **Visits** and **Data file** tables.

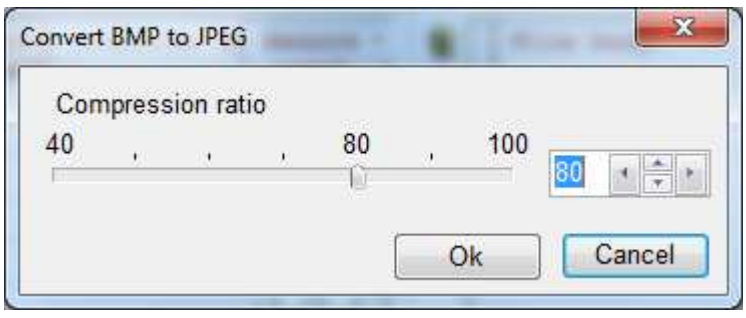
3.5.5 Unit Converter

This option will help to make conversions between units. See the example below.



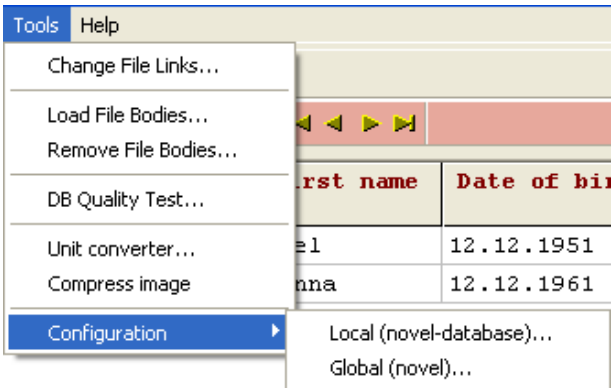
To use the converter, simply choose the units from the pull down lists and the conversion factor will appear.

3.5.6 Compress image



Use this option to compress all images contained within the database.
NOTE: After compression is performed, it is impossible to restore the images.

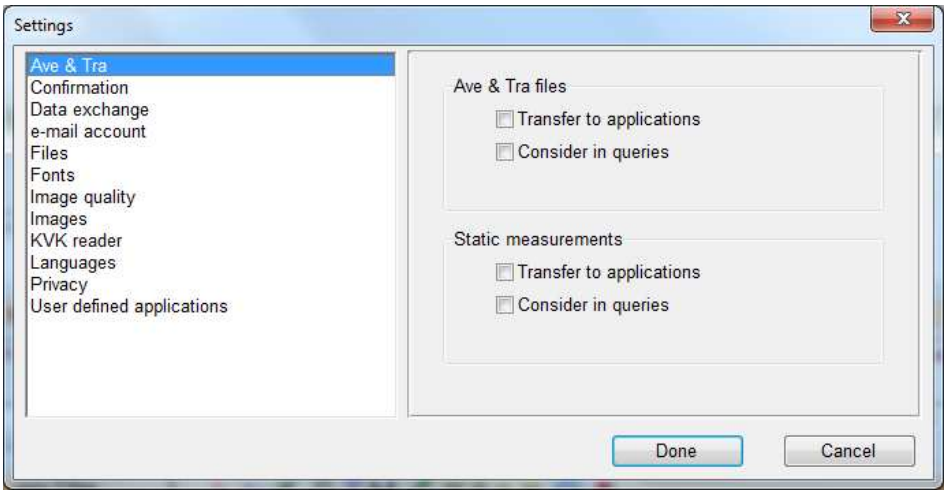
3.5.7 Configuration



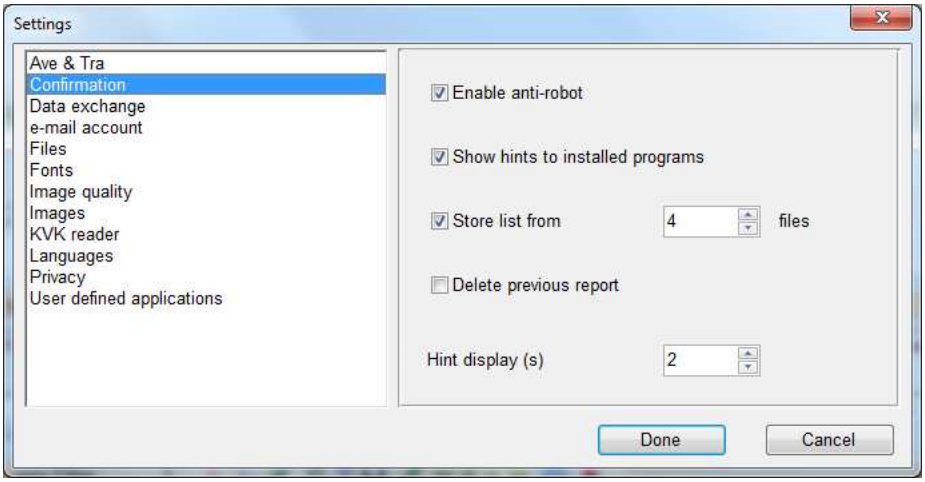
- **Local (novel-database)**

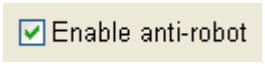
This function is used to define the local settings of the database.

- **Ave & Tra**
  Mark the corresponding check boxes to transfer average and transformed files and/or static measurements to applications, to consider it in queries. By default, no check boxes are marked.



- **Confirmation**





- whenever an action is taken to delete a file from the database, or an action is taken to change the database, a confirmation window will appear where a random number must be entered. This checkbox provides the ability to turn this option off. We recommend leaving this option enabled to avoid accidental changes to the database.



- when the mouse is placed over a program a hint will be provided on the use of the installed program. If this option is not desired, this checkbox turns the option off. Hints may be edited within the nvlsoft.txt file located at c:\novel\novprog\system\National.



- for quick access to previously opened databases the software will display a list. This checkbox turns this option on or off and the number of files desired within the list can be modified.

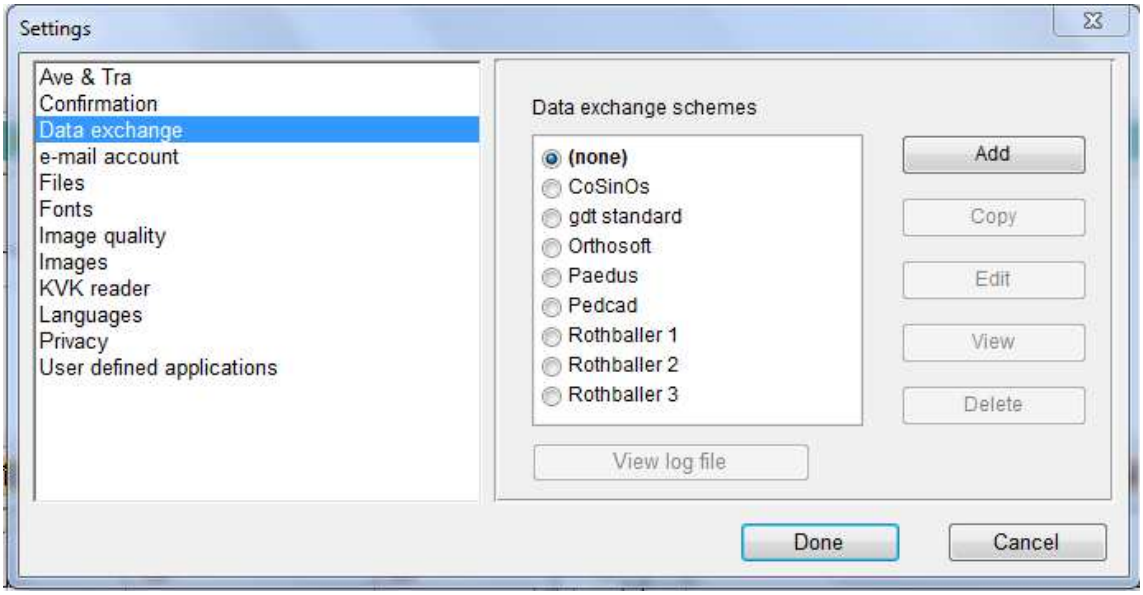


- after a report is edited from the **user Files** table, it is returned back in the database. If this checkbox is marked, the original report will be deleted. Please, select which option is preferable - to store the original and edited reports (box is unchecked), or the edited report only (box is checked).



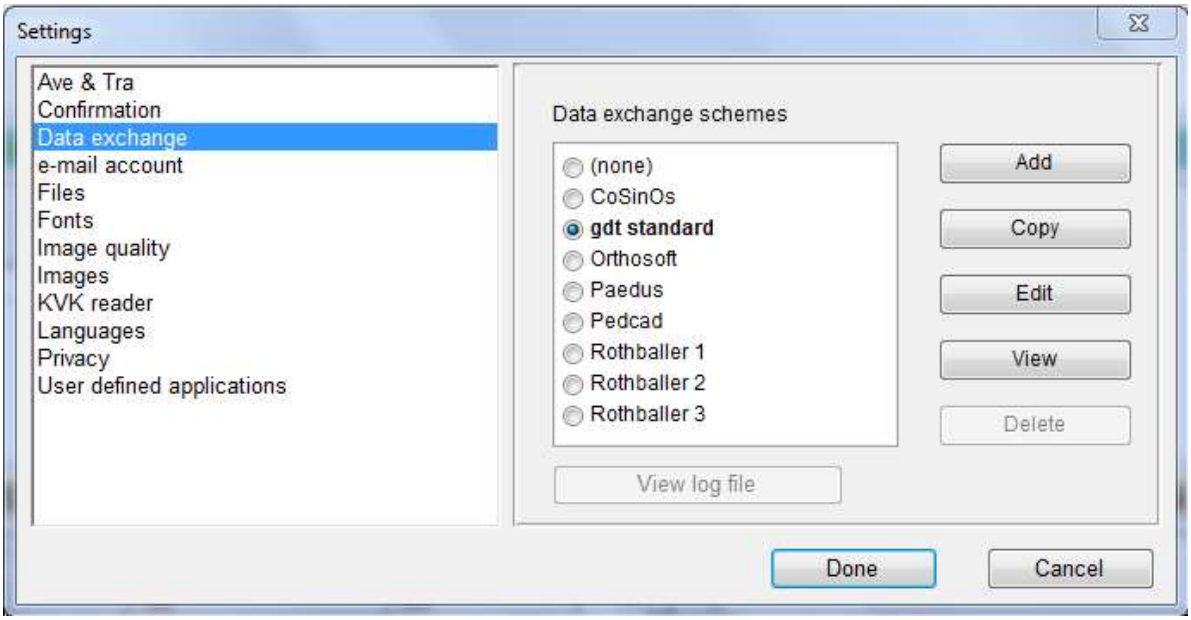
- select the time during for which a hint is displayed.

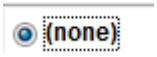
- **Data exchange**



Data may be exchanged between the novel database and other database systems. The database software contains various predefined data exchange schemes (DES). The schemes are listed in the right part of the Settings window.

Select the appropriate DES using the radio buttons. Only one scheme is available at a time. The currently selected scheme is displayed in bold font.



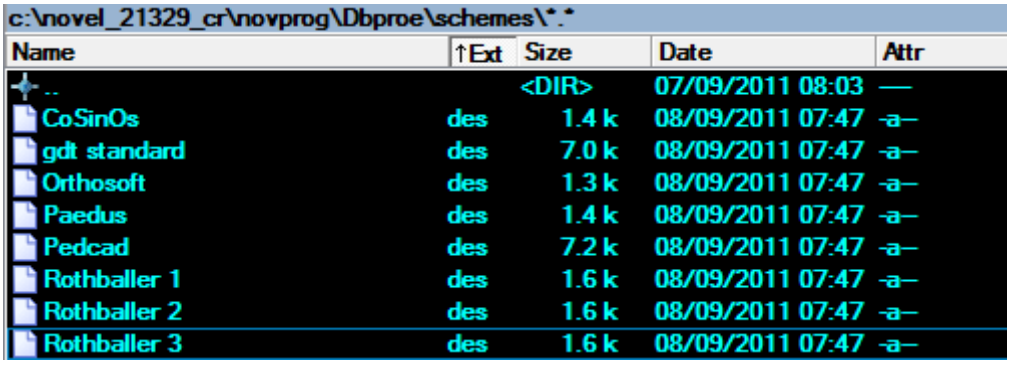
When  is selected, no DES will be used.

Pedcad DES is automatically activated in the presence of the …\novprog\system\dll\nvl2pdcd.dll file.

Use the buttons on the right side of the window Add to add, Copy to copy, Edit to edit, View to view, and Delete to delete the selected DES.
NOTE: Only the last three schemes may be deleted.

Use the View log file button to view the log file with data exchange protocol.

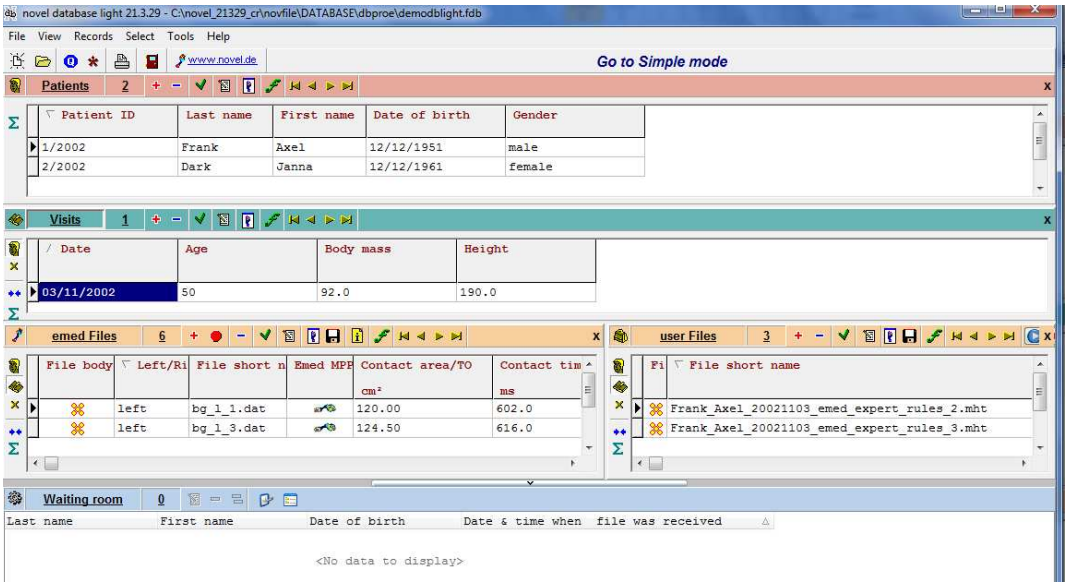
Each DES is stored in the separate *.des file in the c:\novel\novprog\dbproe\schemes\ folder.



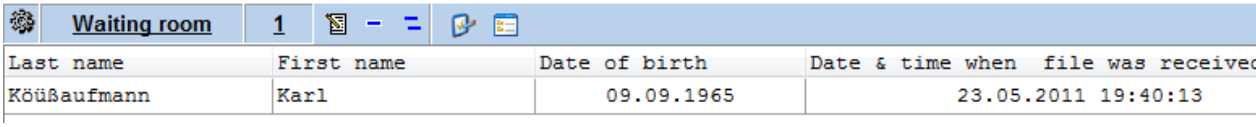
DES description is given in

APPENDIX L: Data exchange (detailed description)

1. Function of the Data Exchange Feature
   When the definite DES is selected the Waiting room table is displayed.



   a. Waiting room table



   b. Buttons description

 - use this button to process the current record;
 - use this button to delete the current record;
 - use this button to delete all records;
 - use this button to open the Local settings at the Data exchange folder;
 - use this button to view the summary of the parameters for the selected DES.

   c. Processing of the files

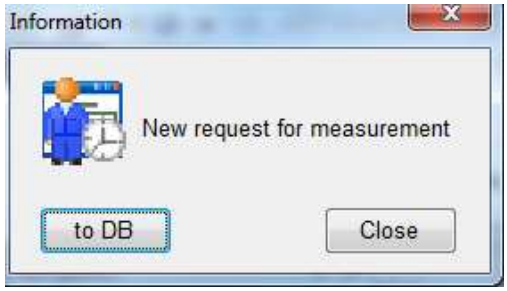
      1. New request for measurement
         - When a new request for measurement occurs, the file is validated for all required fields.
         - If the file is invalid, a warning message appears and the file on the hard disk is deleted.

- If the file is valid, the corresponding record appears in the Waiting room window and the file is renamed as finish.xxx on the hard disk.

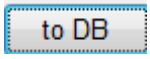
2. New request windows

When new data exchange file is found, a popup window appears. There are three types of popup windows:

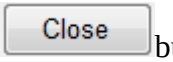
1. New request for measurement



Buttons description

Use the 

button to activate the database.

Use the 

button to close the pop-up window.

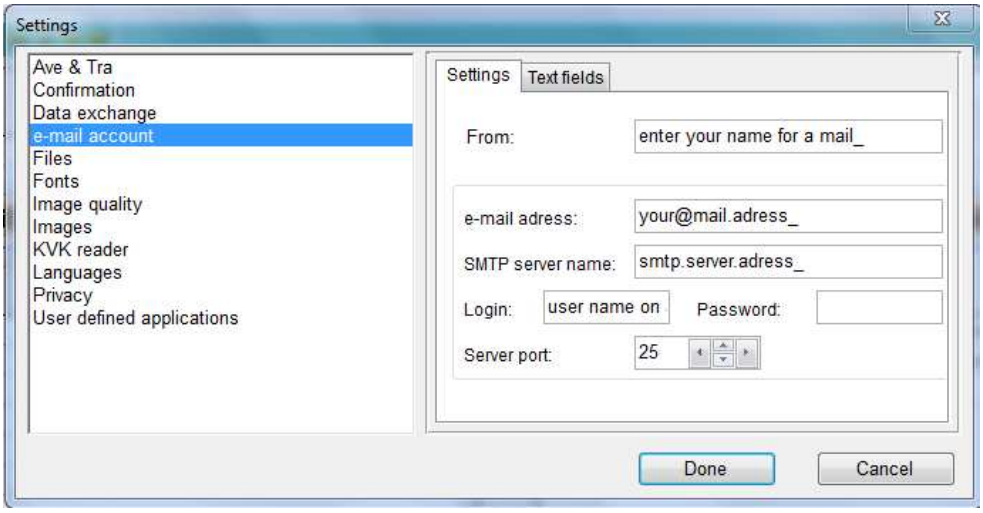
The examples of the input files for several data exchange schemes are listed in

APPENDIX L: Data exchange (detailed description).

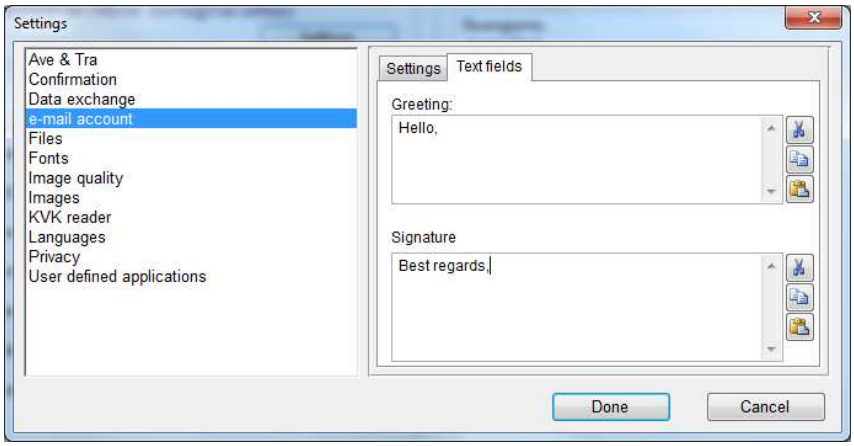
Description of settings for exchange with novel software in MCS-ISYNET format is given in **APPENDIX C**.

The description of the settings for exchange between the novel software and PAEDUS is shown in **APPENDIX D**.

- **e-mail account**

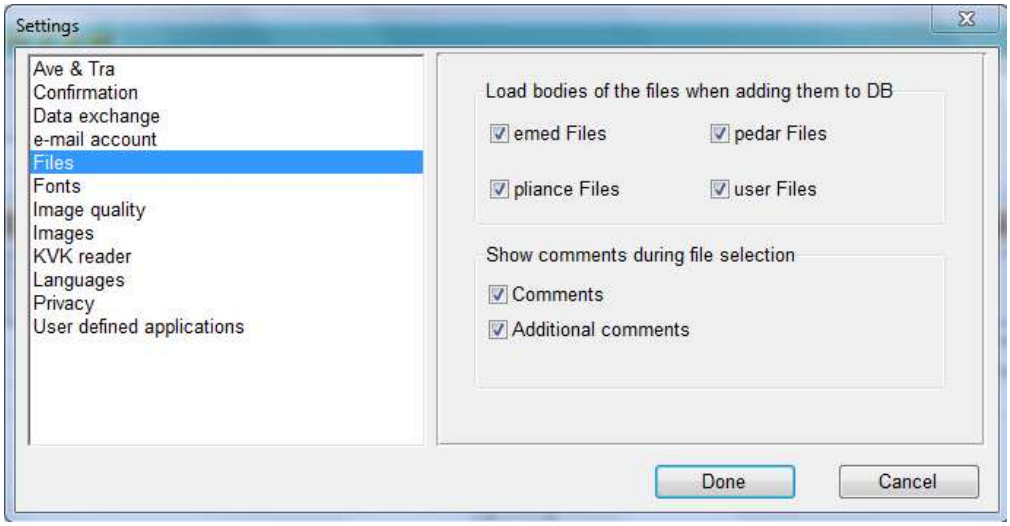


Use these account settings to send e-mails without an external email service (Outlook etc).
Type the default greetings and signature in **Text field**s tab.



Use **Cut**  , **Copy**  , and **Paste**  buttons to add or edit a text.

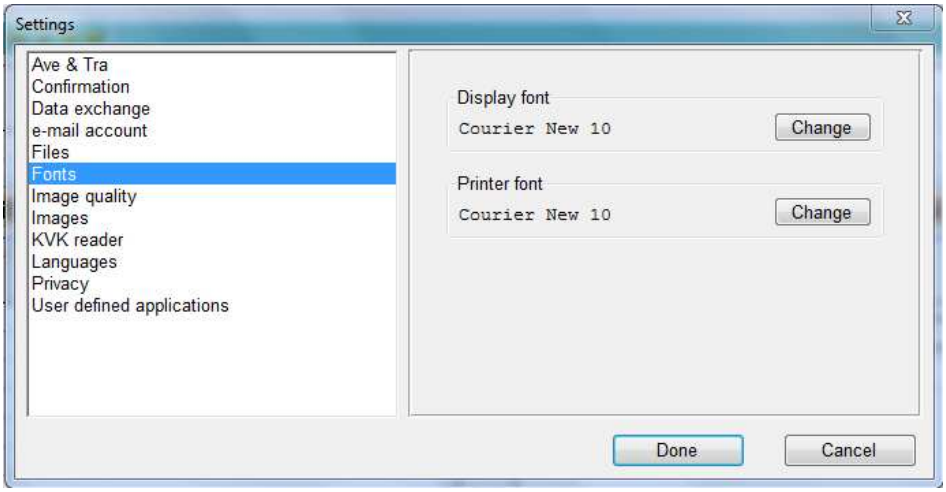
- **Files**



Select the checkboxes to load the file bodies (a copy of the actual data file) when adding it in database. During file selection the user may show comments and/or additional comments for help with file selection.

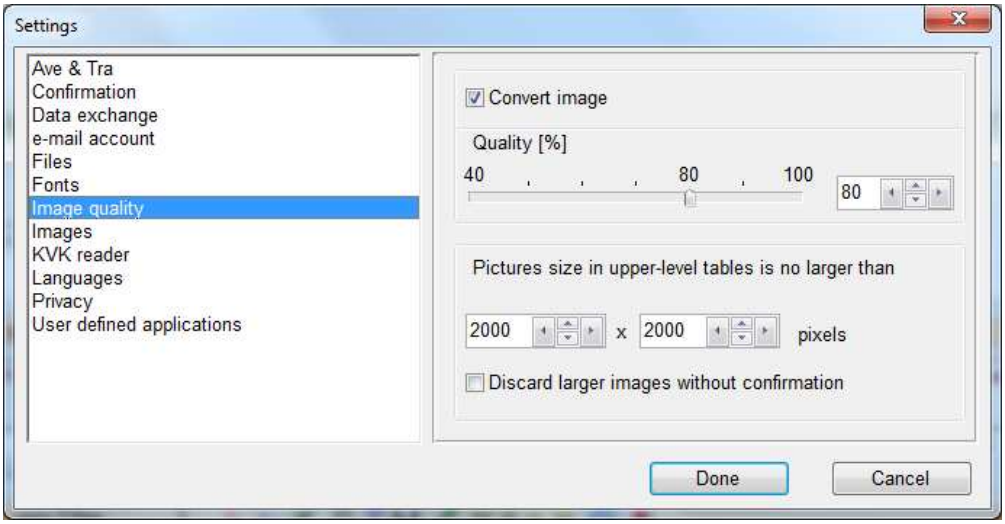
NOTE: COPIES of all data files indicated here will be placed within the database. Therefore, if data is deleted from the database, it is NOT deleted from the original storage folder in which the files exist. Also, if all files are stored within the database the database file will be a self-inclusive file which can move locations with access to all stored data.

- **Fonts**



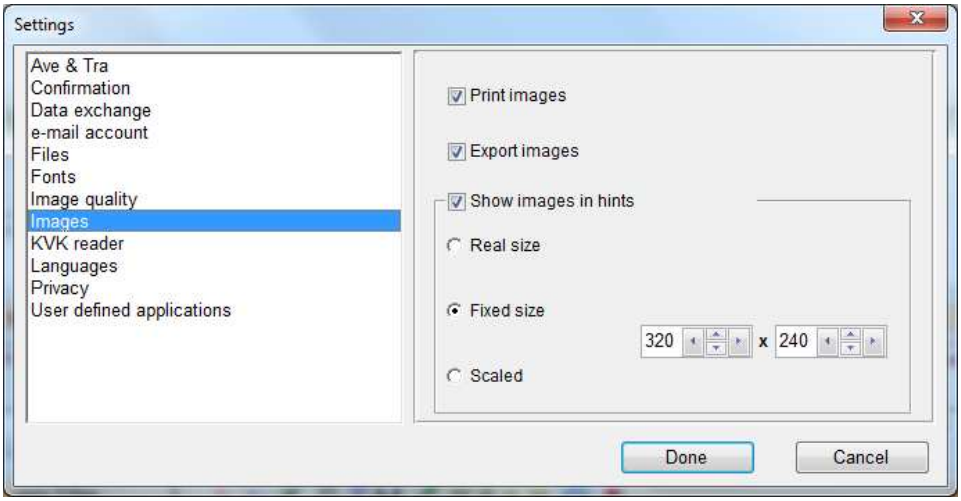
The fonts can be changed for the software display within the software and also on any printout.

- **Image quality**



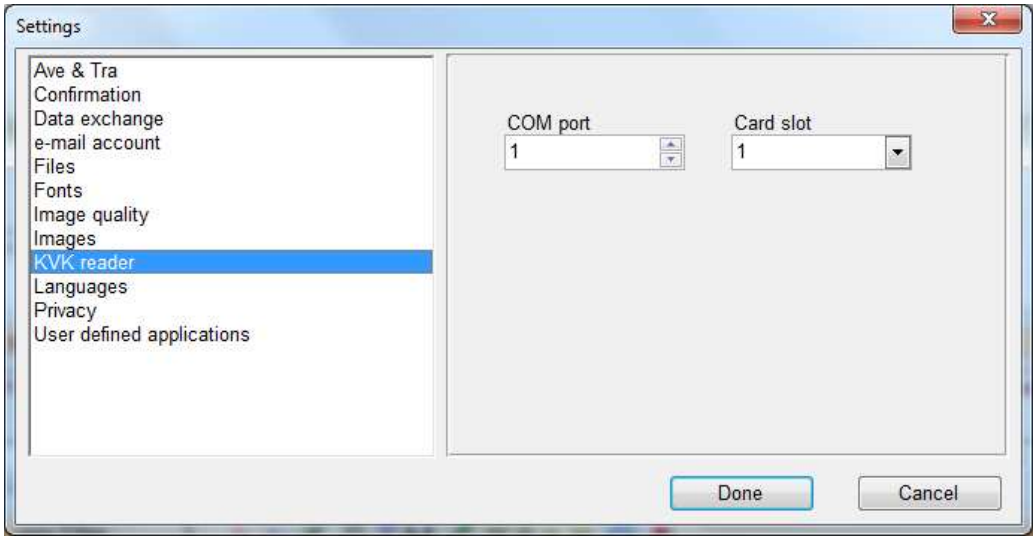
Use these settings to convert images to a defined quality level. The user may also limit the picture size and choose to discard larger images without confirmation.

- **Images**



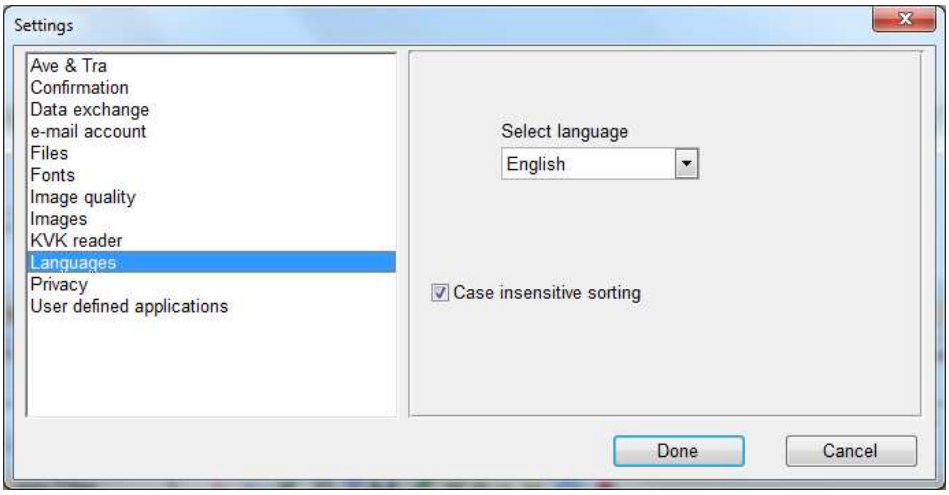
The user may select to print, export, or show images in hints. Also, this may be done with 3 different sizes: real (actual) or fixed or scaled.

- **KVK reader**



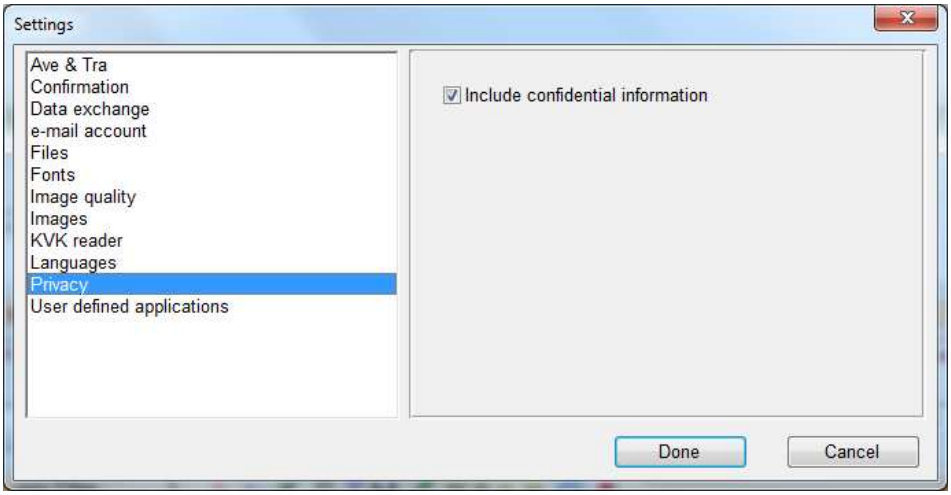
Select the COM port and card slot for the reader. The COM port and Card slot selection is described in the Card reader manual.

- **Languages**



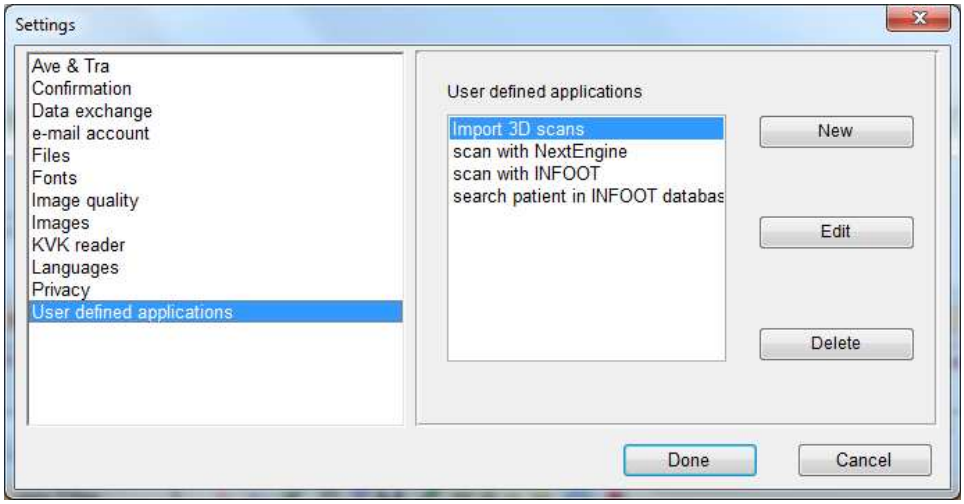
The display language for the software is chosen here. To display all text in English, the **English** option must be chosen. The user may choose to uncheck the option for case insensitive sorting.

- **Privacy**



Private information includes Last name, First name, Phone number, Street, House No., Apartment No., Post index, City, Country. In most cases the Last name and First name fields are transferred to the data collection programs (`emed`, `pedar`, `pliance`) when the **Measure** button is clicked. If this option is turned off, the Patient ID will be transferred to the programs instead of Last name and First name.

- **User defined applications**



This option is used to add new applications to the **Run application** window or to scan the selected folder (s) for a new file (s) with defined mask (s).
A detailed description of this function is given in **APPENDIX E: User defined applications**

- **Global (novel)**

This menu starts the `configuration` program. It allows modification of the global settings that affect all installed `novel` software programs.

IMPORTANT

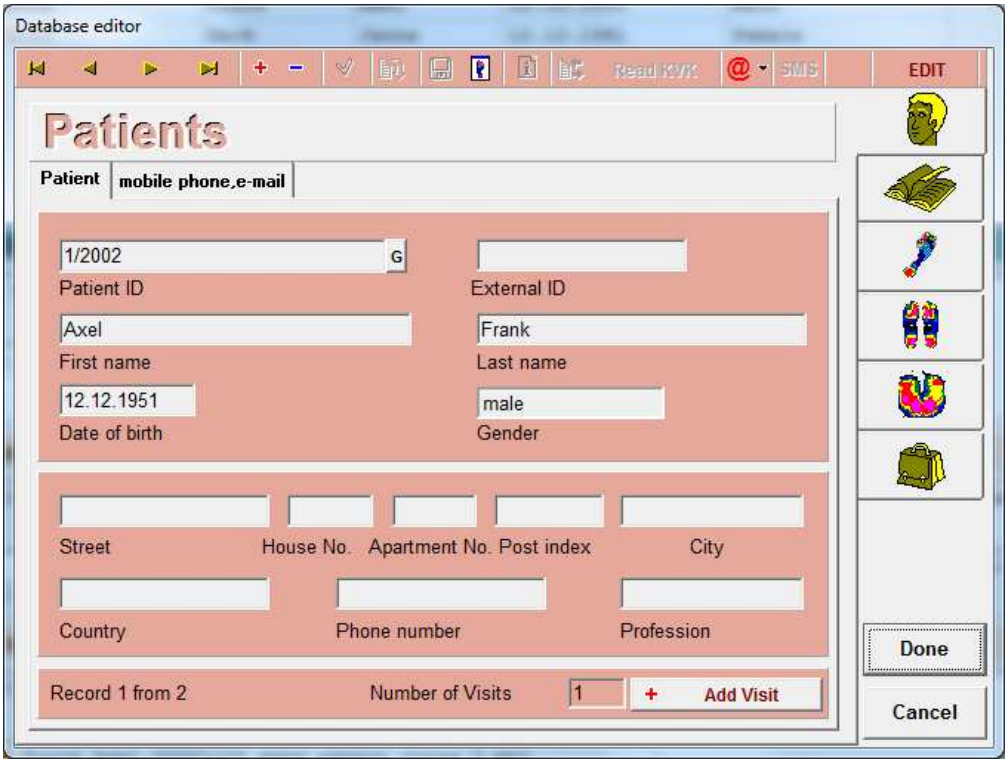
All novel software programs include the configuration program. Changes in the setting of this program will automatically affect
all the programs!

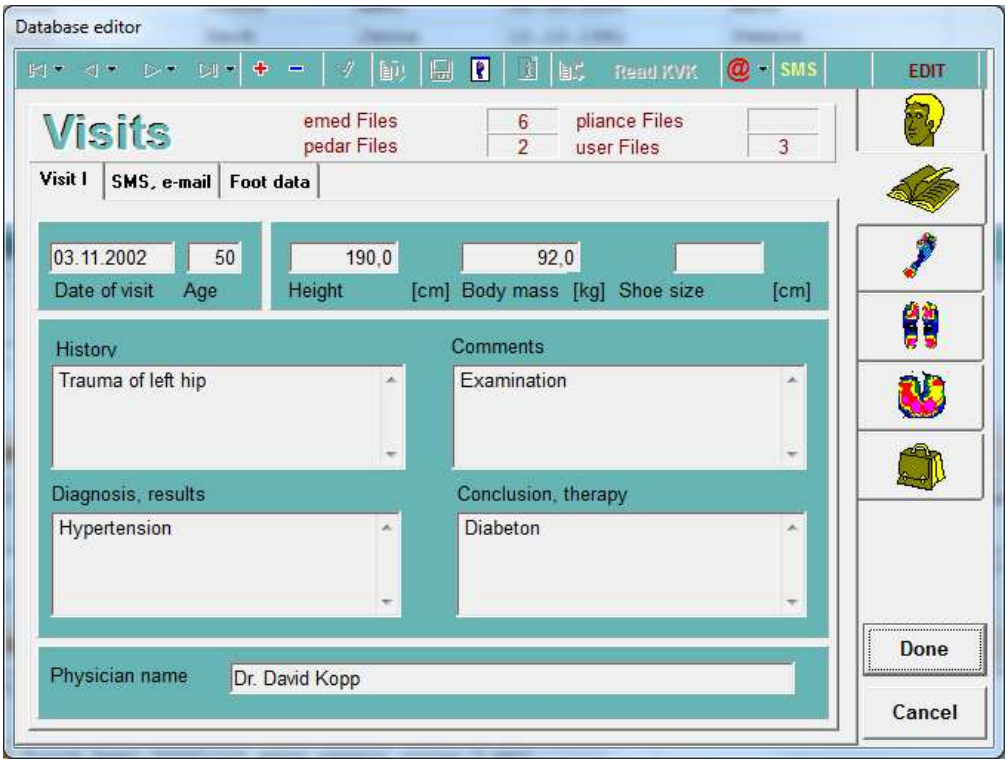
If the settings are changed, click **OK** to apply the changes. Click the **Default** button to restore the standard/default settings.
For more information, see **APPENDIX F**.

##########

4 The Fields of the Database

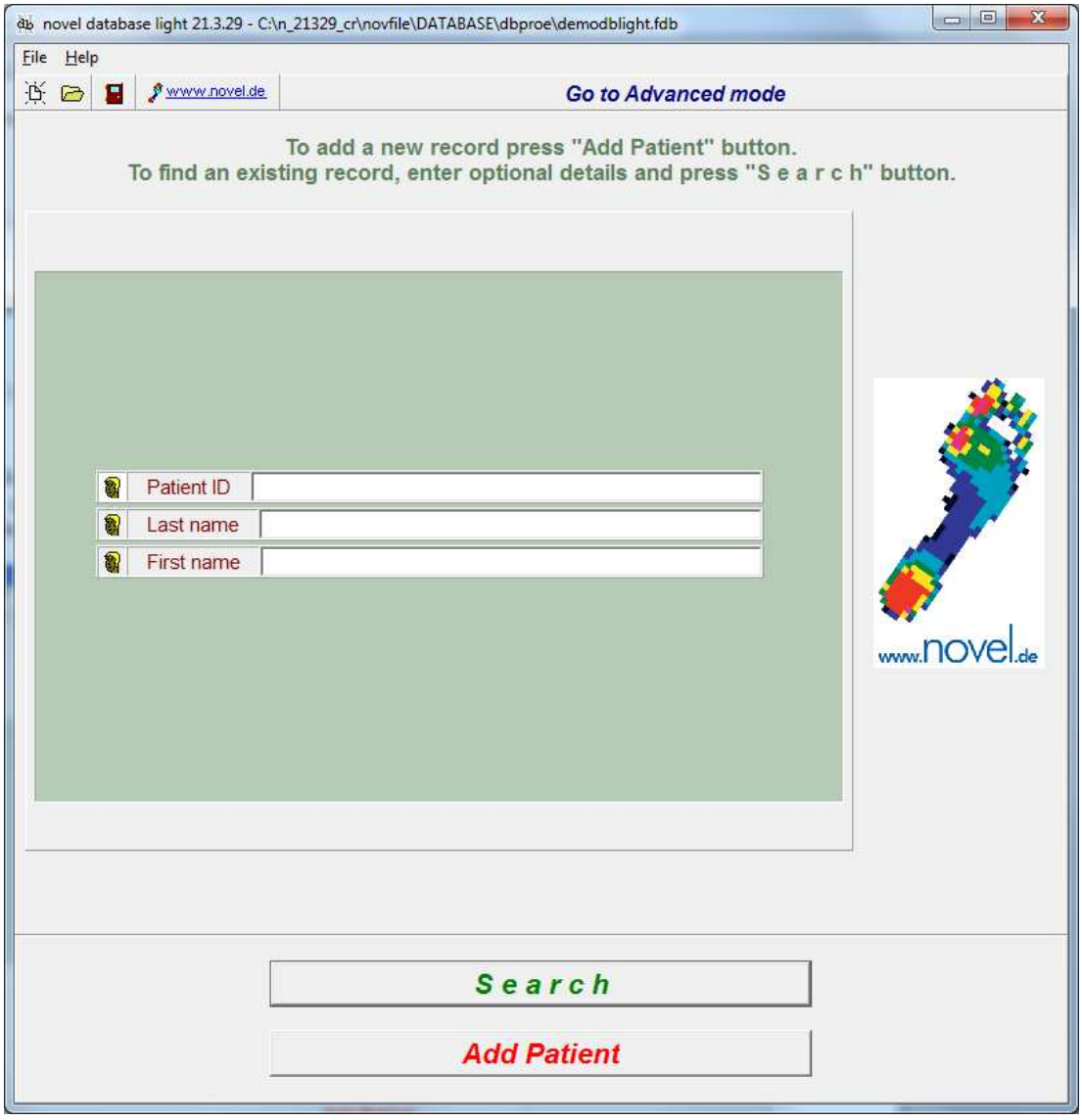
novel database light contains a predefined set of fields that cannot be edited by the user.





An explanation for certain fields is given in detail in **APPENDIX A.**

To display the fields in the **Database editor** window, in the Simple mode click the **Add Patient** button to add a new patient, or type the patient data (e.g. Last name, First name or Patient ID) into the displayed field and click the **Search** button.



In the advanced mode select the patient and/or a specific visit from the hierarchical table structure and double click.

Another way to display the fields is to use the **Records** menu along with the **Add** or **Edit** options for a simple display of all fields in all tables. This display will show all related information for one patient and allow you to add/edit patient information.

NOTE: to edit information the user must click the **Edit** button.

- With a click on the heading tabs you can switch from one tab to the next.
- With the **Tab** key you can move from one field to another.
- In some fields text can be written.
- In some fields a list of possible data can be selected from a drop-down menu.
- When editing a field, the background changes to **yellow**.

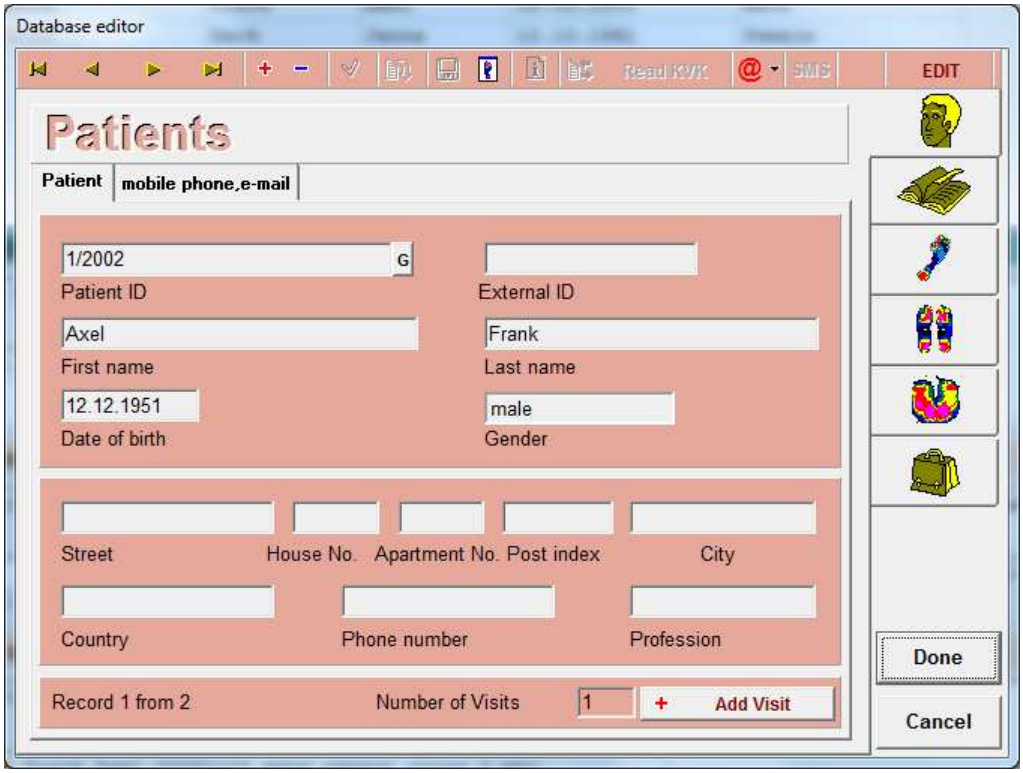
- Some parameters are distinguished between the left and the right foot. Be sure to fill out both sides!

IMPORTANT

Information about amputations is given separately for the left and right foot. Please make sure the data are entered properly.

An example from the novel database light is shown below.

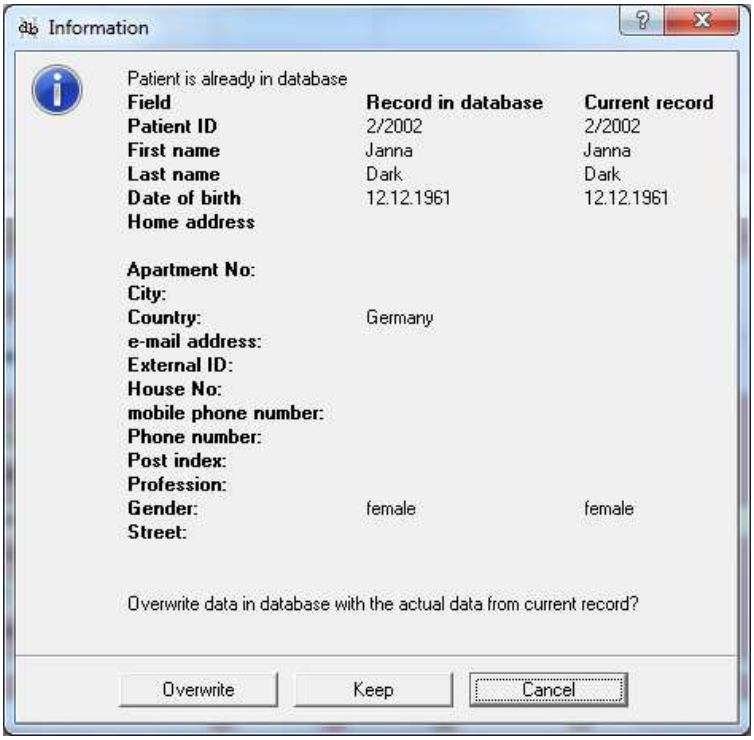
4.1 Patient



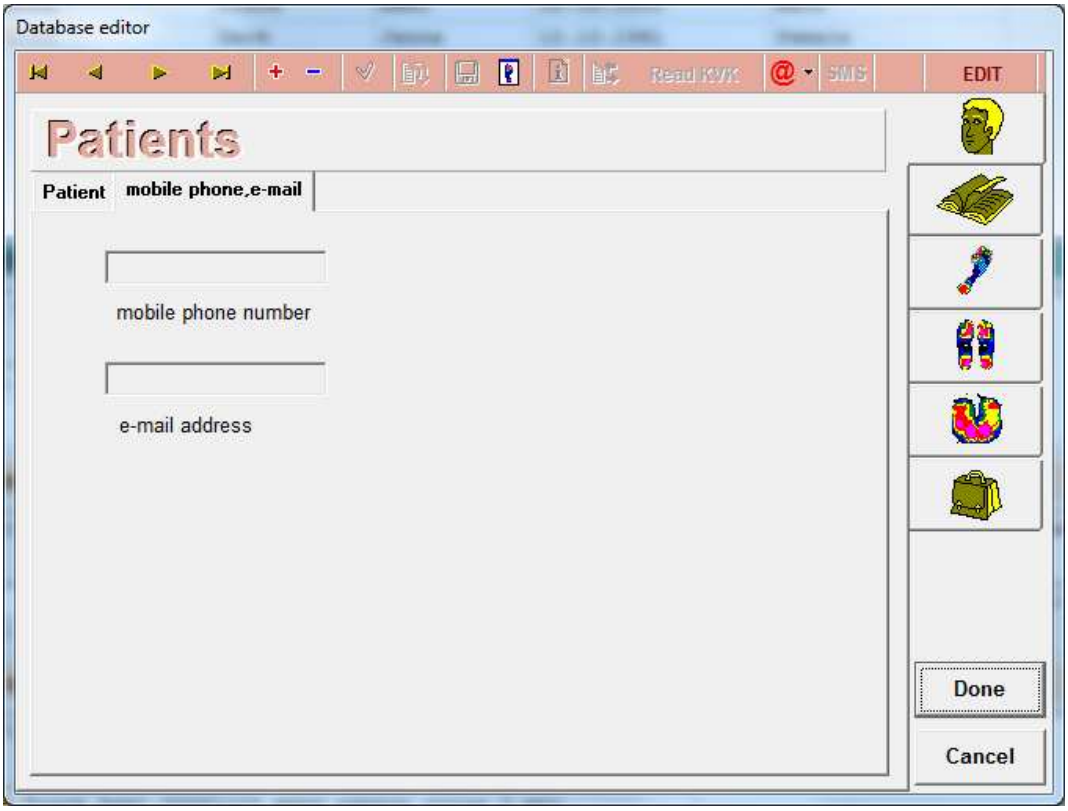
Patient ID, **Last name**, **First name**, and **Date of birth** are the unique fields in novel database light. It means that 2 records with the same Patient ID, Last name, First name, and Date of birth are considered as the duplicate records. There are no required fields in the novel database light.

NOTE: It is recommended to enter the Patient ID in the **Patient ID** field or to generate the Patient ID by clicking the **G** button once First name, Last name, and Date of birth are entered.

If a record already exists with the same **Patient ID**, **Last name**, **First name**, and **Date of birth**, a message will appear. The new data can be ignored (**Keep** button) or saved over the old data with the same ID (**Overwrite** button). Take note when this message appears.



The **mobile phone, e-mail** tab contains mobile phone number and e-mail address.

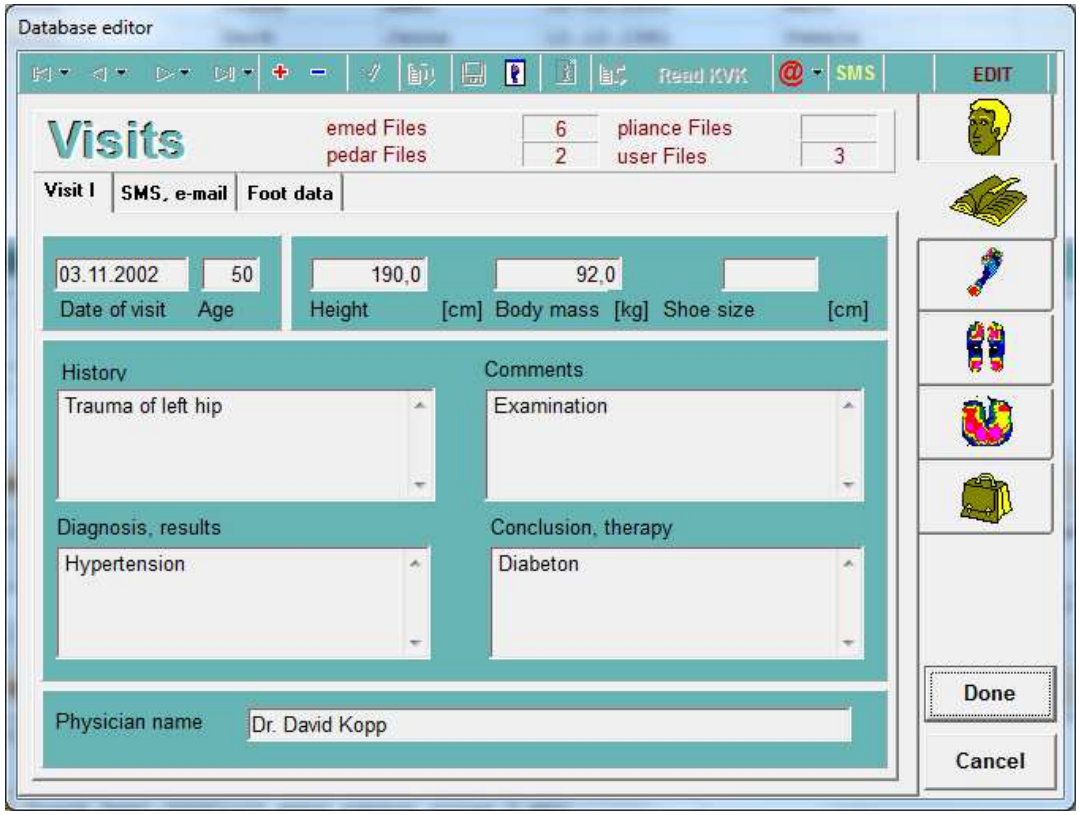


Date and numbers are saved according to the Windows settings defined in the computer's control panel.

The **Add visit** button adds a visit to the patient's record and allows the input of the clinical parameters and foot data examination data. You may also click the Visits tab (on the right side

of the window) to add a visit.

4.2 Visit



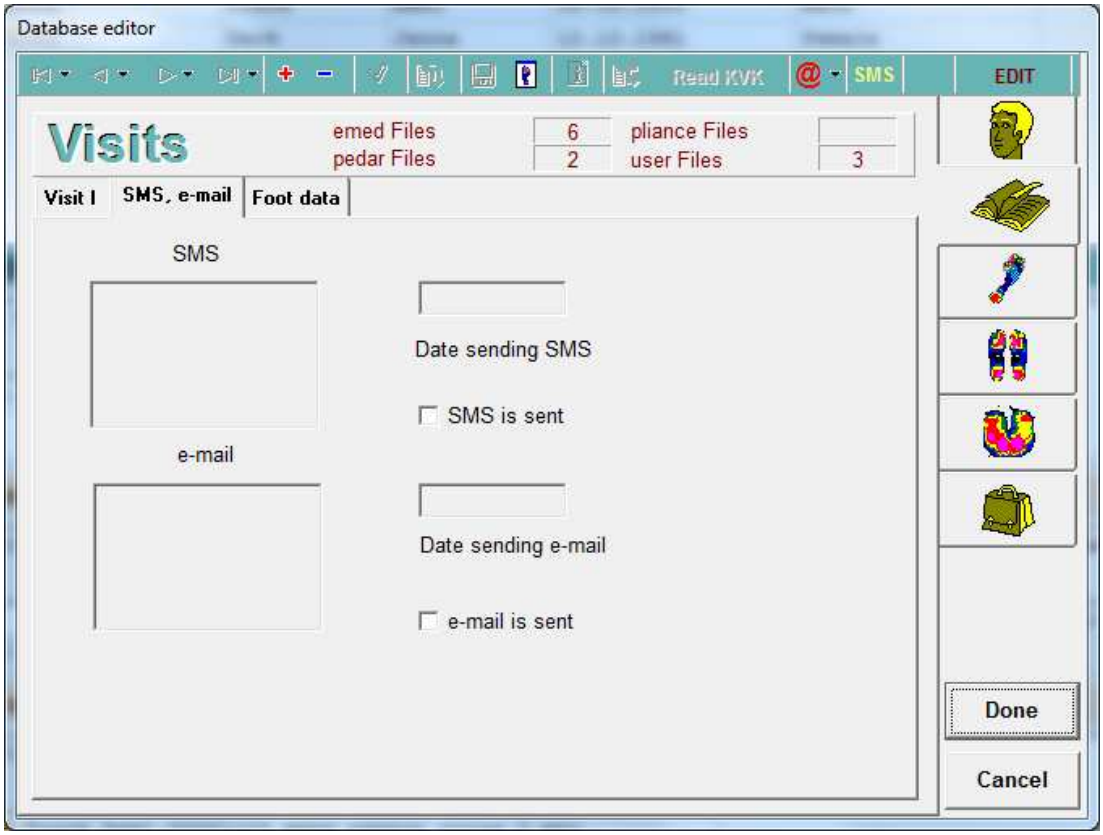
The **Date of visit** and **Age** are automatically displayed. The date is the current date (by default, but this can be edited) and the age is calculated from the **Date of visit** and **Date of Birth** fields. The next are the fields **Height,** in cm; **Body mass,** in kg; and **Shoe size,** in cm.

Use the **Physician name** field to fill in the name of the examining physician, if desired.

In the four available fields you can fill in the anamnesis of the patient **History**, the additional **Comments, Diagnosis** and the **Results** of the pedographic examination**, Conclusion** and/or prescribed **Therapy.**

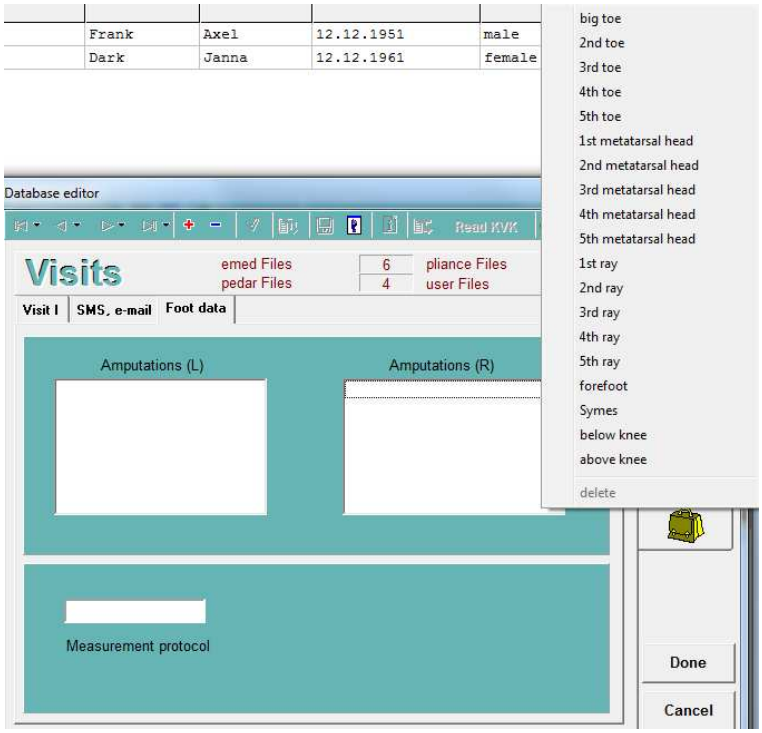
Keep in mind that while these are predefined fields with names, they can be used to enter a parameter of importance to you (i.e. the Comments field could be used to define a pre- or post-surgical visit, or the Physician name field could actually define a condition).

4.3 SMS, e-mail

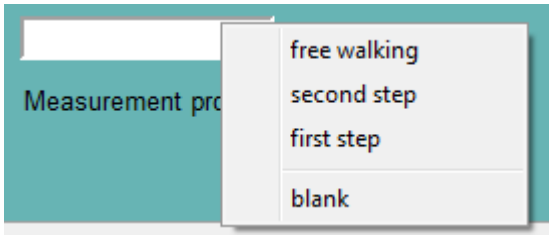


Information about SMS and e-mail is stored in this tab.

4.4 Foot data



Existing amputations can be selected in **Amputations** field. The location can be specified from the menu.



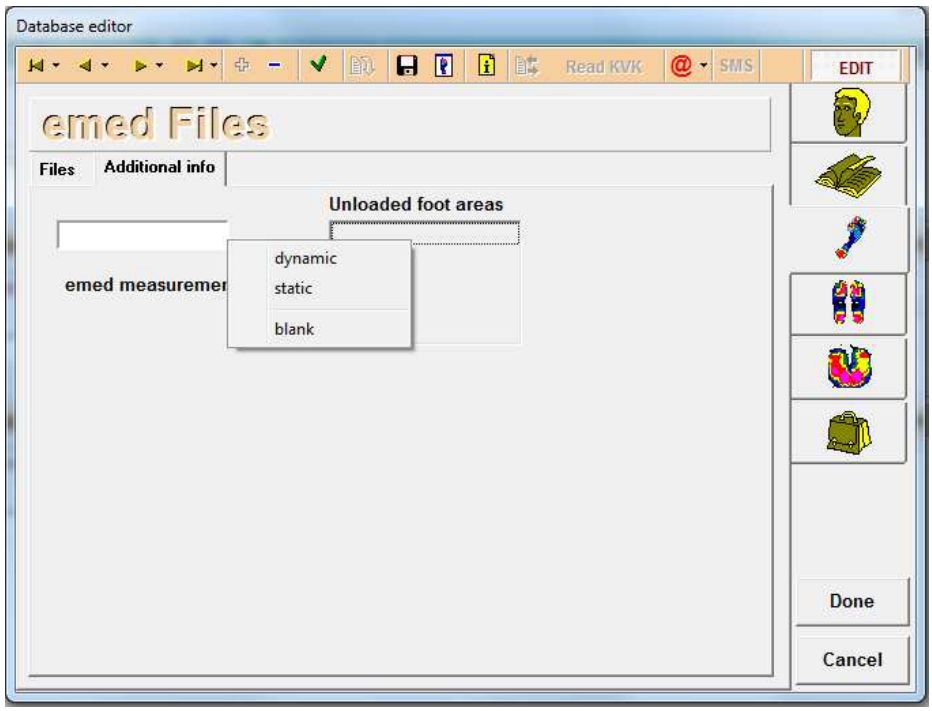
The measurement protocol which was used for data collection can be selected from the drop down menu and is stored within this tab.

4.5 Data files

These fields include information regarding the data files collected from the emed, pedar, and/or pliance systems. Click on the corresponding icon tab on the right side to view the correct data files.

The measure button 

will allow data files to be collected and stored directly in the database. The corresponding data acquisition program will open. When the data file is saved, it will be stored in the database, as well as on the default location on the hard drive (or network drive etc).



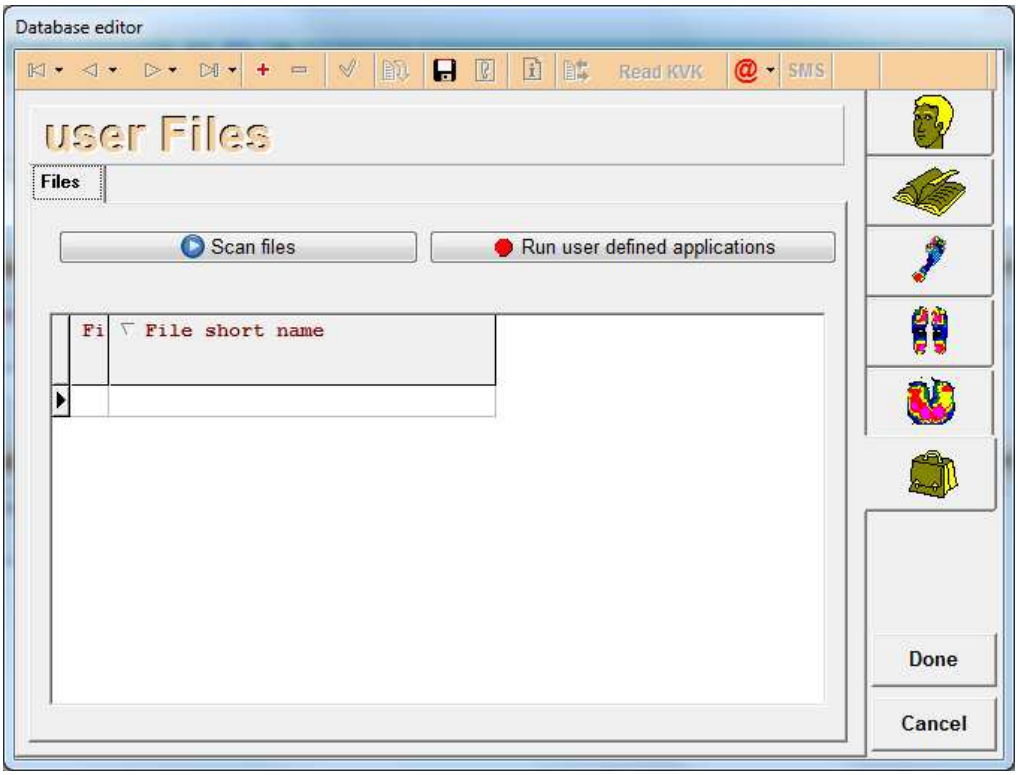
The **Additional info** tab contains information about the measurement procedure, e.g. dynamic or static for emed measurements or without insoles, with insoles type 1 or type 2 for pedar measurements

The **Unloaded foot areas** field displays information about the unloaded foot areas collected during the emed measurement (saved with the comment during the emed measurement). It is also possible to edit this field in **File information** window in **Edit** mode.

4.6 user Files

This table includes any additional files which are stored with the patient information i.e. picture files, ASCII files, MHT or RTF reports, desktop files, or any other file type. Simply click on the briefcase icon tab to view the User files. Use the navigation toolbar to add, delete, and move between the additional files.

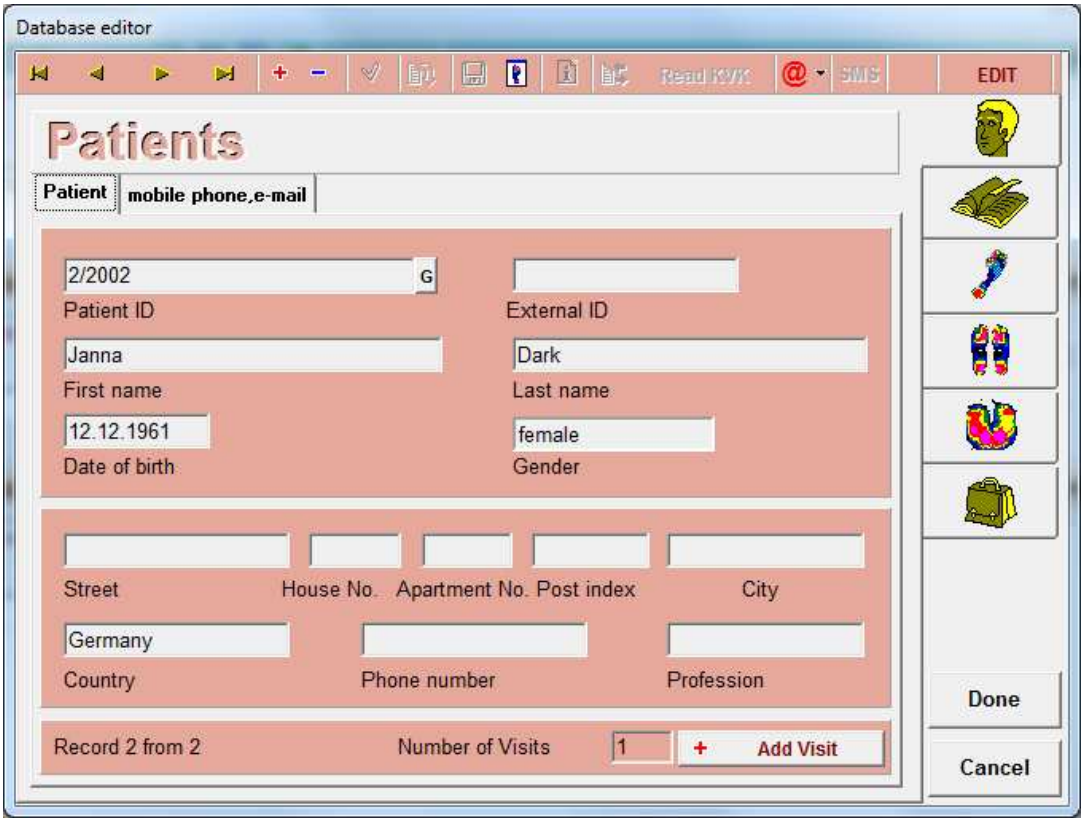
User defined applications can also be started from this window.

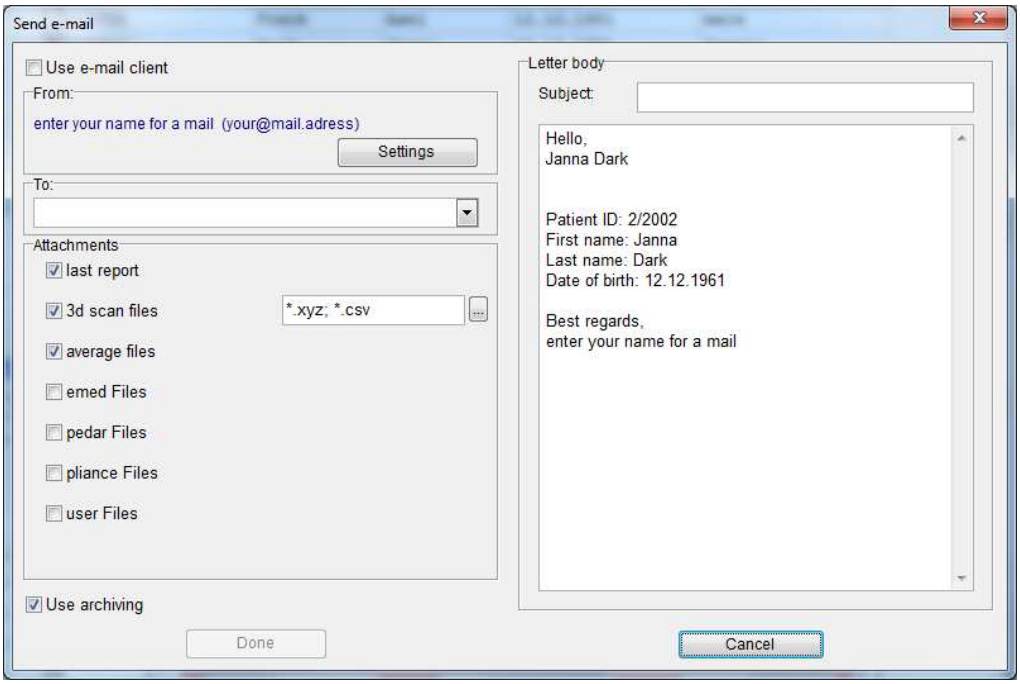


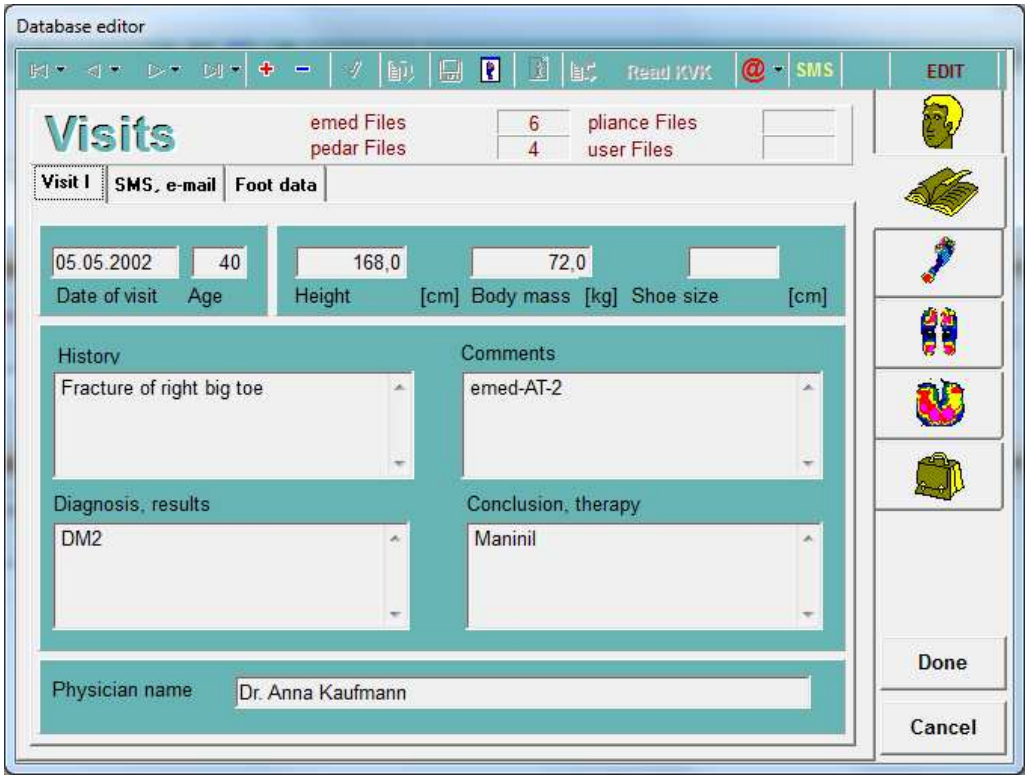
##########

5 Database editor window

Different buttons are enabled in different tables.









- use these buttons to navigate between first, previous, next, and last records on this (**Visits**) or upper level (**Patients**).

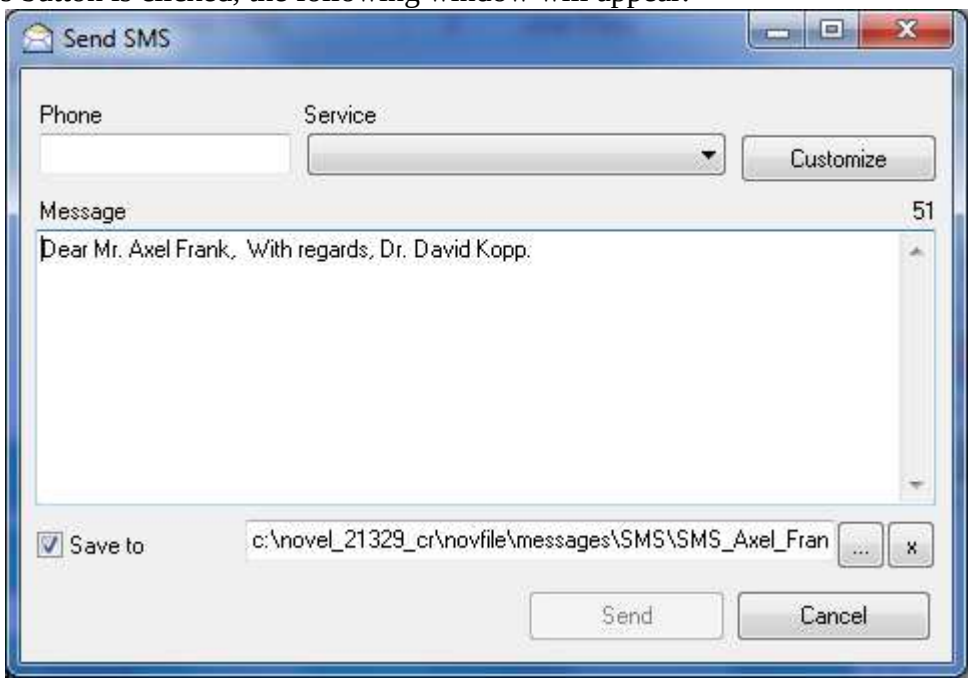


- use this button to copy data from previous visit.



- use this button to send SMS.

When this button is clicked, the following window will appear.



APPENDIX A: Fields

This appendix explains the fields of the novel database light in detail.

Patient ID/Name, Date of Birth, Gender:
It is required to enter the patient identification code or/and Last name, First name, Date of birth, and Gender. It is important to mention that all personal information (includes Last name, First name, Phone number, Street, House No., Apartment No., Post index, City, Country) is excluded when **Include personal information** checkbox is not selected.

Shoe size:
This information is not important but if the shoe size (the foot length, cm) is known then it will be possible to compare it with the foot length calculated from the emed maximum pressure picture (if a proper version of the emed program with the foot length calculation or/and programs in which the foot length is calculated are installed).

Amputations:
Information about amputations is very important in case when the standard masks are used in data analysis and report creation.
Besides, a foot with prior amputation will need a special care because of its altered mechanics.

APPENDIX B: nvlsoft.ini description

General section

[COMMON]
- GroupCount = < *group count* >
- ProgramCount = < *program count* >

Group section

[Group<*GroupNumber*>]
- CAPTION = <*group name* >
- APPLY_TO = < emed;pedar;pliance;user;none > (may include any values, blank is equivalent all except user)

Program section
[Program<*Program number*>]
- SERVICE_NAME – for internal use, unique program name for section definition
- IS_USER = { 1 | 0 } - for internal use, defines user's ability to edit section via database
- EXE_NAME = <Path and name to executable file> (* - necessary field)
- PARAMETERS = < Command line to novel software >
- WORKING_DIR = <Path to working directory>
- ICON = < Path and name to icon file>
- ICON_STOP = < Path and name to icon file> - for 2nd icon, i.e. Start/Stop file scanning
- CAPTION = < Program title >
- DESCRIPTION = <Hint to describe a program>

- PAGE = < Page number >
- ALLOWED_EXT = <dot-comma separated list of allowed extensions>
- FIELDS_2_SEND = <dot-comma separated list of fields that must be send to program>
- TAG = <program tag> - program id, do not confuse with PAGE
- STARTUP_TYPE = { new_instance | connect_existing}
  In case of STARTUP_TYPE = connect_existing:
- SEARCH_TYPE = { window_caption | class_type | broadcast} (how to search existing instance)
  WINDOW_CAPTION = < Window caption >
  or
  CLASS_TYPE = < Class type >
  (in cases SEARCH_TYPE = window_caption or SEARCH_TYPE = class_type we start new instance when cannot find existing one)
- PROTOCOL = { novel | dde | gdt } (*)
  In case of PROTOCOL = gdt (new software):
- INPUT_FILENAME = < Input filename > - dot-comma separated list of file masks, that will be scanned (i.e. C:\novel\in*.pas or C:\novel\in*.pas;C:\novel\in*.dfm)
- ARCHIVE_FILENAME = < ARCHIVE_FILENAME > - if specified, INPUT_FILENAME list will be renamed, in this case in DB added already renamed files. Masks count must coincide with masks count in INPUT_FILENAME section. If parameter not specified, INPUT_FILENAME list will be deleted to avoid duplicates adding.
- OUTPUT_FILENAME = < Output filename > - if specified, system will write FIELDS_2_SEND list in this file (in format USR_FIRSTNAME=Ivan)
  For INPUT_FILENAME, ARCHIVE_FILENAME, OUTPUT_FILENAME, ICON, ICON_STOP sections we can specify template %noveldir%
- SUPPORT_MIRROR = { 1 | 0 } – possibility to transfer mirrored files

NOTE: **Mirror** option is not supported in novel database light.

- CAN_CREATE = { 1 | 0 } – possibility to create new files in program (e.g. emed)
- OPEN_TYPE = { new_measure | open_existing}

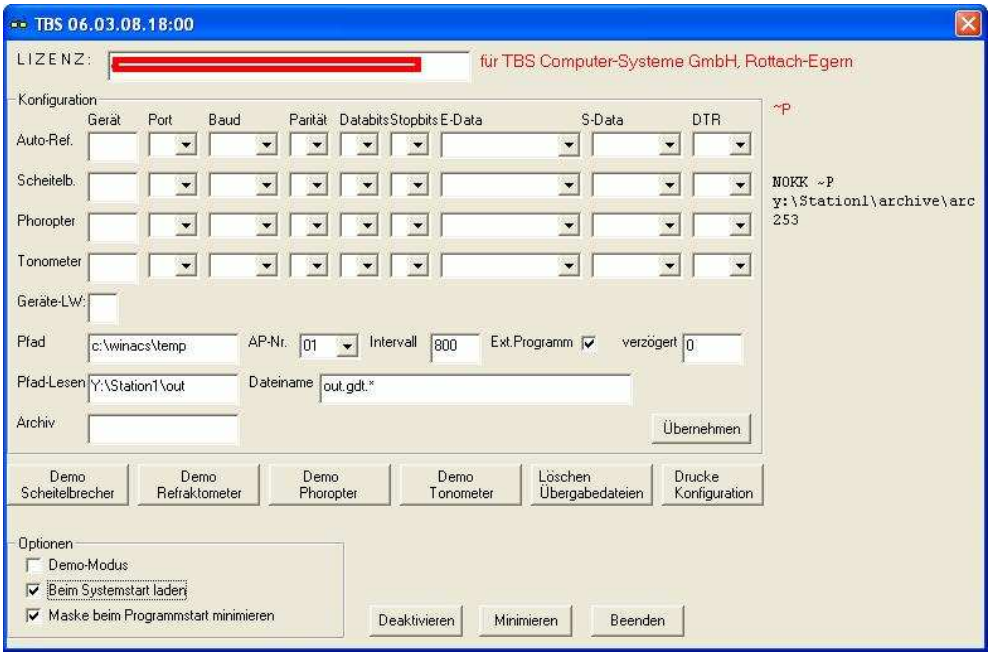
- SCAN_ONLY = { 1 | 0 } – for button to file scan
- TODAY_ONLY = { 1 | 0 } – to scan only today files
- USE_ARCHIVING = { 1 | 0 } - ARCHIVE_FILENAME folder to store the processed files

Extension of protocol to use templates and corresponding directories

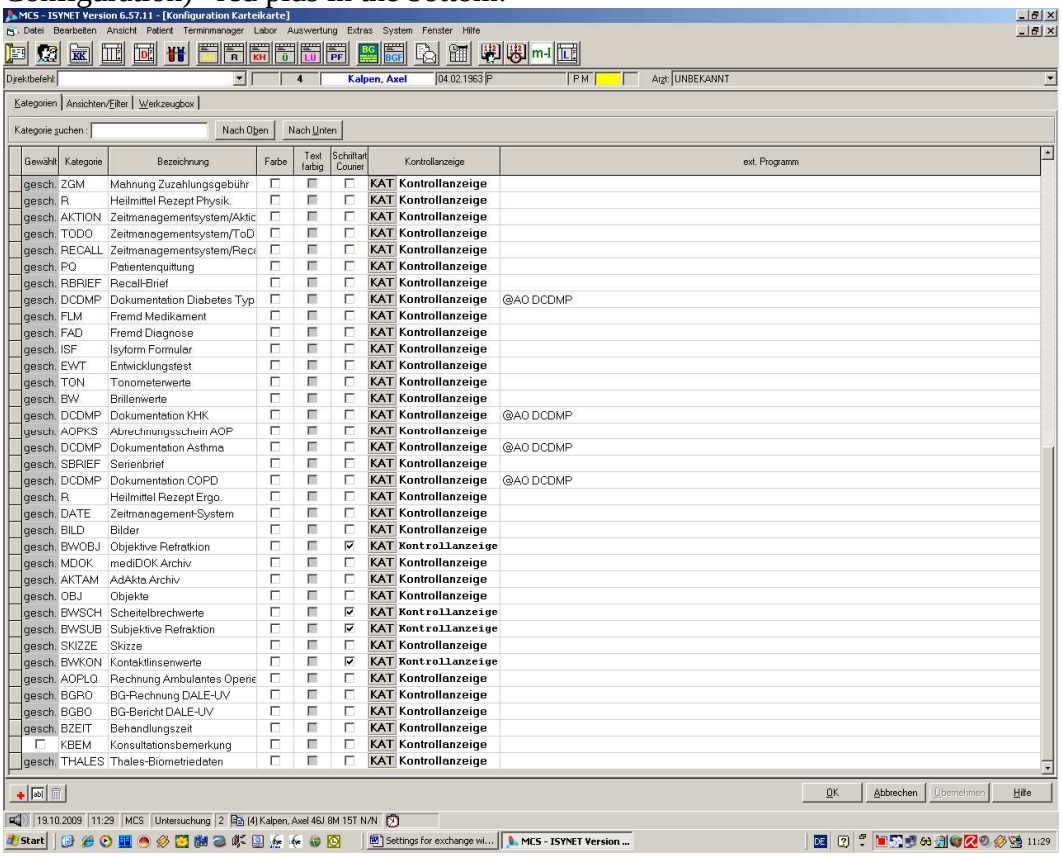
Template	Corresponds to
%noveldir%	C:\novel
%program files%	C:\Program Files
%desktop%	C:\Documents and Settings*username*\Desktop
%my documents%	C:\Documents and Settings*username*\My Documents
%windows%	C:\Windows
%system32%	C:\Windows\System32
%common application data%	C:\Documents and Settings\All Users\Application Data
%application data%	C:\Documents and Settings*username*\Application Data

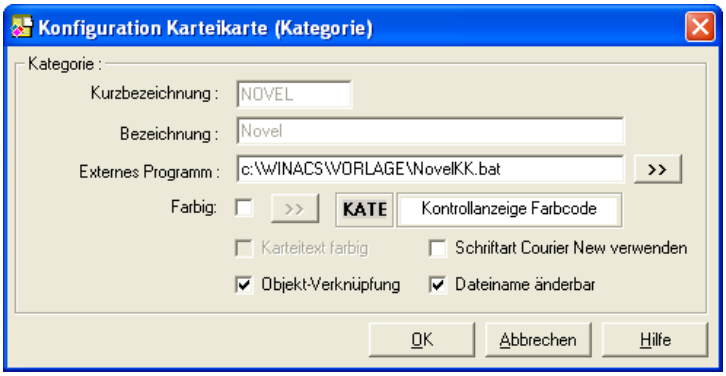
APPENDIX C: Settings for exchange with novel software in MCS-ISYNET

1. TBS.exe should be in C:\WINACS folder with following settings:



2. In WINACS.exe System/Weiteres/Konfiguration Karteikarte (System/Next/Card Configuration)– red plus in the bottom:

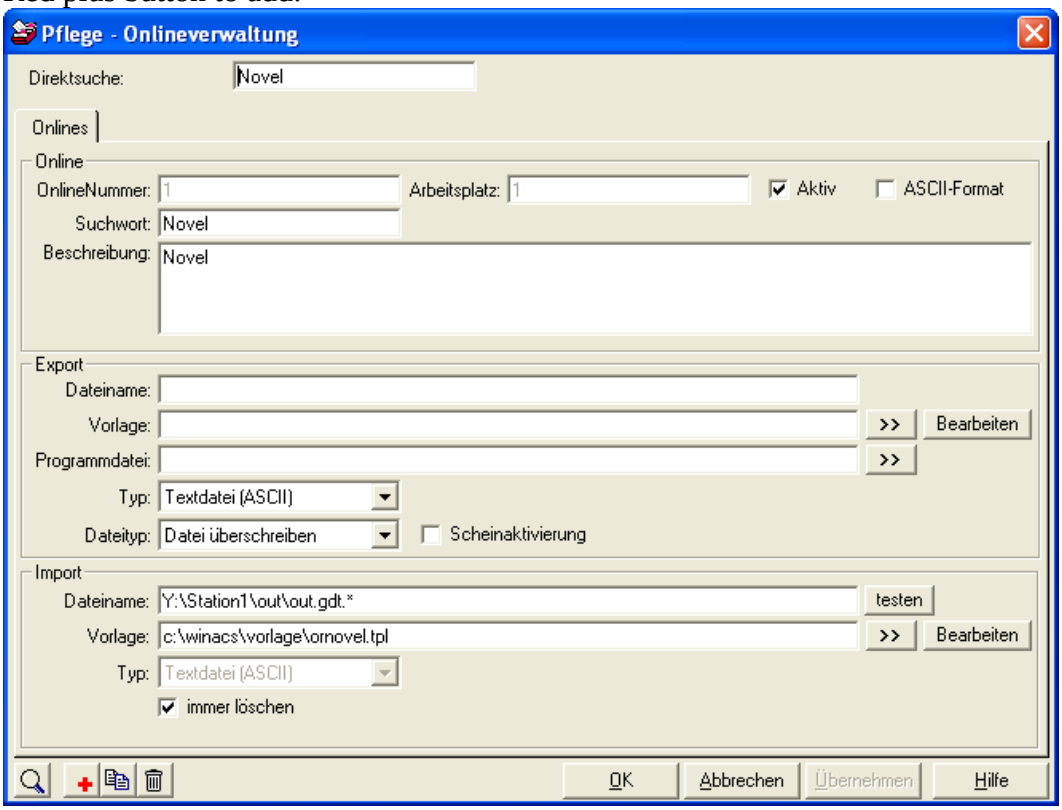




Externes Programm: c:\WINACS\VORLAGE\NovelKK.bat

3. System/Weiteres/Onlineverwaltung:

   Red plus button to add:



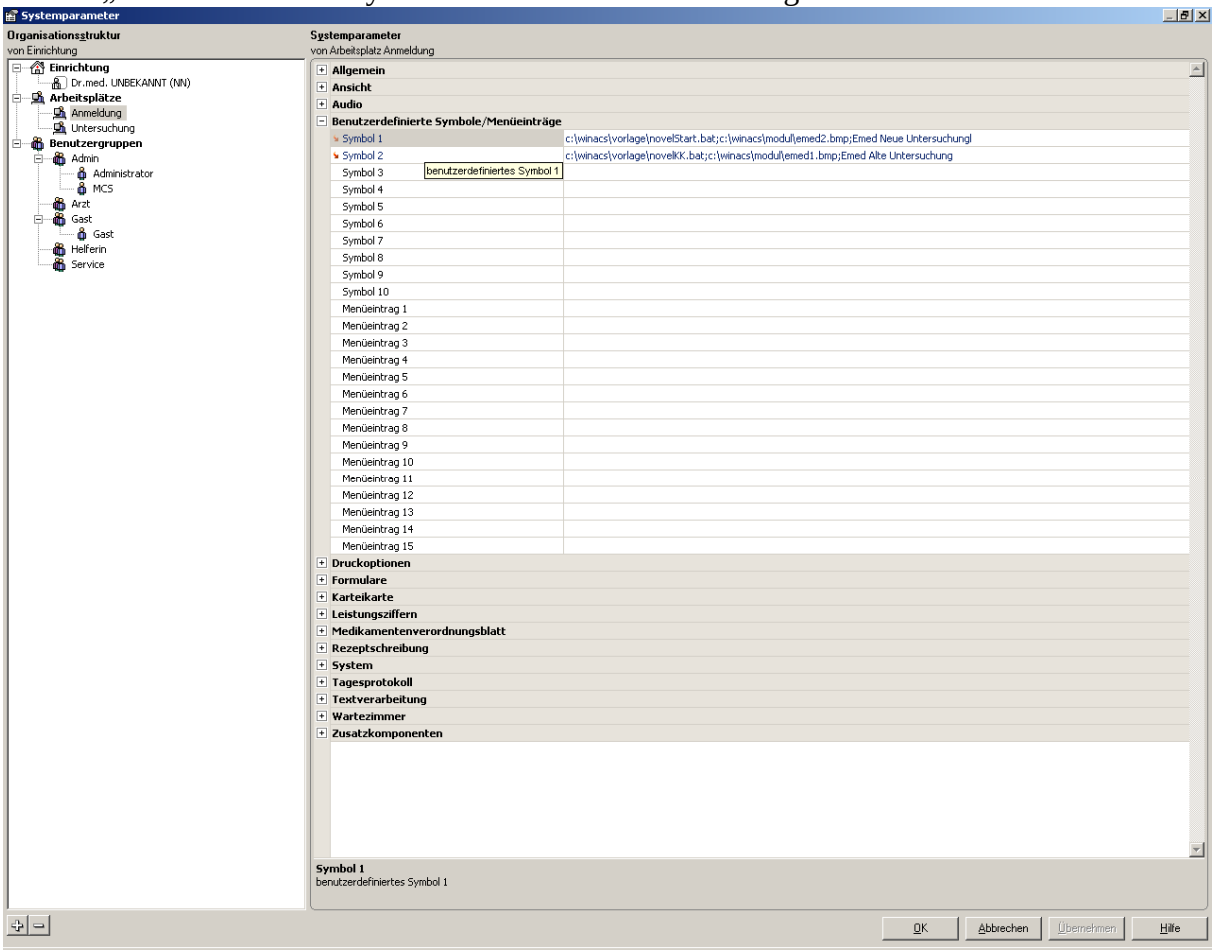
Dateiname: Y:\Station1\out\out.gdt.*
Vorlage:c:\winacs\vorlage\ornovel.tpl

4. Batch files in c:\winacs\vorlage:

   In file novel.kk.bat modify path to archive file like:
   @echo off
   c:\winacs\temp\kkinf NOKK ~P Y:\Station1\archive.gdt~D %1 ~M

   In file novelstart.bat modify path to input file like:
   @echo off
   C:\winacs\temp\kkinf NOWL ~P Y:\Station1\in\novel~~.*~D ~M

5. System\verwaltung\Parameters:
   Click on Arbeitzplatz\Anmeldung on left side
   Extend „Benutzerdefinirte Symbole“ and edit 2 lines finishing with Enter



Symbol1: c:\winacs\vorlage\novelstart.bat;c:\winacs\modul\emed2.bmp; emed proceed with new measurement
Symbol2: c:\winacs\vorlage\novelkk.bat;c:\winacs\modul\emed1.bmp; emed view old measurements

6. Additional files (not from Demo CD):
   a. C:\winacs\Tbs.exe (with valid licens)
   b. C:\winacs\temp\kkinf.exe
   c. C:\winacs\modul\emed1.bmp
   d. C:\winacs\modul\emed2.bmp
   e. C:\winacs\vorlage\novelkk.bat
   f. C:\winacs\vorlage\novelstart.bat
   g. C:\winacs\vorlage\ornovel.tpl

(files are stored T:\install_allgemein\MCS-ISYNET\)

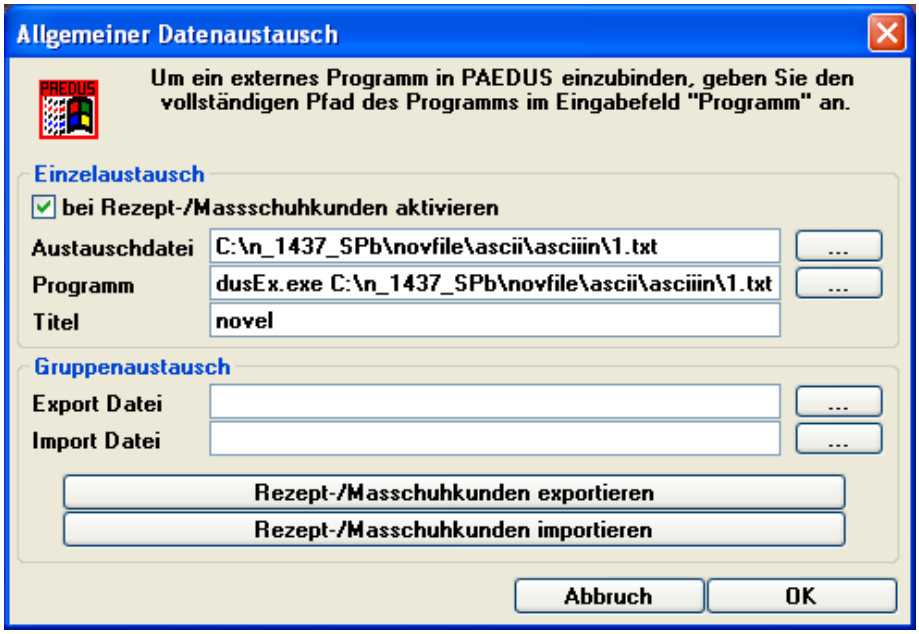
APPENDIX D: Settings for exchange with novel software in PAEDUS

1. Start of paedus.exe
2. Start of novprom.exe
3. Settings in Paedus.

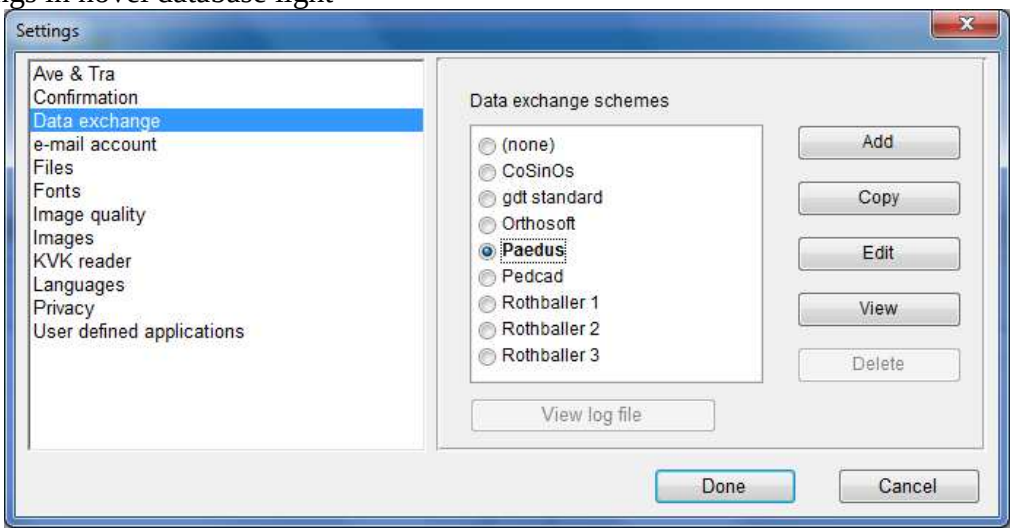


Enter Datenaustausch and programm.

NOTE: In «Programm» edit box necessary to set path to PaedusEx.exe file and after space path to Paedus export file.



4. Settings in novel database light

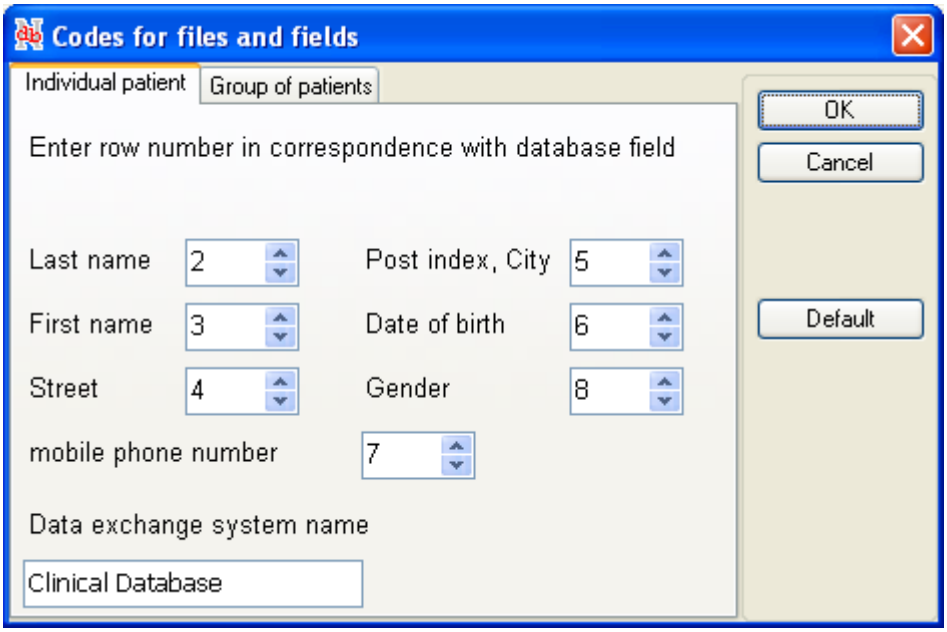


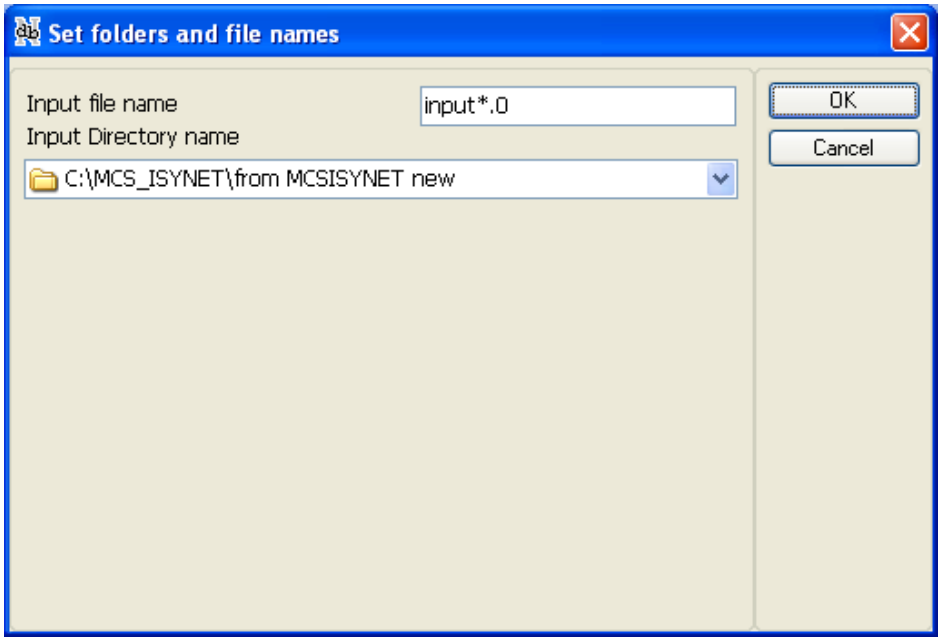
A **Clinical database** folder must always exist. The data exchange is available only if the file …\novprog\system\dll\PaedusEx.exe is included within the setup.

If the gdt standard radio button is selected, then the user must select a field from the novel database to use for data exchange.

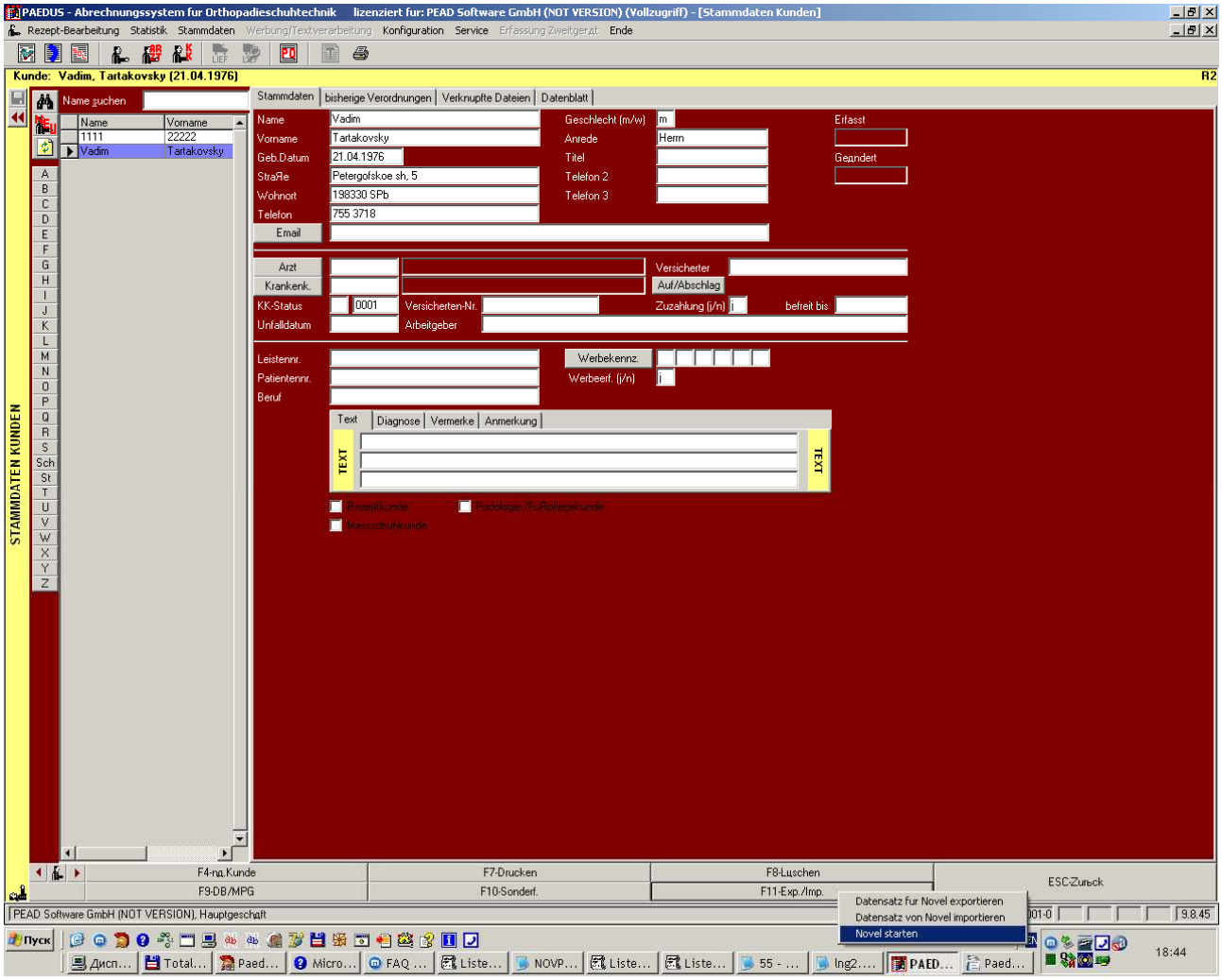
If the Paedus radio button is selected, then the user does not need to select a field from the novel database to use for data exchange.

The user must confirm or enter data codes and file names and directories.





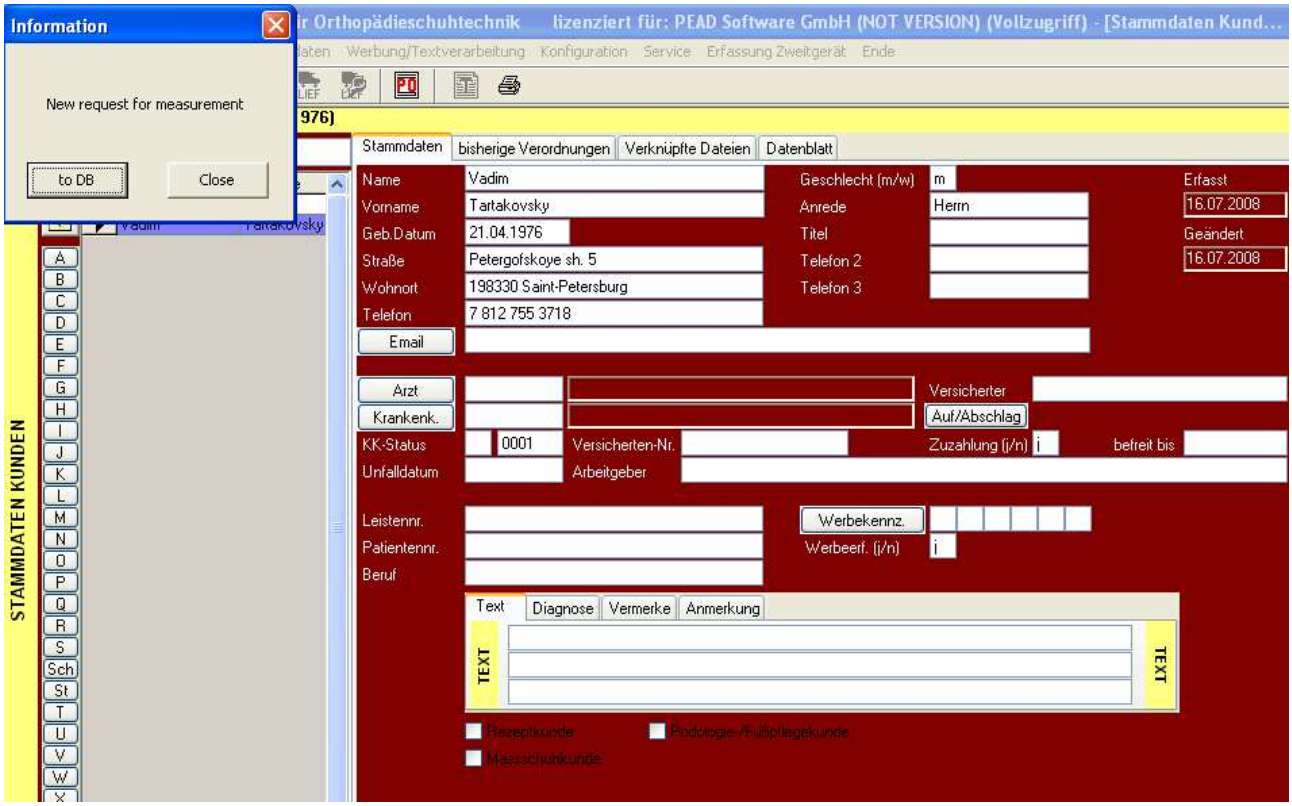
5. Enter the patient data (for demo) (Stammdaten (Basic data) menu \Neu (New) option), then press the F11 button and select novel starten (novel start) option.



A confirmation window will appear.



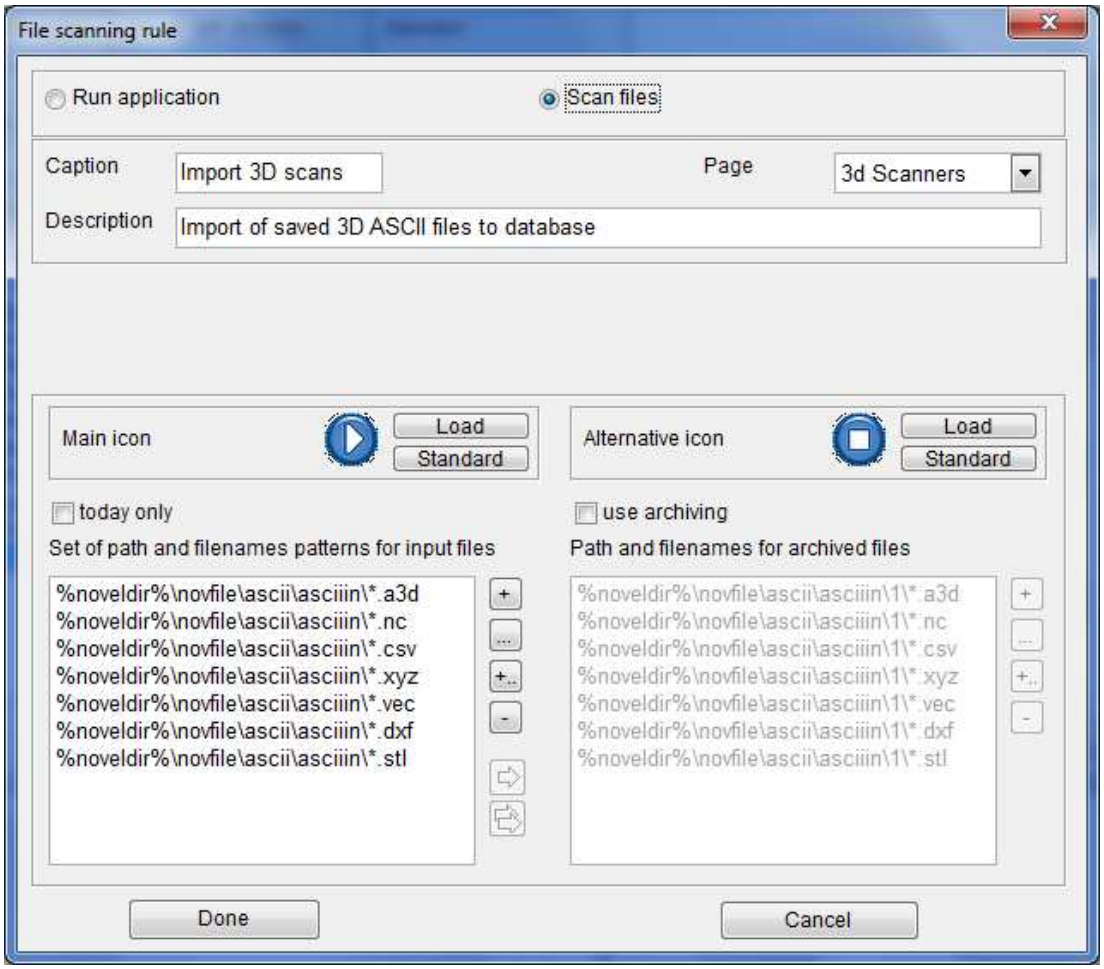
6. A pop-up window will appear after clicking **OK**.



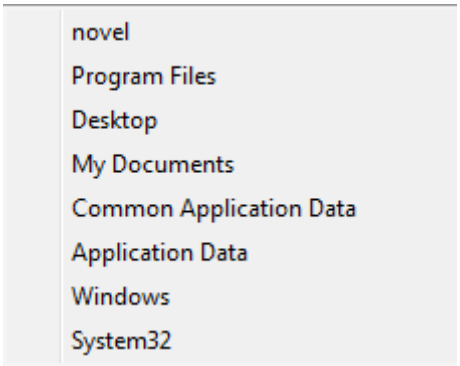
NOTE: The name and path of the export file from Paedus should not coincide with name and path of the input file for the database.

APPENDIX E: User defined applications

Two options are available.
1. It is possible to include (to add) new applications to the **Run application** window.
1.1. The following must be input: Selection of page (group of programs), caption with description, and application file name. When the existing application file name is entered, the applications default icon will be shown on the right side.

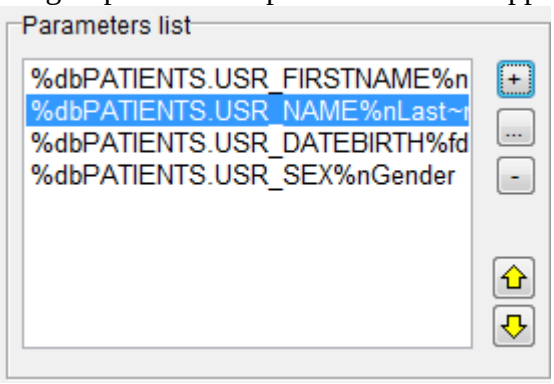


1.2. Use the **Insert template** button to insert the path to the application system one of the predefined folders:



If the **replace templates automatically** check box is selected, this replacement will occur during the application selection.

1.3. Use the **Parameters list** group to transfer parameters to the application.



To edit the parameters, use the buttons below.

 - use this button to add the parameter

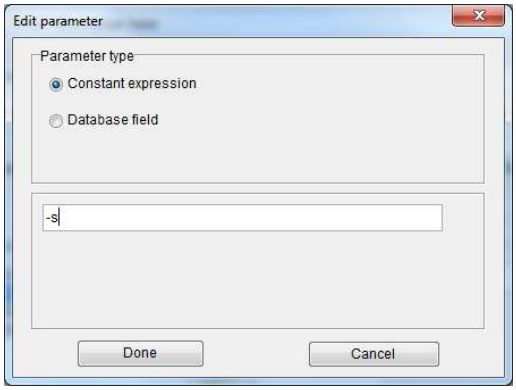
 - use this button to edit the parameter

 - use this button to delete the parameter

 - use this button to move the parameter upwards

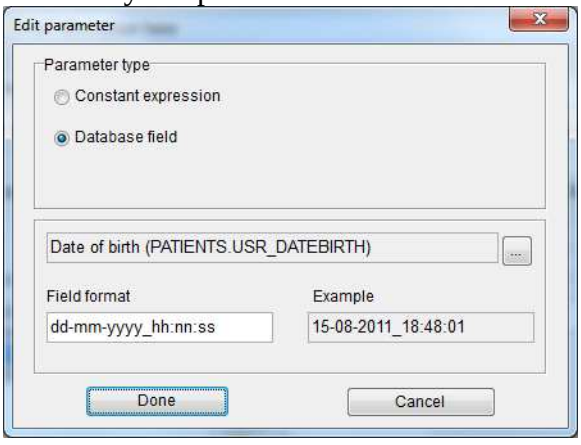
 - use this button to move the parameter downwards

When the **Add parameter** or **Edit parameter** button is clicked, the following window will appear. This window is different for different parameter types.

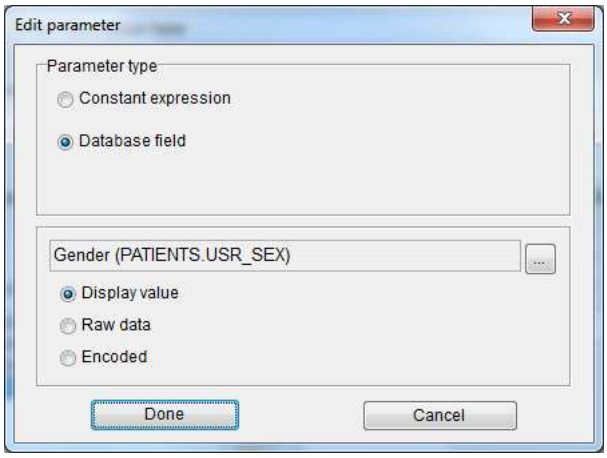




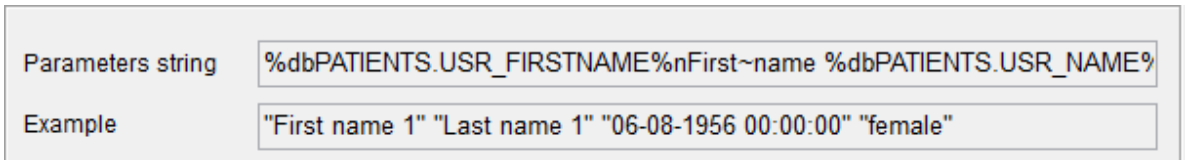
For date/time fields the format may be specified.



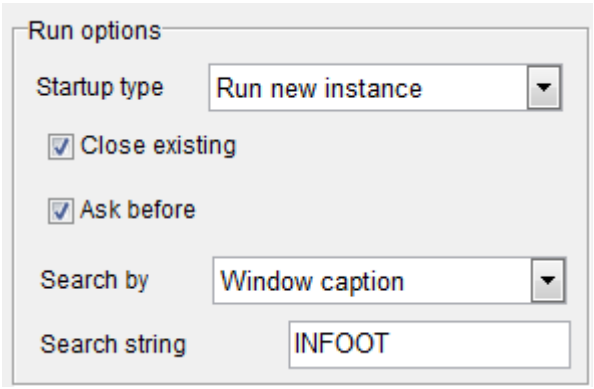
For gradation and multi gradation fields the display values (as they are displayed in database) or raw data (as they are stored in database) may be transferred to the application.



The **Parameters string** textbox shows a parameter string as it is stored, and the **Example** text box shows what will actually be transferred to the application (current record values).



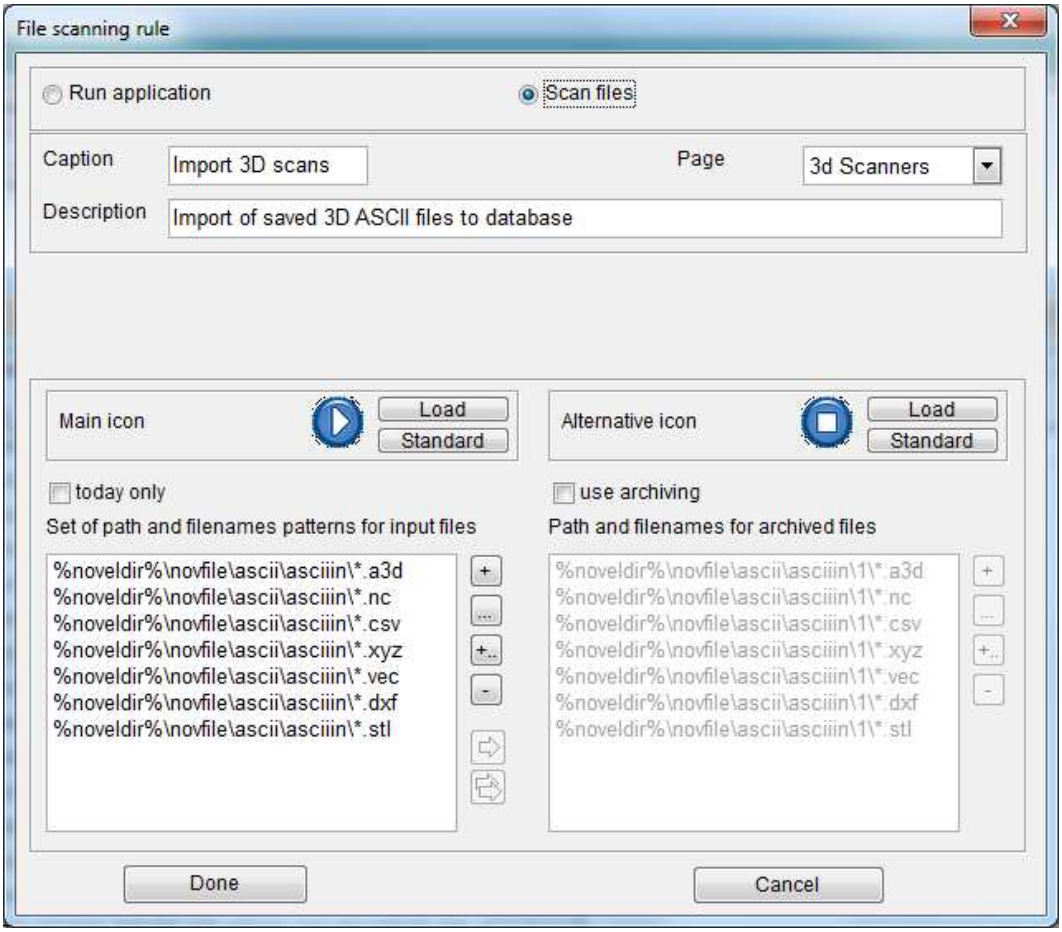
1.4. The **Run option** group controls the application's start up.



The **Startup type** box defines whether to run a new instance of the application or connect to an existing one (if no running application exists, a new instance will run).
If the **Close existing** checkbox is selected, the existing instance of the application will be closed before a new instanced is started (a confirmation window will pop up if the **Ask before** checkbox is selected).
The **Search by** and **Search string** boxes defines how to find the existing instance of the application – by window caption or class type.

1.5. It is possible to run the application with the defined settings by clicking the **Test** button.

2. It is also possible to scan the selected folder (folders) for a new file (files) with a defined mask (masks).



2.1. To edit the mask for file, use the buttons below.

 - use this button to add the path and file name pattern.

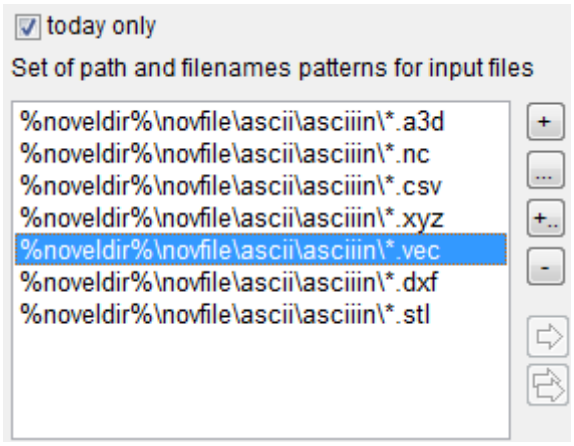
 - use this button to add the path and file name pattern.

 - use this button to copy the path and file name pattern.

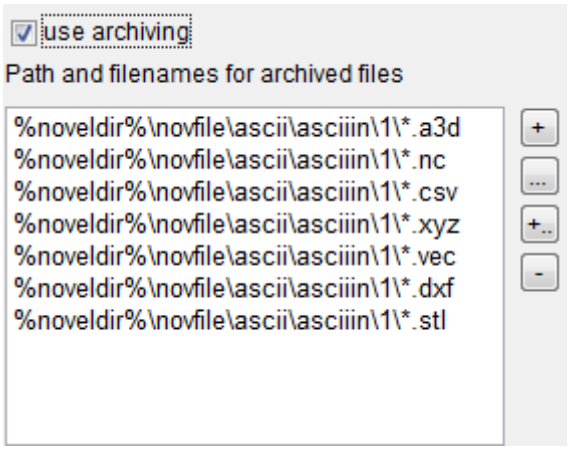
 - use this button to delete the path and file name pattern

 - these buttons are disabled when the **use archiving** checkbox is not selected.

2.2. Select the **today only** checkbox to scan the files created today only.

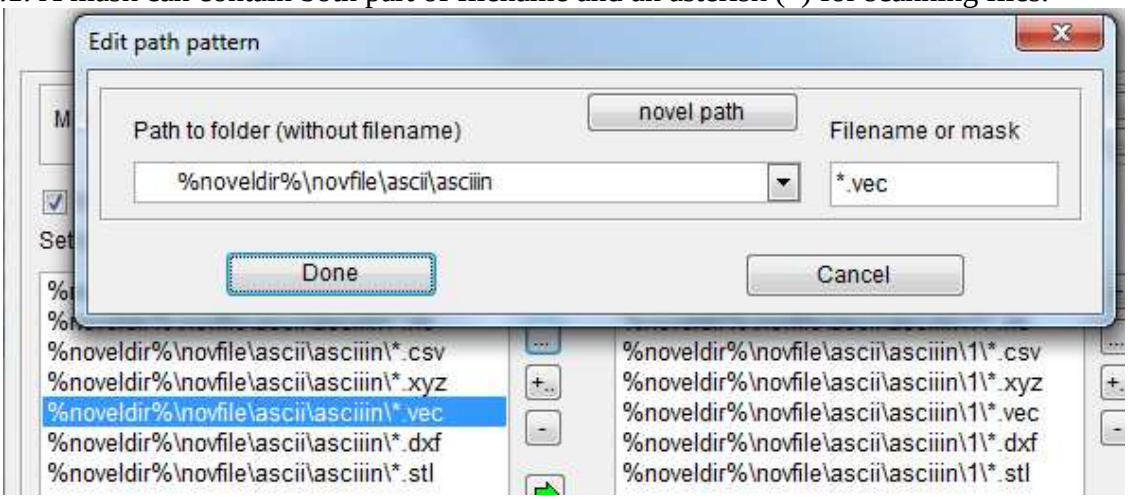


2.3. Select the **use archiving** checkbox to rename and move the files in the archive.
Do not select the **use archiving** check box to move the file (after adding it in to the database) to predefined **proceeded** folder, which is located in input folder.

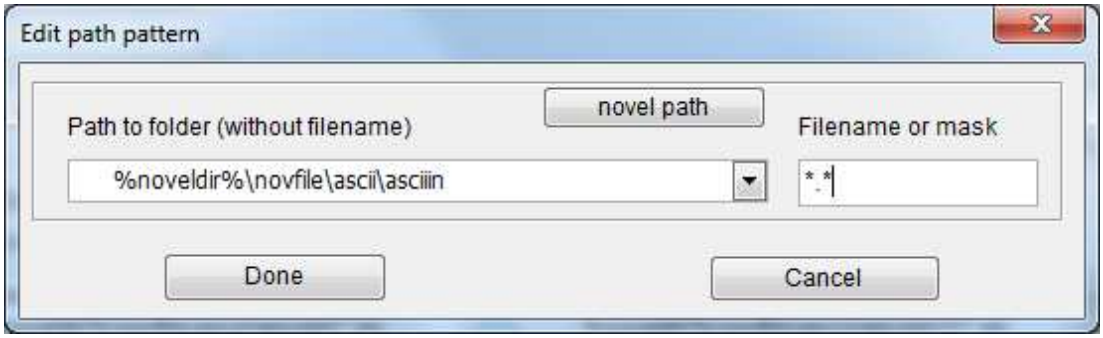


2.4. Please note that the input and archive folders may be different or they may be the same. Extensions for both types of files do not have to be different if you use different folders, but should be different (e.g. *.gdt and *.arc) if you use the same folder.

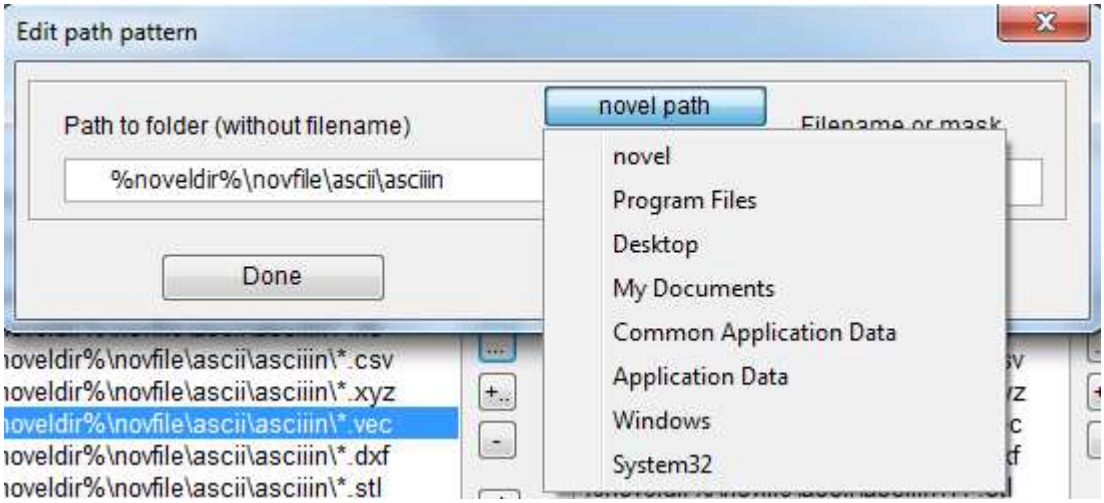
2.4.1. A mask can contain both part of filename and an asterisk (*) for scanning files.



2.4.2. A mask can contain either a filename or an asterisk (*) for files in the archive.



2.2.5. Use the **Insert template** button to insert a predefined system folder system in the path field.

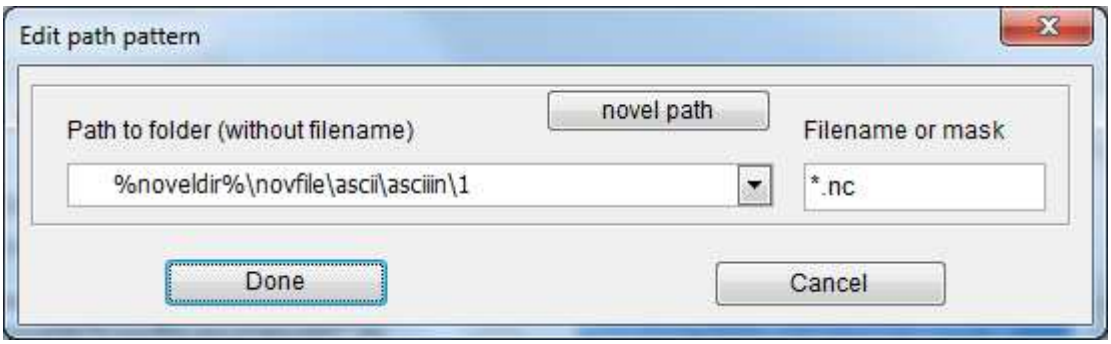


2.2.6. If archiving is used, the number of input and archive patterns which are entered must be the same.

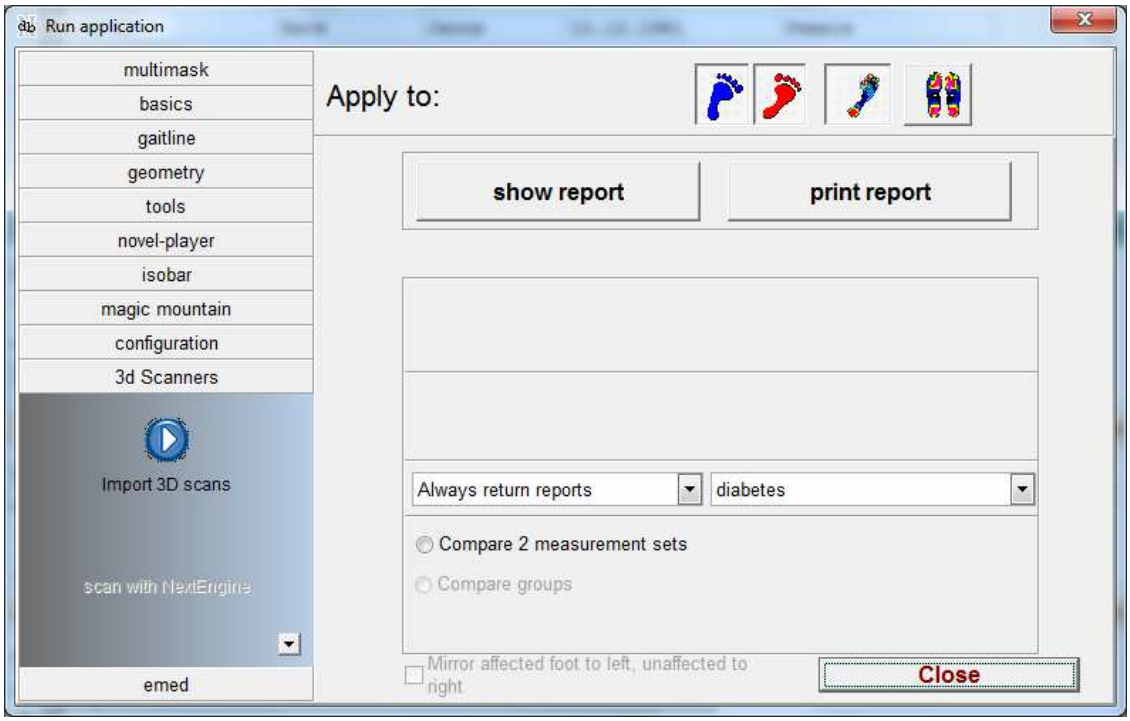
 - use this button to add the pattern (based on current input file) to archive files.

 - use this button to add all patterns (based on current input file) to archive files.

Enter the path to the archive files (must be different from the input path).



1. Start the user defined applications from the **Run application** window, e.g. Import 3D scans.



Appendix F. THE PROGRAM configuration

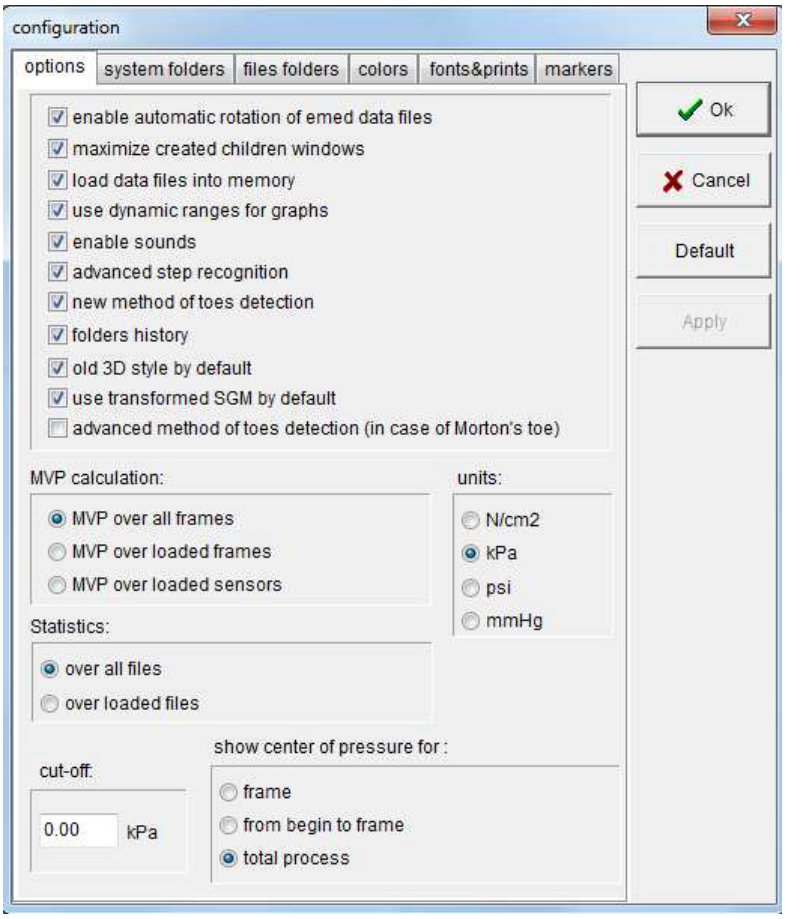


This program is used to edit program settings.

IMPORTANT

Changes in this program affect all novel software programs.

After starting the program, the following window appears:



- To activate any changes and exit from program made to the settings, click **OK**.
- To activate any changes made to the settings, click **Apply**.
- To cancel any changes made to the settings, click **Cancel**.

To restore default settings, click **Default**.

F.1 Options

F.1.1 Default Options

To (de)activate options, (un)check the boxes. A checkmark indicates an activated function.

Enable Automatic Rotation of emed Data Files
With this function the pressure picture is automatically rotated and displayed in a consistent position with the heel at the bottom of the screen and the toes at the top. This is helpful, if the platform is crossed in different directions.
NOTE: This option assumes that the roll over process starts with heel strike. Troubles can occur if the data do not begin in this way or when not studying feet.

Maximize created children windows
When activated, this option allows all windows in a program to be displayed in full size.

Load data files into memory
When activated, data files currently being worked with are loaded to the RAM of the computer to increase the speed of the calculations.

Use dynamic ranges for graphs
This option is only relevant to scale the y-axis of the graphs in the programs `multimask evaluation` and `groupmask evaluation`. It is activated in the default settings. If it is activated, the length of the y-axis is defined by the maximum value of the data and it is then divided into equidistant sections. When deactivated the 100 (or 20) value closest to the maximum value of the data is used as maximum of the y-axis.

Enable sounds
Enables audio description of the commands.

Advanced step recognition
The first and last step for left and right foot will not be displayed and analyzed if this check box is marked**.** See below the description of step recognition algorithm.

New method of toes detection
Allows for a new algorithm of toes detection. This check box is marked by default. It is recommended to deactivate this in order to compare the results received when another algorithm for toes detection was used.

Folders history
When opening or saving files, the last directory that was used will be opened if this option is selected. If this option is not selected only predefined subfolders will be used.

Old 3D style by default
When deactivated, 3D picture is displayed using OpenGL (Open Graphics Library).

Use transformed SGM by default
When deactivated, original SGM (sensor geometry matrix) is displayed. Correspondingly, when activated, transformed SGM is displayed.

Advanced method of toes detection (in case of Morton's toe)
When activated, an updated default curve separating the forefoot area from the toe areas is used.

Advanced step recognition algorithm description.

Different values used in the algorithm are displayed **in bold**.
Actions which are taken are marked with an underline.
Groups of actions are surrounded by brackets { and }.
The algorithm is applied independently for the left and right feet (the same procedure occurs first for the left, then for the right foot).

The detection algorithm uses 2 parameters: **Minimum step length in frames** and **Force limit**. A step considered a step if there is a continuous number of frames with **force** greater or equal to the **Force limit**.

Action number	***Description***
1	Set **Threshold percentage = 5%**
2	Set **Minimum step length = 0 frames** and **Force limit = 0.** Scan foot steps with these values.
3	Calculate **Fmean = Mean max force over steps**
4	Set **Force limit = Threshold percentage* Fmean** and **Minimum step length = 5 frames** Scan foot steps with these values.
5	If **number of steps >= 3** then delete first and last steps
6	Recalculate **Fmean = Mean max force over steps** (without deleted first and last)
7	Repeat once the procedure of scanning and deleting steps using ALL the initial data, including possibly deleted steps (**action numbers 4,5, and 6)** and Proceed to **action 8**

Action number	*Description*

8 If **Threshold percentage is 5%**, then calculate transitions from below 25% to above 75% in this way:

Set **ForceLim25 = 25 % * Fmean**

and **ForceLim75 = 75 % * Fmean**

Set **BelowFlag = TRUE**

Set **Transition counter = 0**

For all frames in the total file do

{

Set **F_i = force value in the i-th frame, if i-th frame belongs to non-deleted steps and otherwise** Set **F_i = 0**

If **F_i < ForceLim25** then Set **BelowFlag = TRUE**

If **F_i > ForceLim75** and **BelowFlag is TRUE** then

{

Increase **Transition counter**;

Set **BelowFlag = FALSE**

}

}

At last, if **Transition counter > Number of steps in the file** (not counting deleted), then Set **Threshold percentage = 10%** and proceed again to **action 2** (repeat the whole detecting procedure using new **Threshold percentage**).

9 Last time rescan foot steps with **Force limit = Threshold percentage* Fmean,** after that completely delete 1st and last steps (if **steps count >= 3**) and clear (set values to zero) all frames which are not in steps

F.1.2 MVP Calculation

This section allows three choices for calculation of the mean value picture (MVP).

- **MVP over all frames**
  The mean value is calculated for all frames, even if the sensors are loaded or not.
- **MVP over loaded frames**
  The mean value is calculated only from frames where at least one sensor is loaded.
- **MVP over loaded sensors**
  The mean value is calculated for each sensor using only the frames where the sensor was loaded.

  NOTE: Beware that these three methods yield different results demonstrated in the following example:

  Let us assume that the sensor array consists of four sensors and that the data collection contains four frames.

1	2	0	0	0	1	0	0
3	4	0	0	0	0	1	0
Frame 1		Frame 2		Frame 3		Frame 4	

  We want to calculate the mean value for the upper left sensor and obtain:

- **MVP over all frames:** 1/4 = 0.25 12/16 =3/4 = 0.75
- **MVP over loaded frames:** 1/3 = 0.333 12/12=3/3 = 1
- **MVP over loaded sensors:** 1/1 = 1 12/6 =2/1 = 2

F.1.3 Units

Select the pressure units.

- N/cm^2
- kPa 1 N/cm^2 = 10 kPa
- psi 1 N/cm^2 = 1.4504 psi
- mm Hg 1 N/cm^2 = 75.006 mm Hg

F.1.4 Statistics

- **Over all files**

All files are used in statistics.

- **Over loaded files**

Files with loaded sensors only are used in statistics.

F.1.5 Cut-off

Enter a threshold. Only pressure values above the threshold will be displayed.

Note: The definition of a cut-off can be important for clinical analysis of diabetic feet, where it is necessary to control peak plantar pressures.

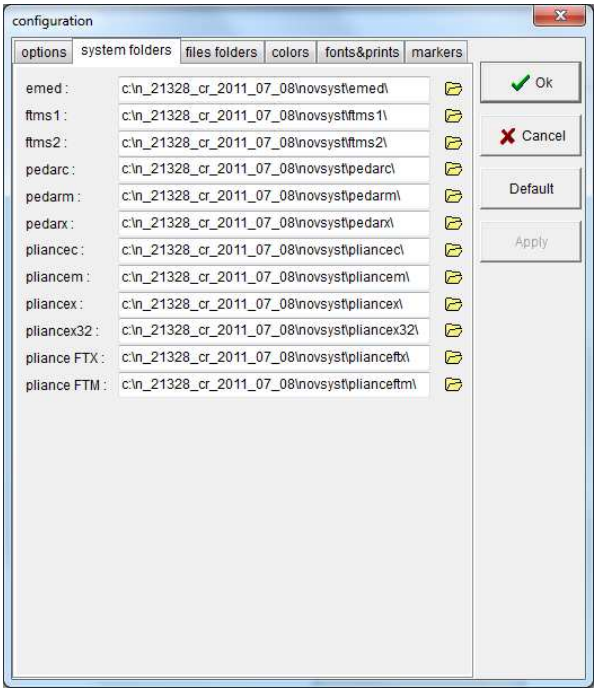
F.1.6 Show center of pressure for:

There are three options for displaying the gait line.

1. **Frame** shows the gait line only to the current frame
2. **From begin to frame** shows the gait line from begin to the current frame
3. **Total process** displays the entire gait line.

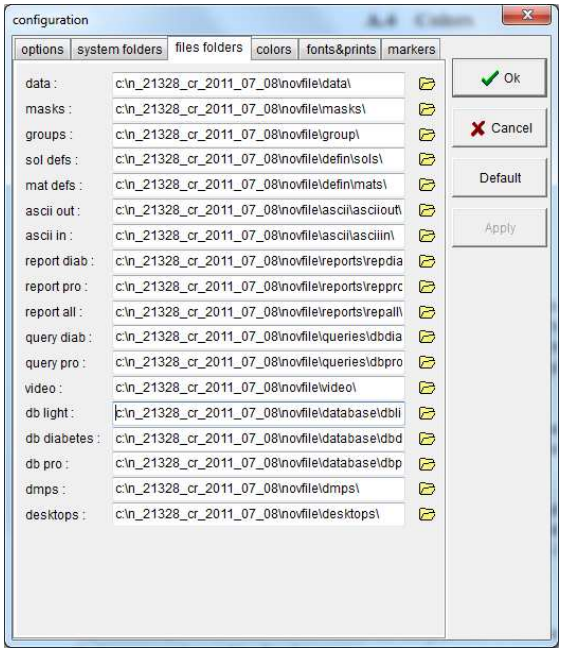
F.2 System folders

System folders show the default settings for the paths and directories where novel pressure distribution measurement systems are saved and loaded from. If paths other than the defaults are desired, enter them here.

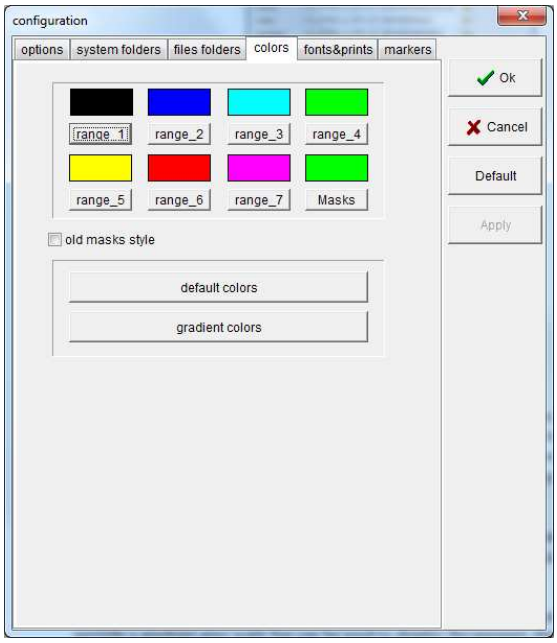


F.3 Files folders

Files folders show the default settings for the paths and directories where data, mask, group, text files et al. are saved and loaded from. If paths other than the defaults are desired, enter them here.



F.4 Colors



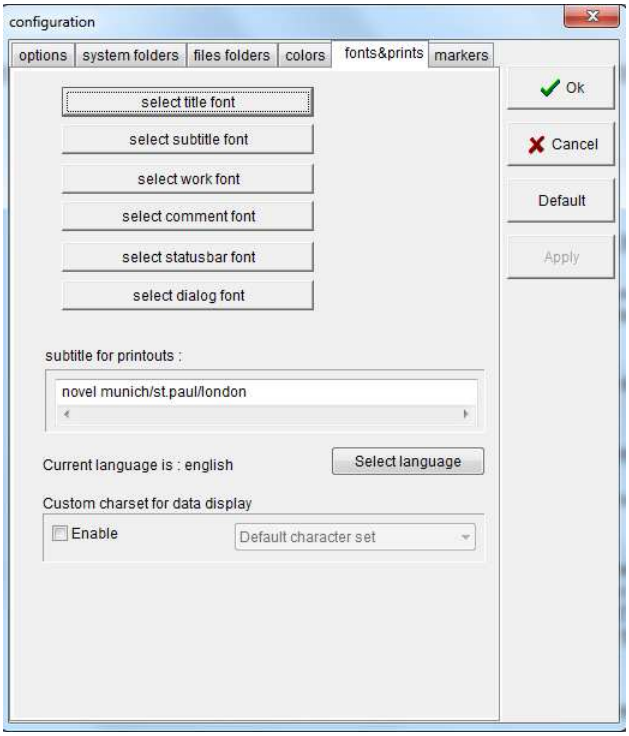
The pressure ranges for the seven color regions are defined before data collection and can only be modified in the program value master. It is however possible to change the colors used for displaying the pressure range. Simply click on the button below the colored field and select a color. However, novel recommends using the default settings as these are commonly used in published literature.

The colors of the lines defining each mask can also be modified by changing the color in the field **Masks**. By selecting **Old masks style**, the lines will be thin. If this option is not selected, the lines will be thick.

Gradient Colors is an option most usually used for black-and-white printing in order to provide a gradient gray scale but can be used to display the pressure data as any gradient color. For a gray scale gradient, choose white in the window **From** and black in the window **To**.

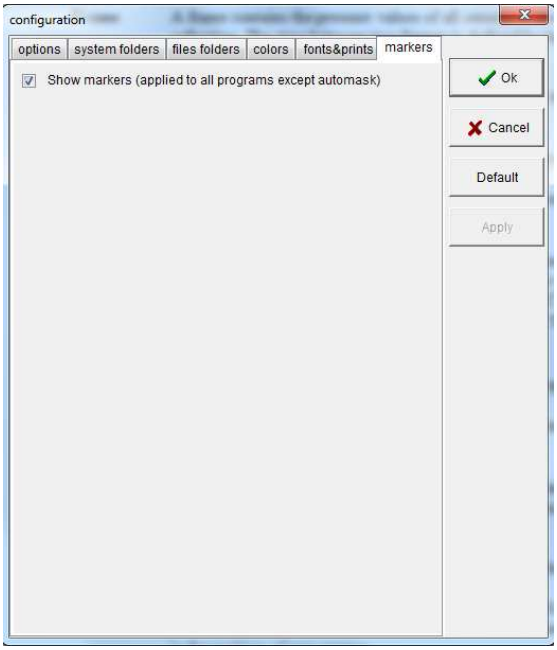
F.5 Fonts & Prints

This section allows the fonts for the screen and the printouts to be selected as well as to select the language to be used in all software.
Language selection in all programs is synchronized with language selection in configuration or in the other program. (Translation of database fields is not synchronized. It is required to translate the database fields in window “Field database”).

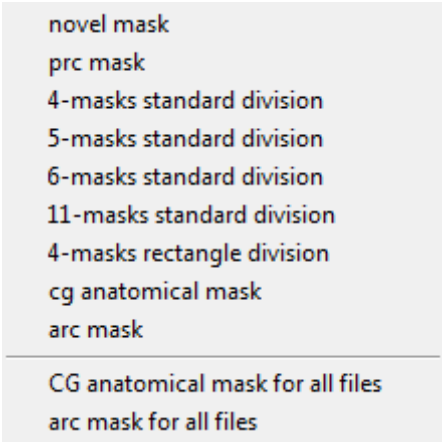


When **Enable** check box is activated, it is possible to select the customer character set for data display.

F.6 Markers



When the **Show markers** (applied to all programs except `automask`) check box is enabled, markers will be displayed in the corresponding programs (creation of any and percent masks, multimask and groupmask evaluation, ofoot, regional velocity of the COP, report) and one of the following masks can be opened (the mask will be created using markers).



IMPORTANT

Standard masks based on markers (for `emed` files) are displayed with the presence of a *.c3D (*.csv) file with the same name in the same folder as the path to the file.

Display of the markers in the `automask` program in the presence of a *.c3D (*.csv) file with the same name in the same folder as the path to the file does not depend on whether the above mentioned check box is activated or deactivated.

#####

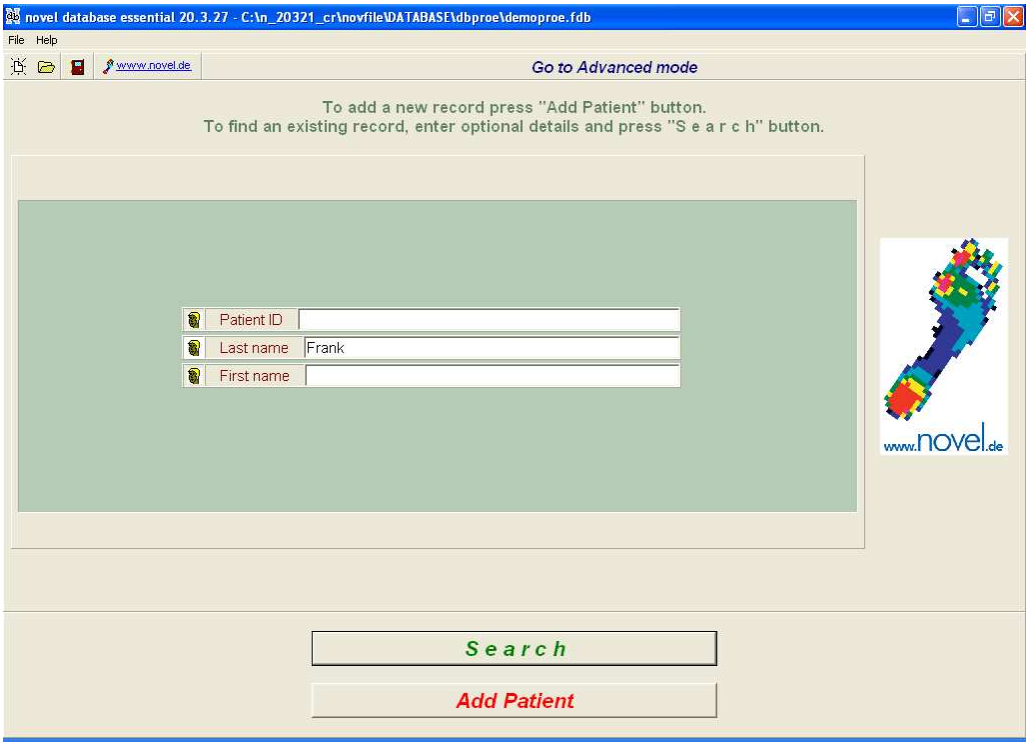
APPENDIX G: Example 1: Daily procedure of collecting data

Simple mode is active

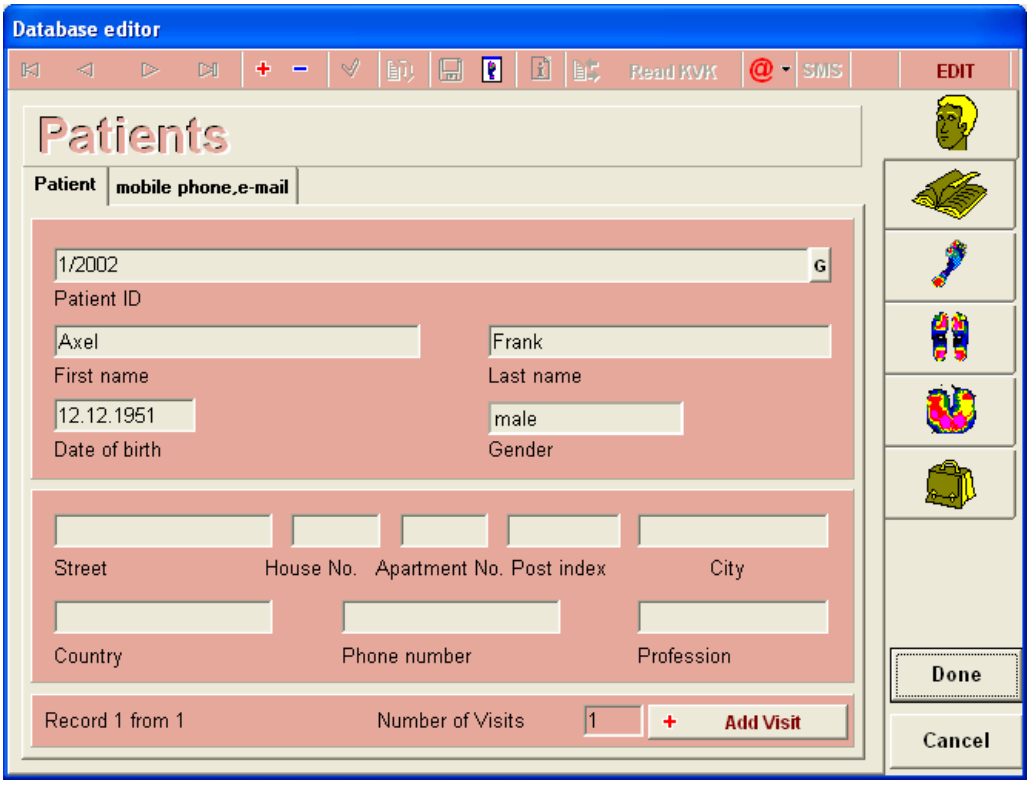


To search for an existing patient:

1. Type Last Name or/and First Name or/and Patient ID data (the exact information must be entered. Partial words will not display results).

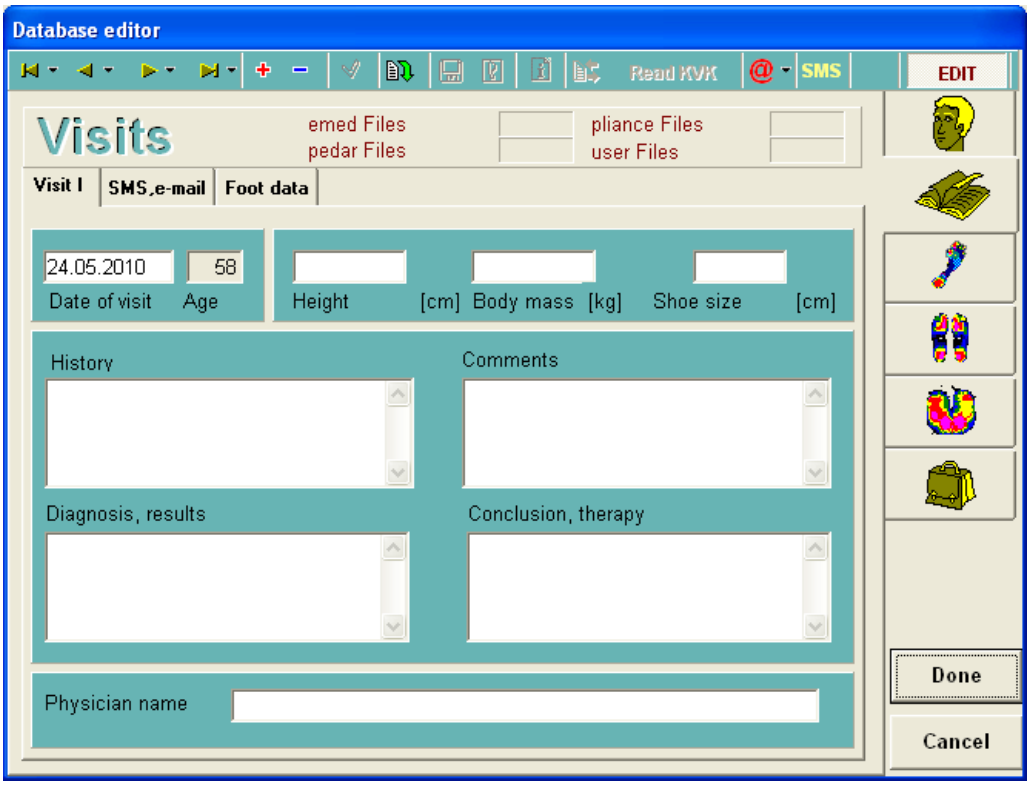


2. Click on the **Search** button.
3. The **Database editor** window will open.



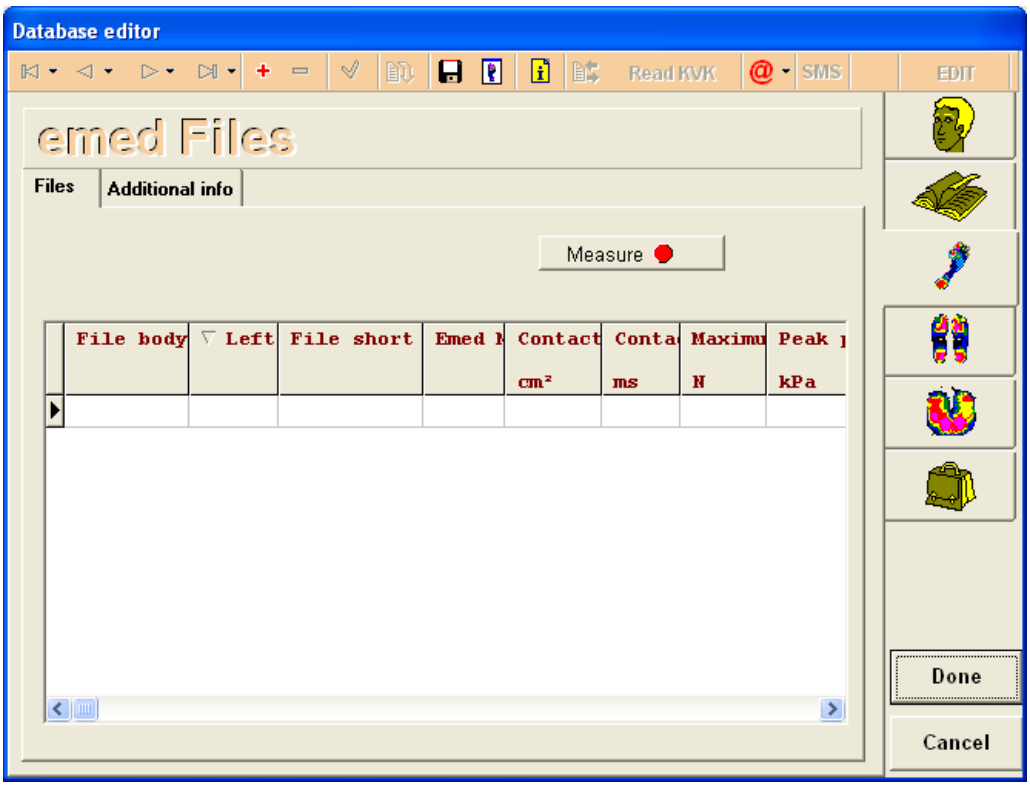
To edit existing patient related information, click the **Edit** button in the top right corner of the Editor Window and add or change the appropriate information (e.g. home address or patient photo or a new profession).

4. Click on the **Add visit** button.

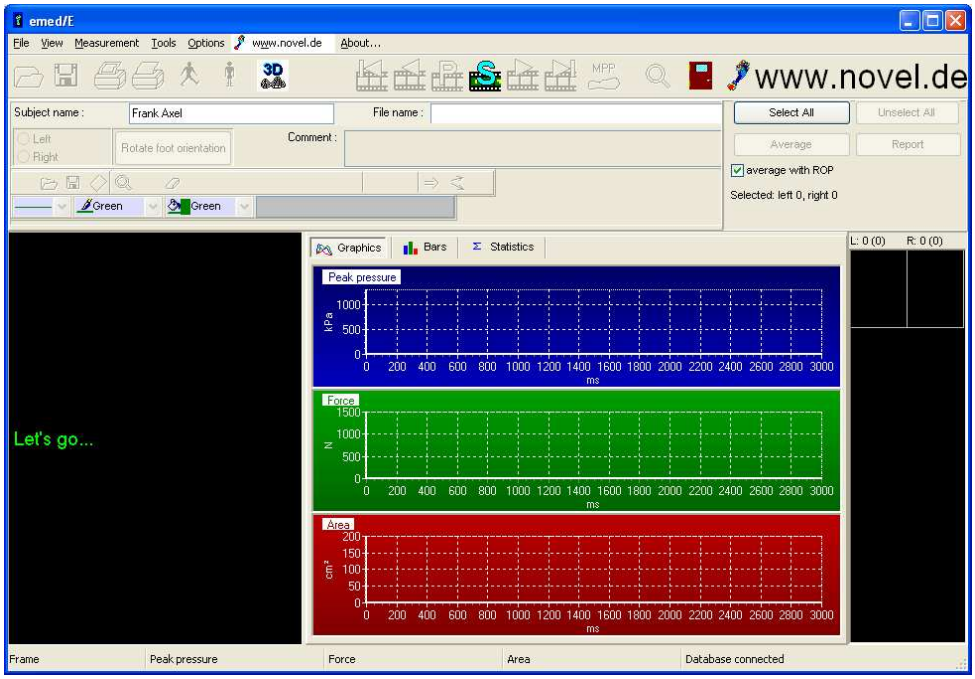


Click the **Load from last** button  to copy information from the latest visit prior to the current visit and change the appropriate information. Add the visit related information if it is different from existing visit information. Don't forget to fill in the fields History, Comments, Diagnosis and Results, Conclusion and Therapy in **Visit I** tab; Amputations - in **Foot data** tab.

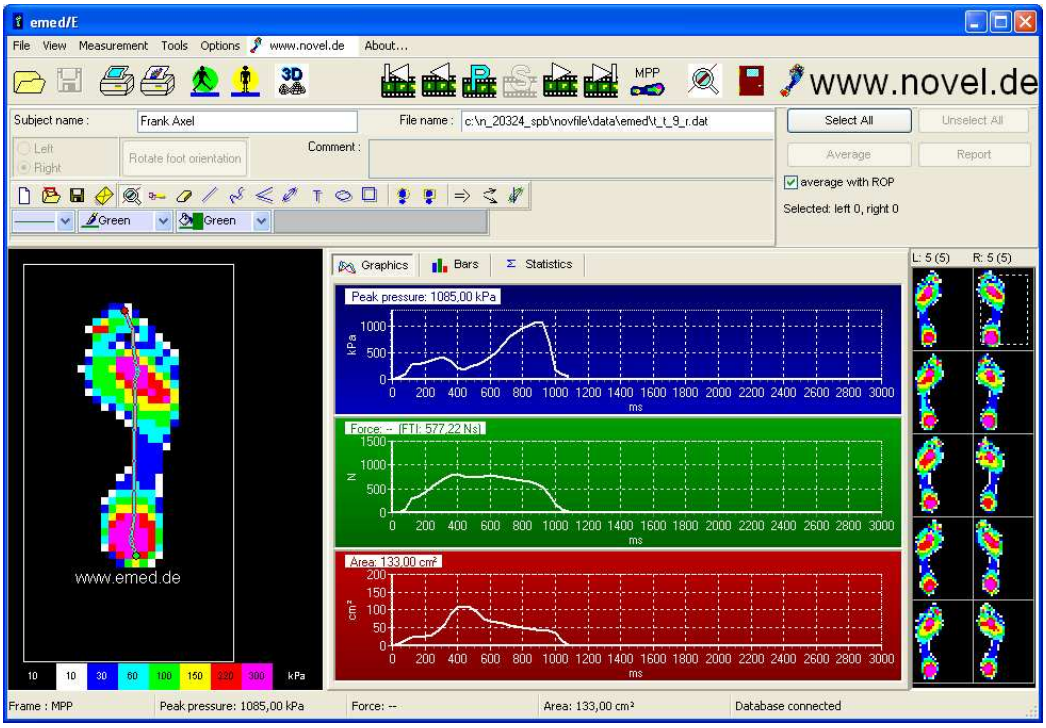
5. Click on the appropriate Data file tab (`emed`, `pedar`, or `pliance`). New files can be collected using the M**easure** button.



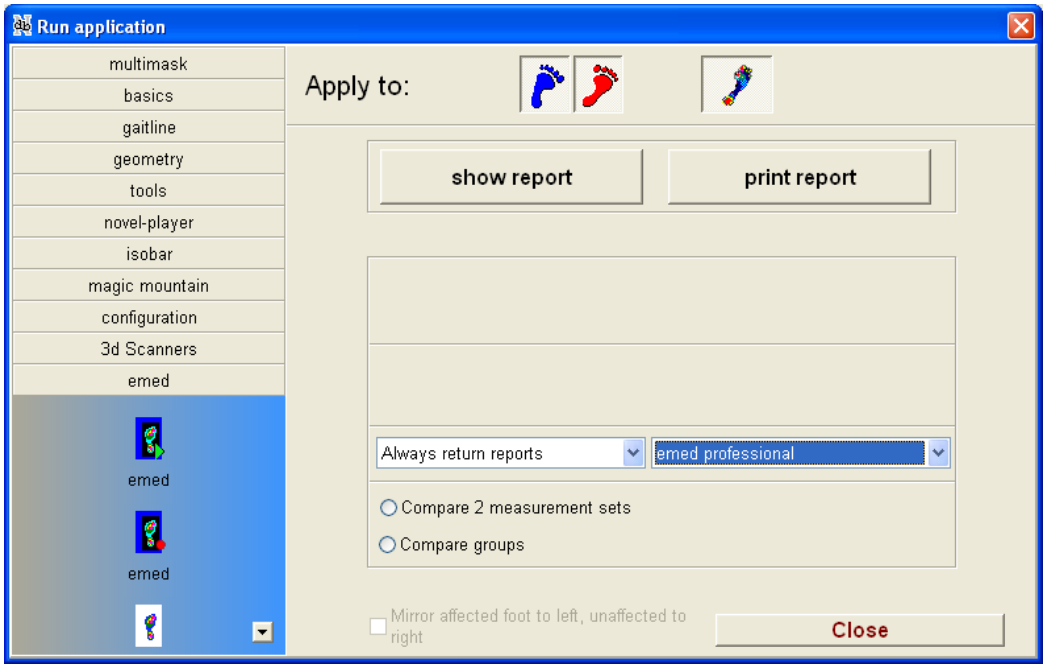
6. Click the **Measure** button.
7. The **emed** program window will be opened.



Collect the necessary data, e.g. 5 left foot and 5 right foot files.



8. Click the **Report** button.
9. The **Run application** window will be opened.

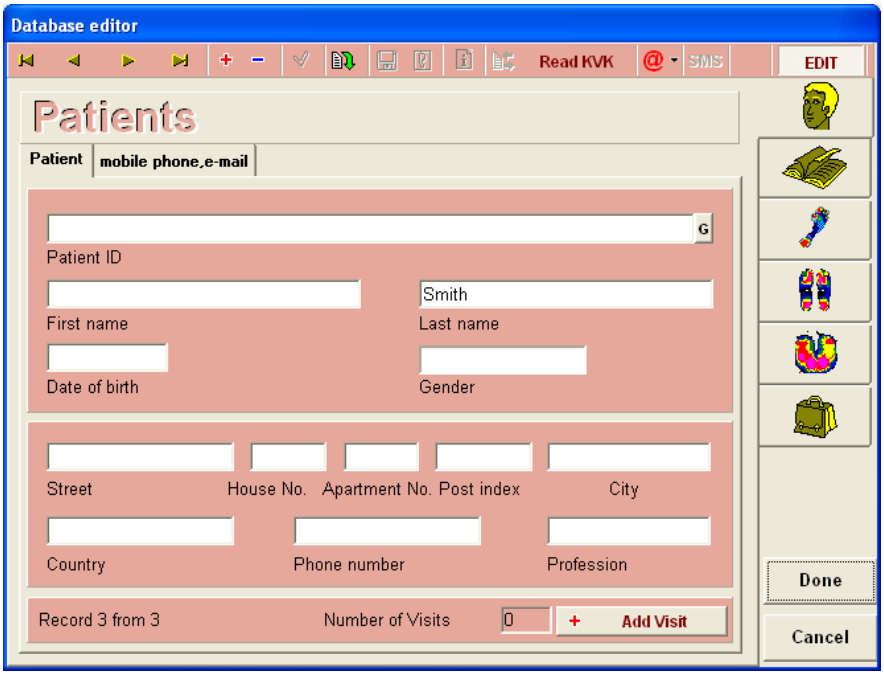


10. Click one of the buttons **Show report** or **Print report** or **novel-projects** to create or to print the report.

When the **Search** button was clicked and more than one Patient matches the search criteria, a message will appear stating multiple records were found. You will need to scroll through the results to find the correct Patient using the scroll buttons: .
When a proper patient is found, repeat the steps 4 – 10 typing the visit related information.

When the **Search** button was clicked and no records match the query, click the **OK** button and then click the **Add Patient** button to add a new patient.

The **Database editor** window will be opened.



Information typed in Simple mode will be displayed. Add any information you need: Patient ID, and/or Last name, and/or First name, and/or Date of birth. Patient ID is not a mandatory field. However, we recommend either to type or to generate it on the base of the fields selected when Patient ID field was created (e.g. Last name, First name, Date of birth).

Repeat the items 4-10 typing the visit related information.

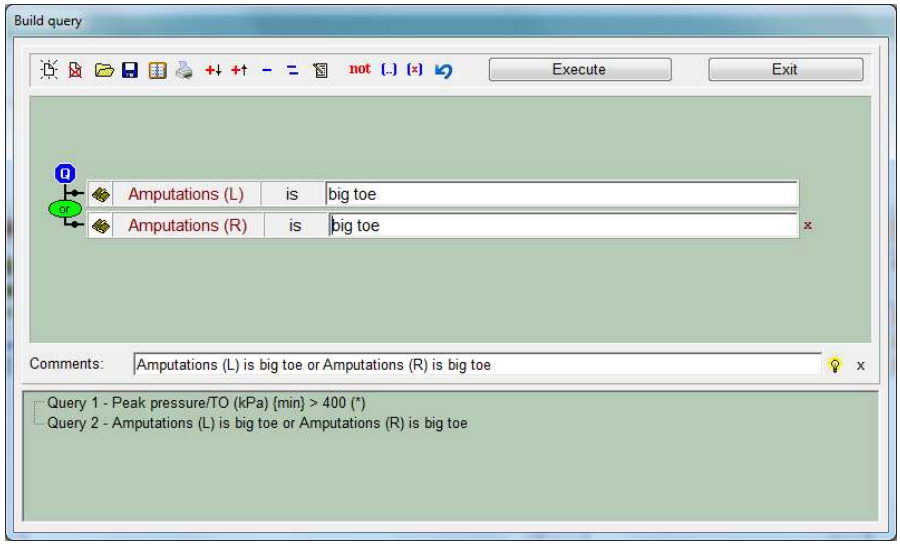
APPENDIX H: Example 2: Data export

Data export is a routine procedure that is often used for data exchange between the different databases, exporting the data for analysis in other programs, etc.

The entire database may be exported, or only a subgroup may be exported (typically more common). This can be as simple as a group of patients between two dates, or specific to parameters contained within the database.

To select patients based on parameters of interest, the **Build Query** function may be used.

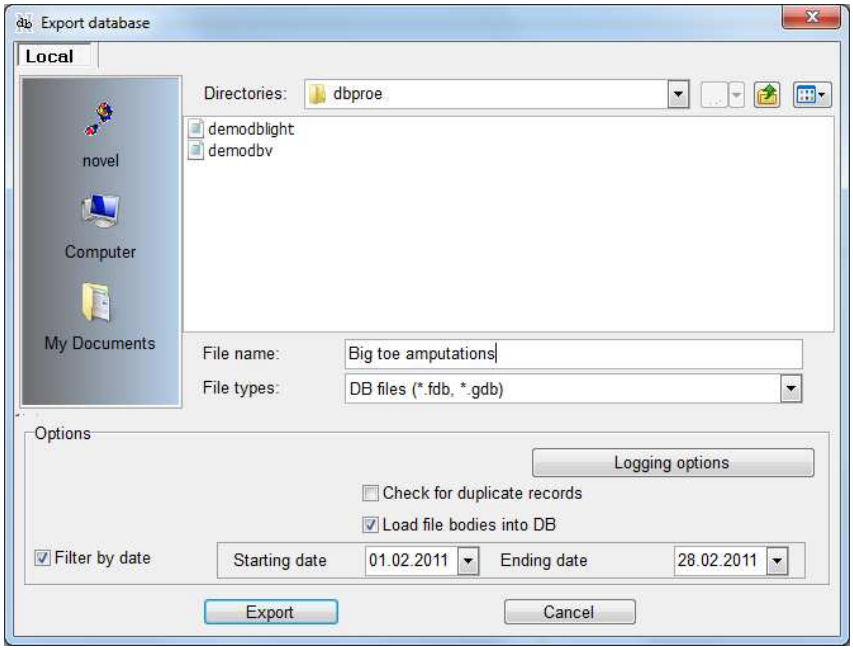
- Click the **Build query** button  or select the **Build query** menu item.



- Select a predefined query, or create a new Query. In the screen capture above a query will display all patients with amputated big toe either on the left or on the right foot.
- Click the **Execute** button. The database is now filtered.
- Select the **Export database** option.

There are two Export options: novel database and ASCII. When the **Export database** option is selected, information from all fields will be exported to a new database.

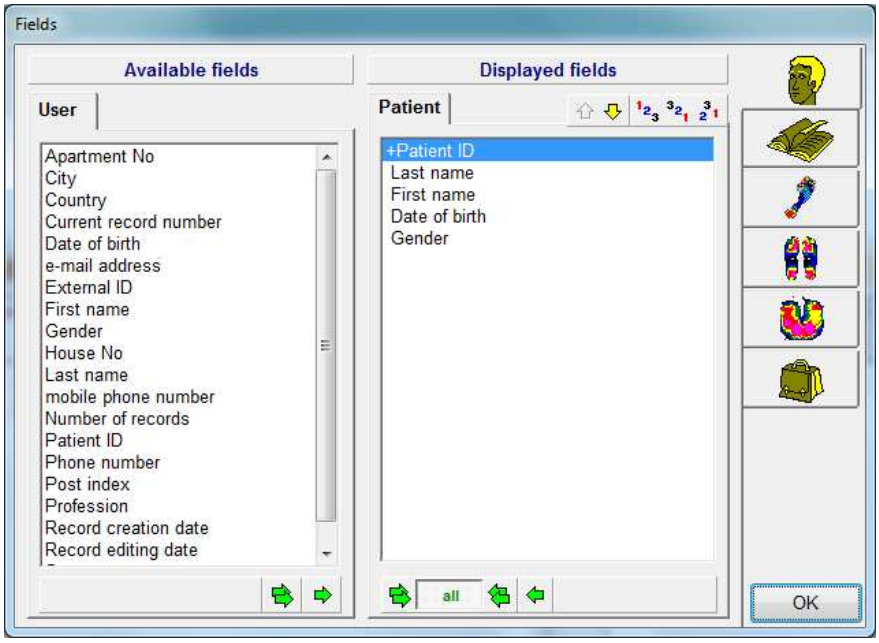
- Click on the **Export | novel database** menu item.



- If required, filter by date (Date of visit) by selecting the **Filter by date** checkbox and typing in the corresponding dates. In the above screen capture the patients with Date of visit between 01.02.2011 and 28.02.2011 will be exported to a new database called Big toe amputations.fdb.
- Select the **Check for duplicate records** check-box to prevent the export of duplicate records.
- Select the **Load file bodies into DB** check-box to prevent the loss of collected data files.
- Click the **Export** button to create the new database.

- When the **Export ASCII** option is selected, information from the fields currently displayed only will be exported. Therefore, before the **Export ASCII** option is selected, the user should select the fields for the display, and correspondingly for the export.

  - Click the **Fields to display** button or select the **Fields to display** option.



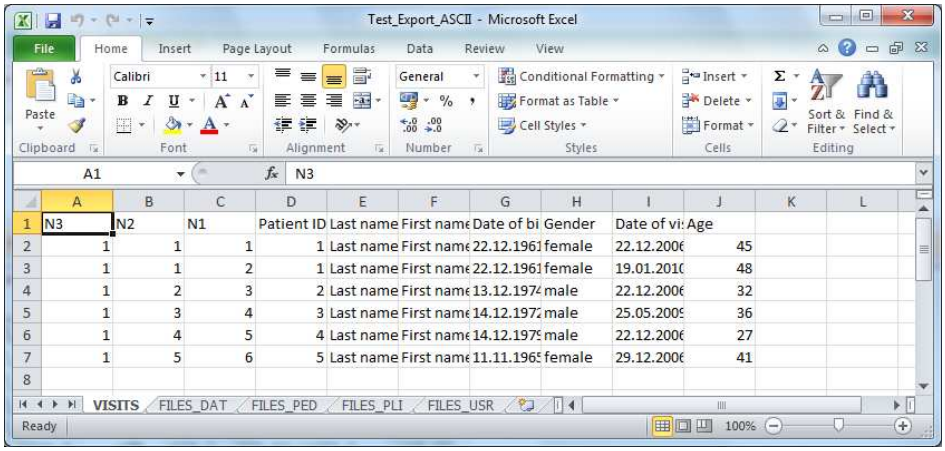
To export the database Select the **Export ASCII** option.



- If required, filter by date (Date of visit) by selecting the Filter by date checkbox and typing in the corresponding dates.
- The database may be exported to ASCII in two file types: ASCII DB files, and Excel Files:
  - The exported ASCII DB files are saved in the directory "c:\novel\novfile\ascii\asciiout" and are saved in several different files:
    - ***.avs**: Includes the data from the Patients and Visits tables
    - ***.adt**: Includes the data from the emed files table

- ***.apd**: Includes the data from the pedar files table
- ***.apl**: Includes the data from the pliance files table
- ***.aur**: Includes the data from the User files table

o When the Excel files type is selected, the following window will be opened in background (Windows 7) or foreground (Windows XP).



▪ A page will be created for each of the tables from the database

- **VISITS**: Includes the data from the Patients and Visits tables
- **FILES_DAT**: Includes the data from the emed files table
- **FILES_PED**: Includes the data from the pedar files table
- **FILES_PLI**: Includes the data from the pliance files table
- **FILES_USR**: Includes the data from the User files table

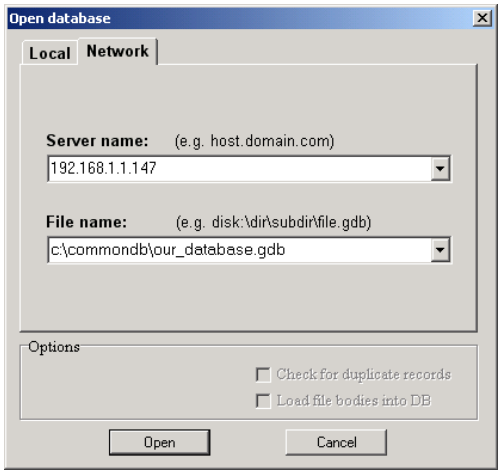
IMPORTANT

MS Excel 2003 or later should be installed.

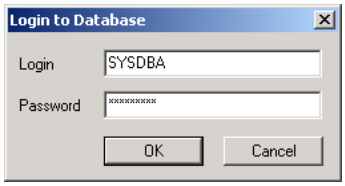
o Click the Export button to create the new database

APPENDIX I: Example 3: Installation of the novel databases in a network

1. **Firbird SQL server version 1.5** must be installed on the server.
2. The database software should be installed on each client station (station).
3. On one of the stations a new database (*.fdb file) should be created using the menu item **File|New DB**. If necessary to change database structure (i.e. create user fields etc) on this station.
4. The newly created database file should be stored on the server hard drive. (It is not necessary to share this folder on the network!)
5. Copy the dbproudf.dll file from the “Drive\Program Files\Firebird\Firebird_1_5\UDF” folder on the local station to folder the folder with the same name on the server.
6. On the local station, go to the menu **File|Open**, click on the **Network** tab. Specify the server name or the network address and the path to the server where the database is stored.



7. Click **Open** and the following login dialog will appear. Please type the login name and the password:

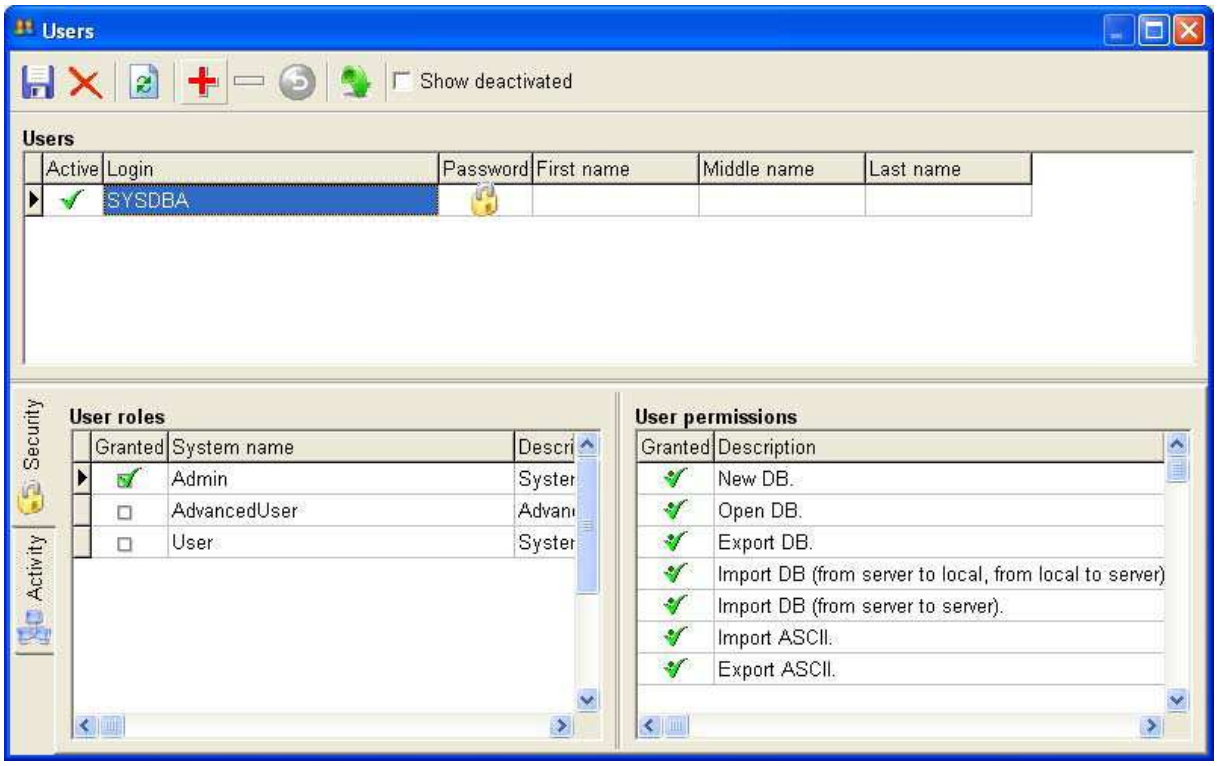


   By default the login name is: “SYSDBA” and the password is: “masterkey” .The PC will now open the database located on the network.

8. **VERY IMPORTANT: This user name and password is common to all users. If you do not change password for the SYSDBA user – anyone can gain access to the data! Anyone could change or destroy database structure or delete important data.**

   PLEASE CHANGE THIS PASSWORD FOR YOUR DATA SECURITY AND SAFETY! PLEASE STORE THIS PASSWORD IN A SAFE PLACE!

9. If required, use the File|Administration menu option to create one or more users with individual passwords and individual access rights.

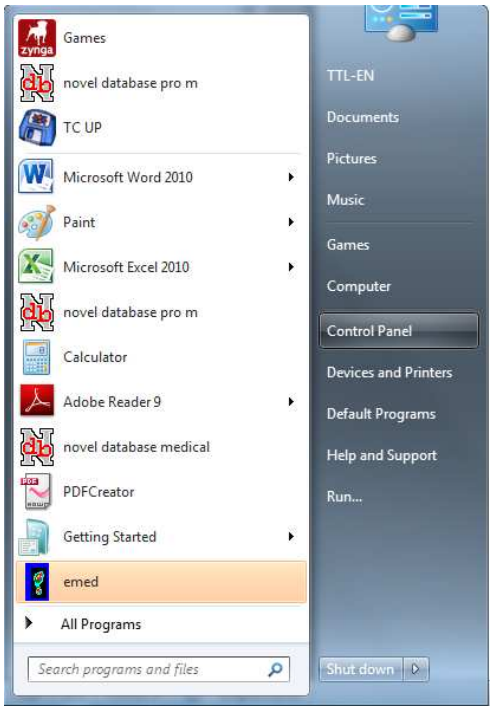


10. To create a new user click the "+" sign. Click on the lock button to change the password for a user (including the SYSDBA user). By clicking on the different user roles listed, it is possible to give different permissions for each user (the permissions are listed on the right side). After you finished creating the appropriate users click the **Save** button.

#####

APPENDIX K: Example 5: Opening port 3050 for the incoming connections to Firebird server via TCP/IP protocol (Windows 7, Firewall)

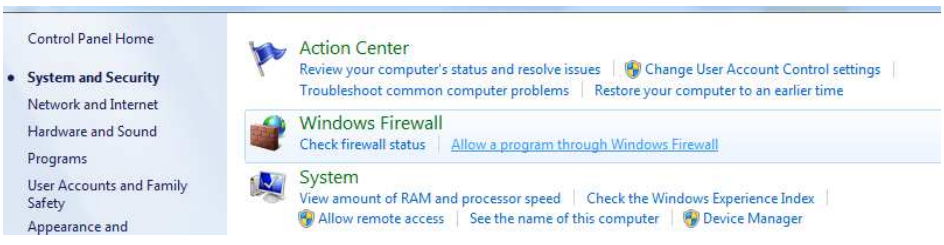
1. Go to Program/Control Panel



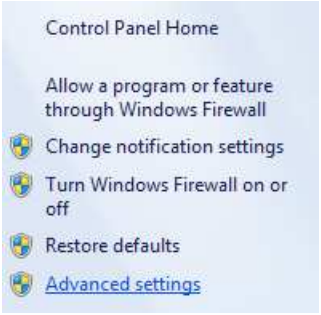
2. Go to System and Security.



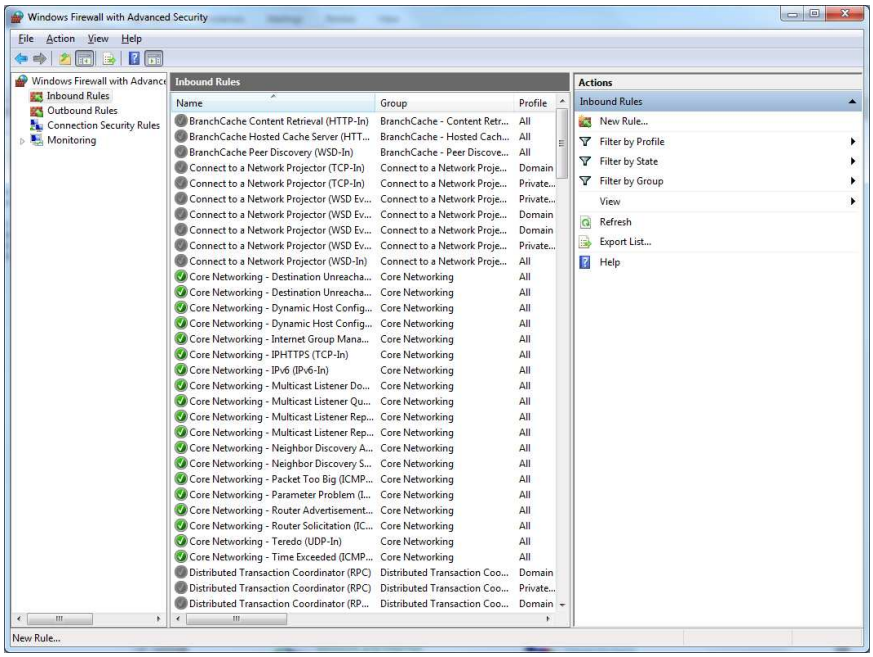
3. Go to Windows Firewall.



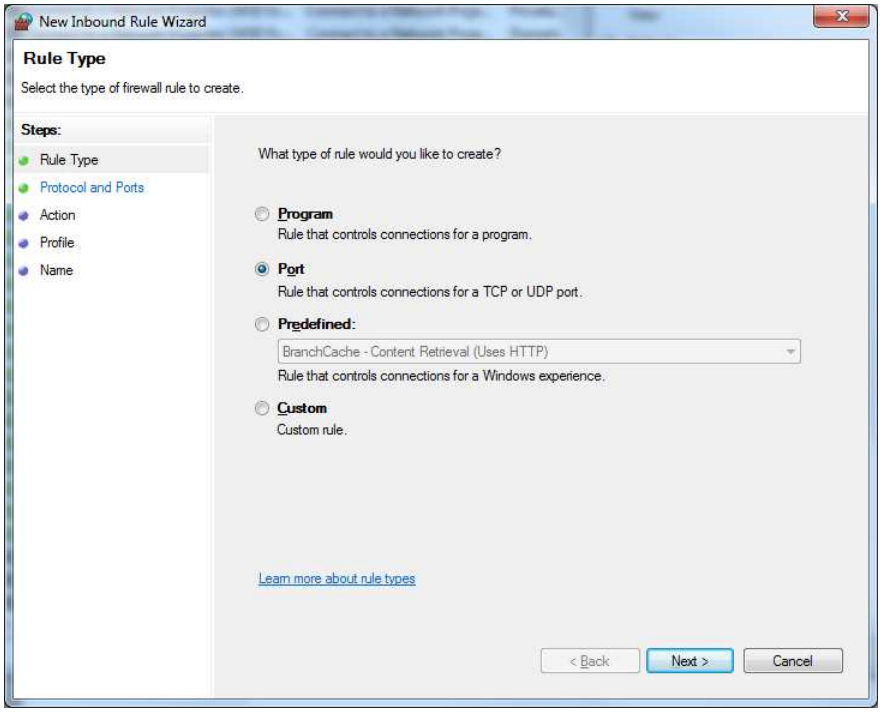
4. Go to Advanced settings.



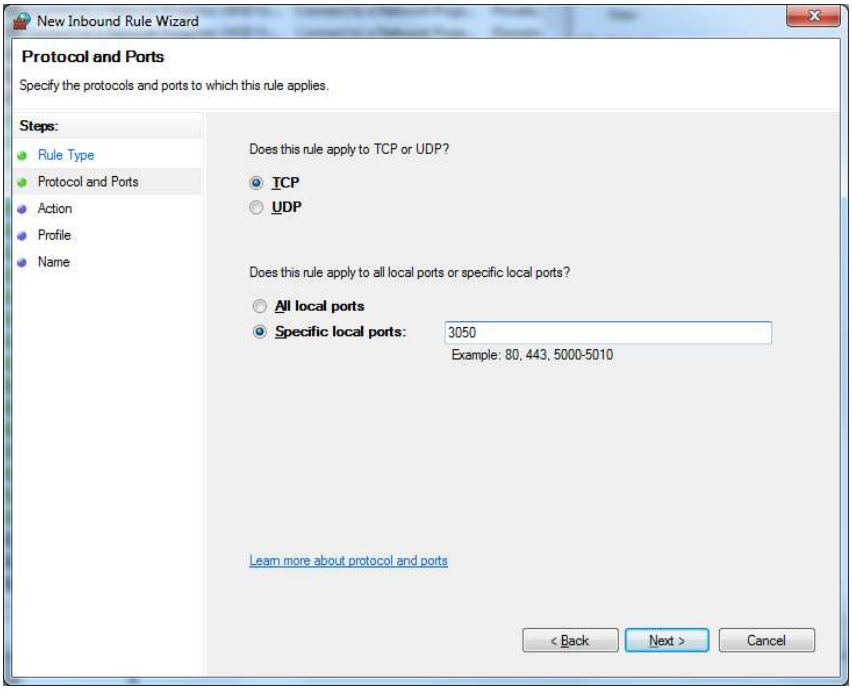
5. Select Inbounds Rules in the left side of the window and New Rule … in the right side of the window.



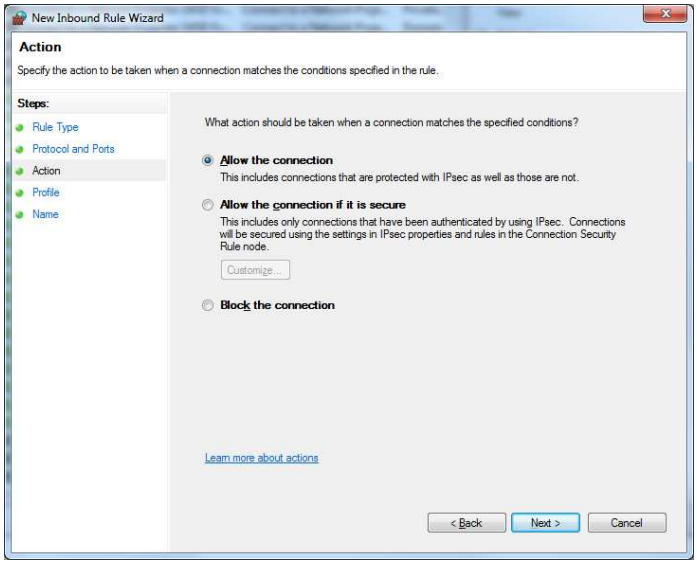
6. Select Port rule type.



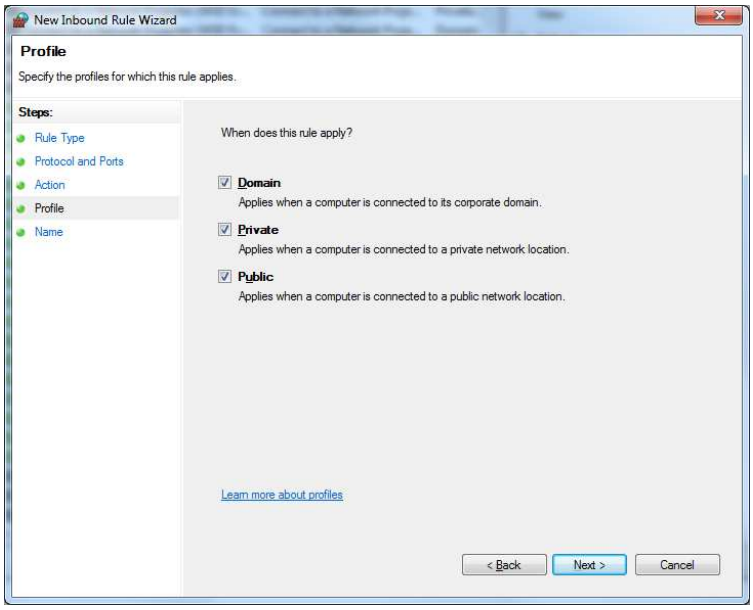
7. Apply the rule to TCP protocol and enter 3050 in the Specific local ports box.



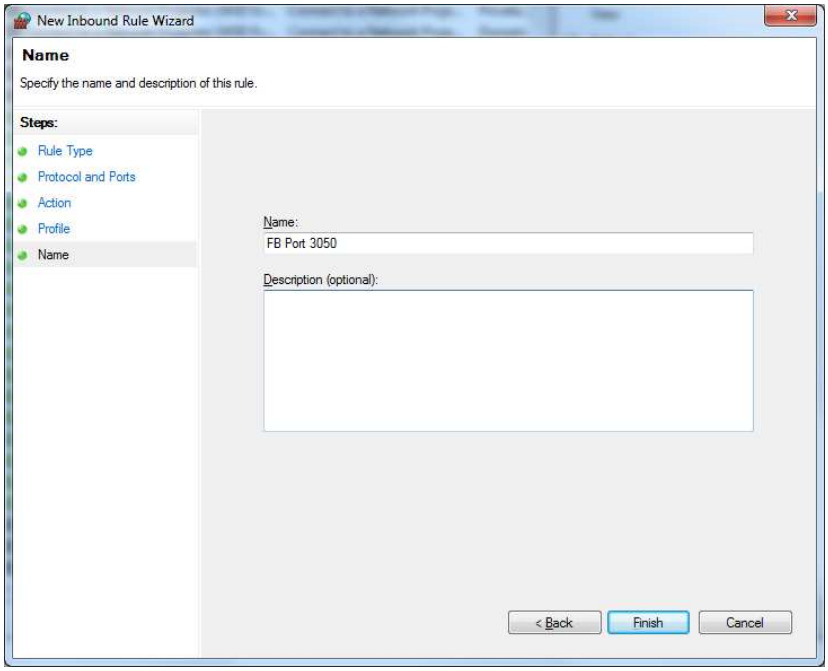
8. Click the Next button.



9. Click the Next button.



10. Enter the Rule name, e.g. FB Port 3050, and click the Finish button.


New Inbound Rule Wizard
Name
Specify the name and description of this rule.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
Name:
FB Port 3050
Description (optional):
< Back
Finish
Cancel

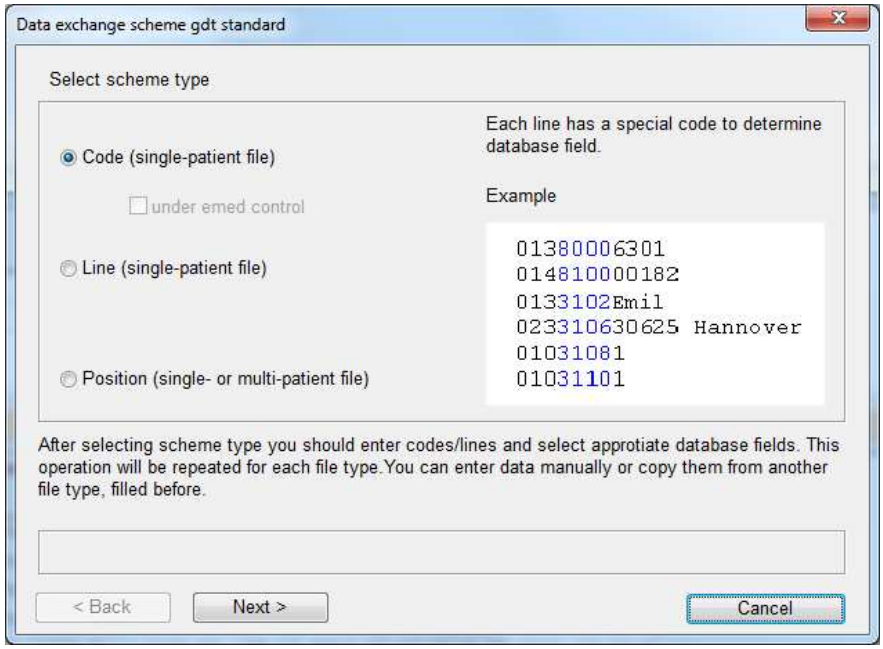
#####

APPENDIX L: Data exchange (detailed description)

The DES file includes:

- scheme type
- file types (paths and file name masks)
- options
- data in each file type
- scheme name

1. Scheme type



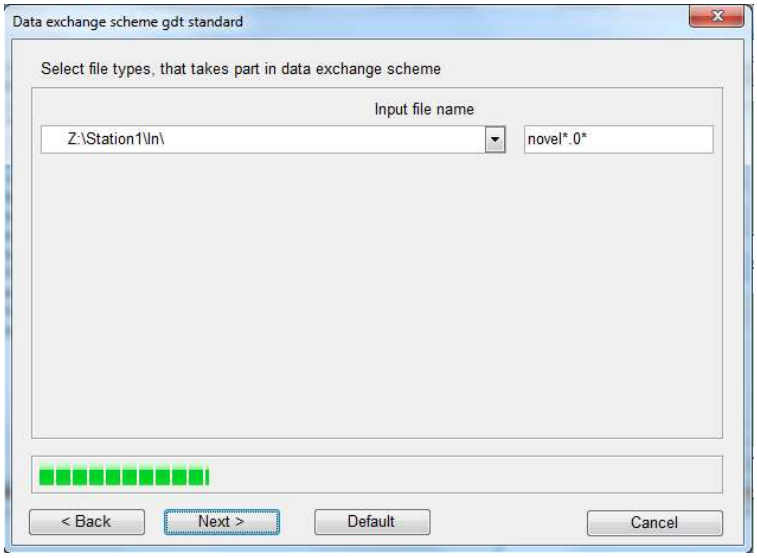
Selection of the scheme type is required for the input file only.

- Code - each line has a special code, determining the database field
- Line - each database field is located on the defined line number
- Position - each database field is located on the defined position.

NOTE:

- Code and Line types are used for one patient in the input file only; Position type – for one or several patients.
- Code type is used in gdt standard DES (MCS-ISYNET, Medistar etc.), Line and Position types are used in shoe makers databases (i.e. Paedus, CosInOS, Rothballer, Orthosoft etc.).
- Code type is used for all below listed file types, Line and Position types are used for the input file only.

2. File types



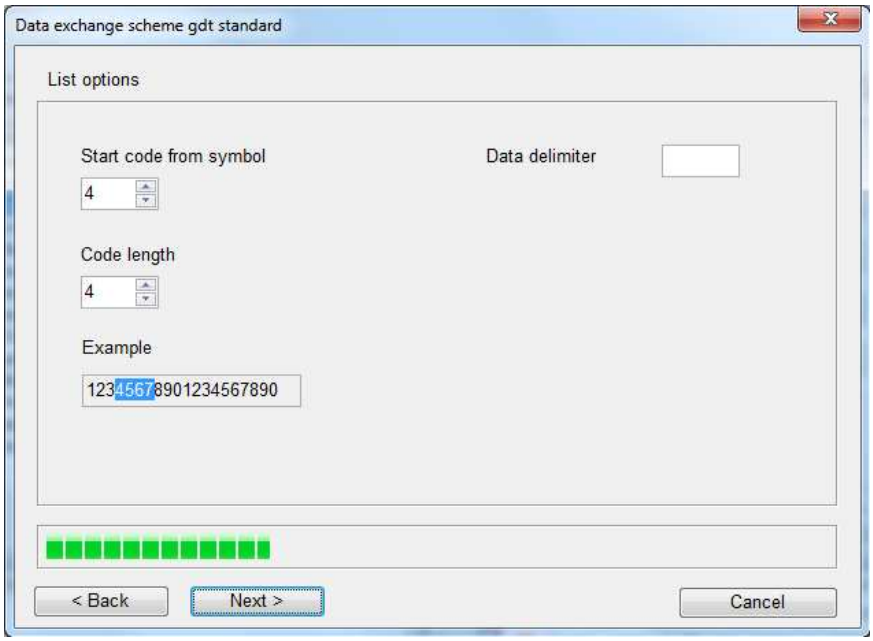
Request for a new measurement (input file) is available only.

NOTES:

- The path to the input file and the mask for the input file name exist in each DES.
- The file mask (e.g. novel*.0*) is valid for the input file. It is not recommended to specify the *.* mask.
- Use the Default button to return to the paths and file names in previous versions of the software (stored in Novpro.ini).

3. Options
   The options which are displayed depend on the selected Scheme type.

   Code type of the DES

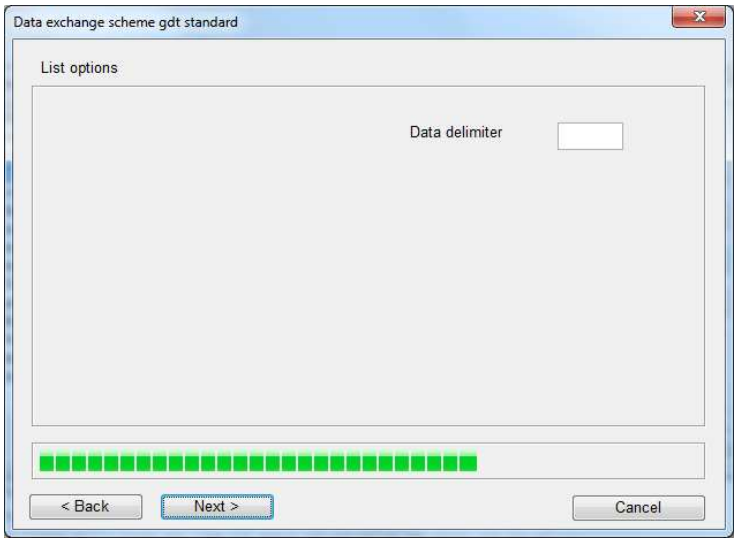


The following information is stored in the DES:

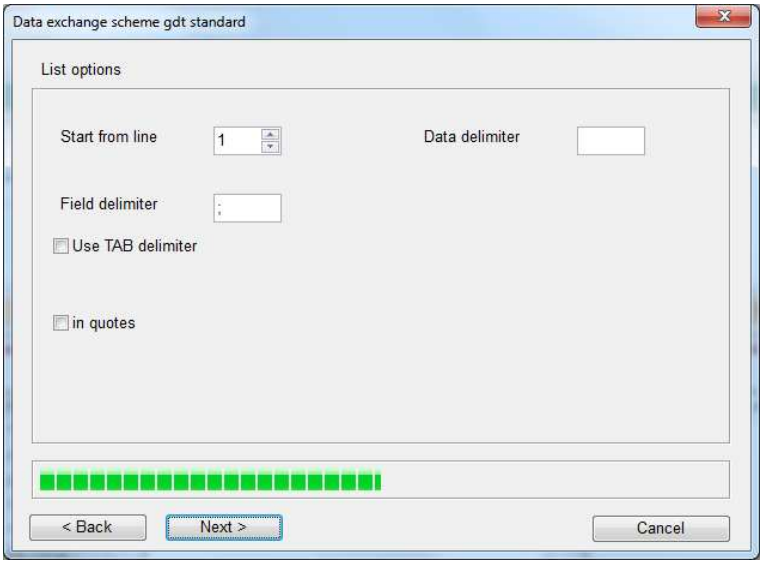
- the current number of the symbol from which the code is stored
- the code length
- delimiter for data separation

NOTE: The presence of the Data delimiter symbol is not required.

The code position is displayed in the Example text box.
Line type of the DES



4. Position type of the DES



The following information is stored in the DES.

- line number, from which the data reading will start
- delimiter for data separation
- delimiter for fields separation

NOTES:

- If the field names are absent then the line number is equal to 1.

- If the field names are present then the line number is equal to 2.
- If the Field delimiter is empty, then the Use Tab delimiter is used.
- If data are in the quotes, then the **in quotes** check box is selected. Additionally the quotes symbol is stored.

5. Data in the input file (request for a new measurement)
   Code type of the DES

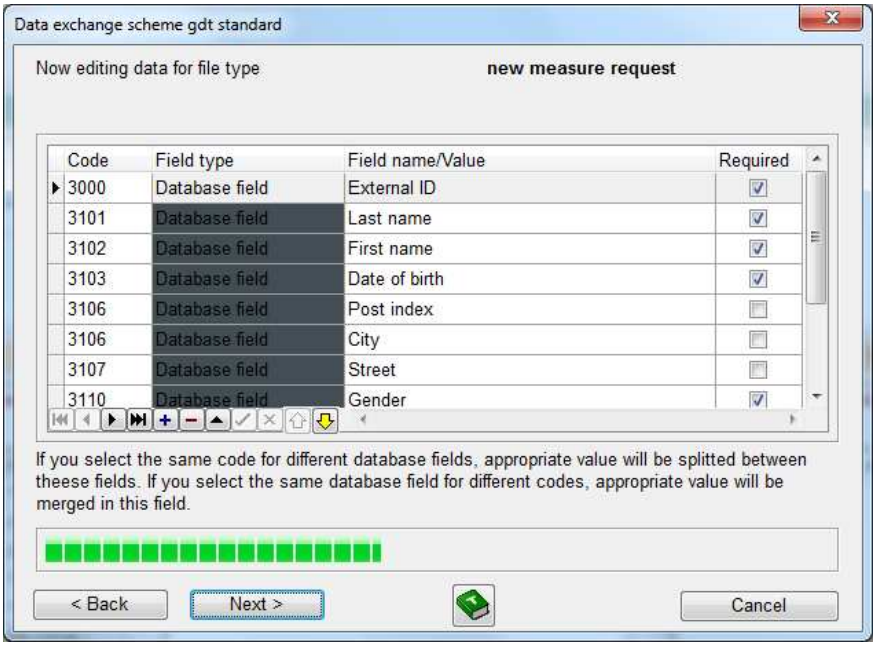


Table columns

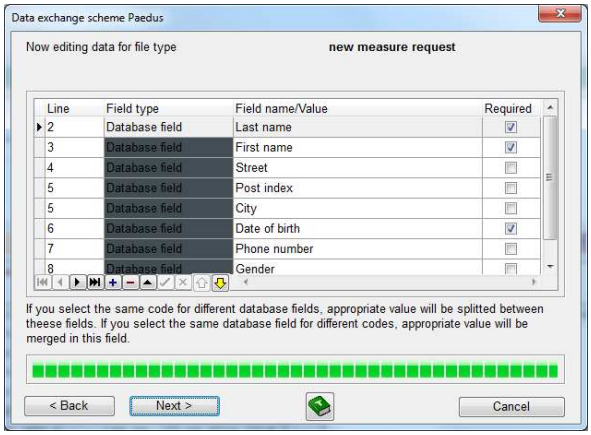
- The **code** column contains the code, corresponding to the defined data in the input file
- The **field type** column contains of the following types: Database field or Service field or Constant expression.
- The **Field name/Value** column contains the data in correspondence with the code.
- The **required** column stores the mandatory data indication (absence of this data will stop the procedure of the data exchange).

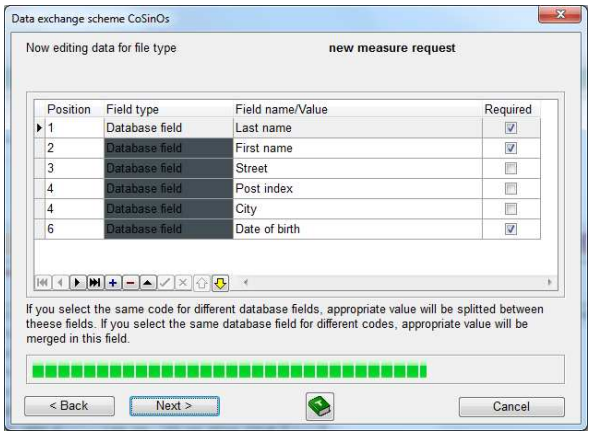
Use the buttons on the bottom of the window

- to go to the first, previous, next, and last records

- to add a new record, to delete and to edit the current record
  
- to apply or to cancel the changes
  
- to move the selected record up and down
   .

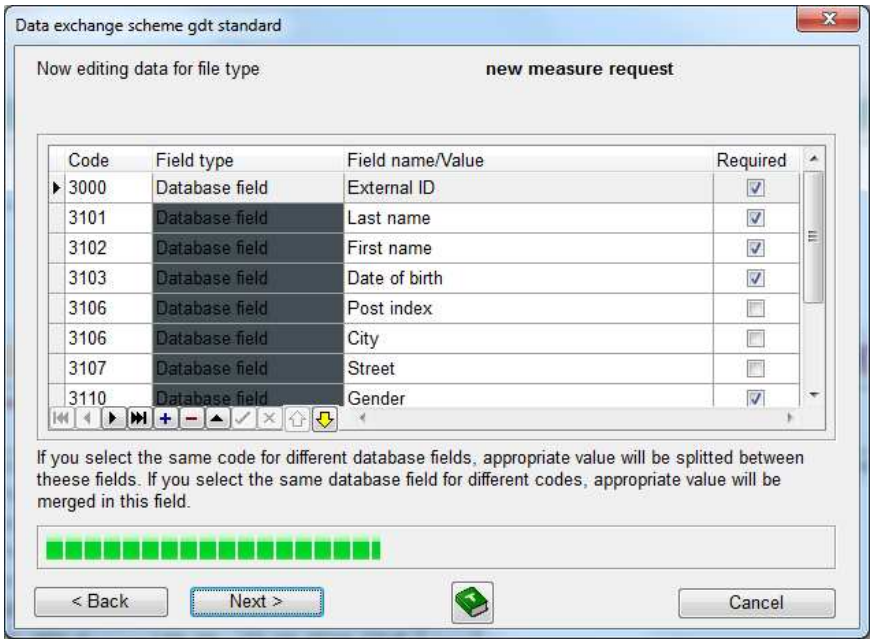
6. Line and Position types of the DES



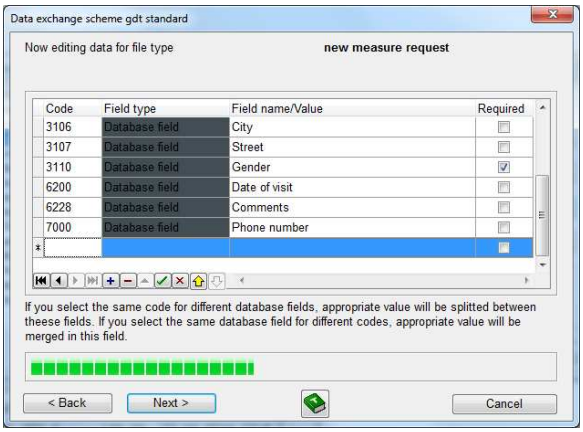


The same table is used for Code, Line, and Position types of the DES. The first column will be Code or Line or Position accordingly.

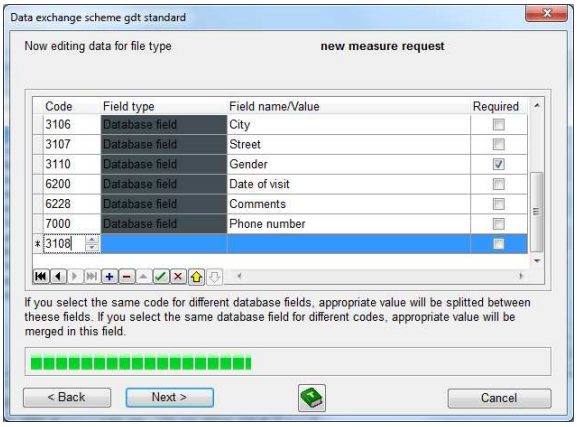
7. Adding a new row in any table (for all file types)
   The input file for code type DES is given below.
   Example: add the **Body mass** field with code 3108.



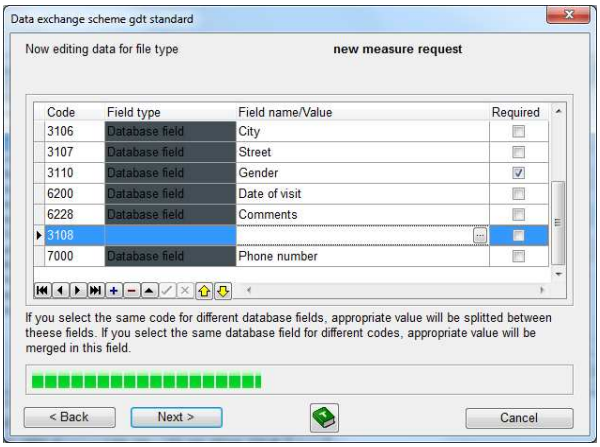
a. Use the + button to add a new row.



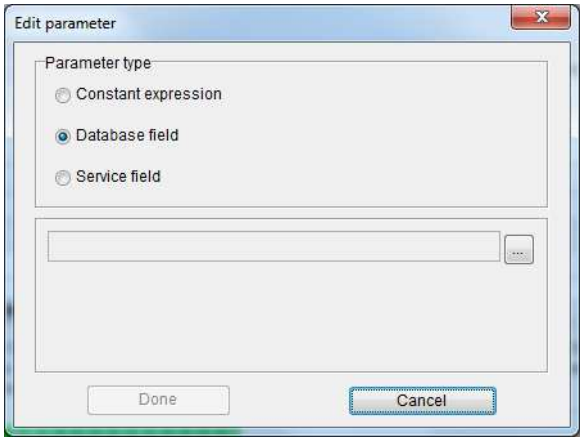
b. Enter the code 3108 in the Code column.



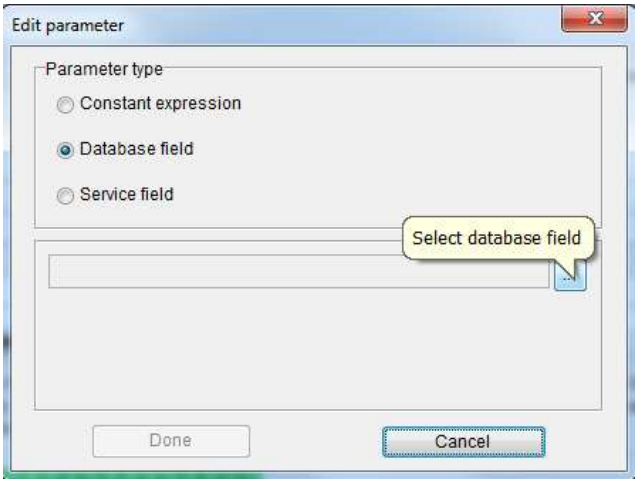
c. Click the **Field name/Value** column in this row to activate the ... button.



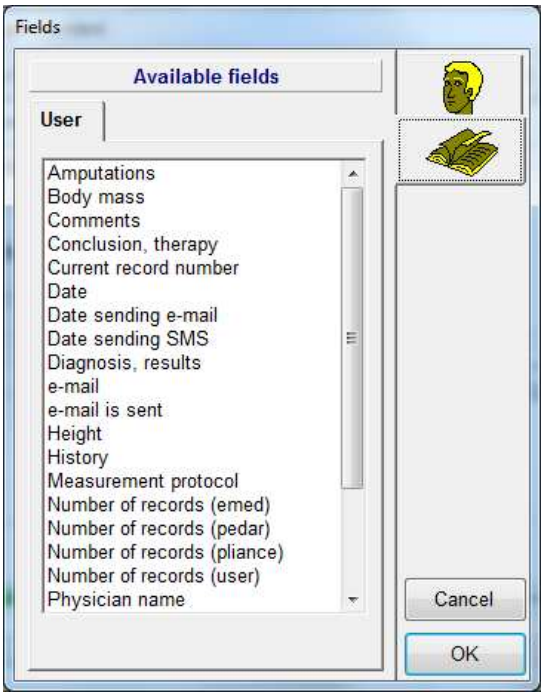
d. Click the ... button and select the parameter type, i.e. Database field for Body mass.



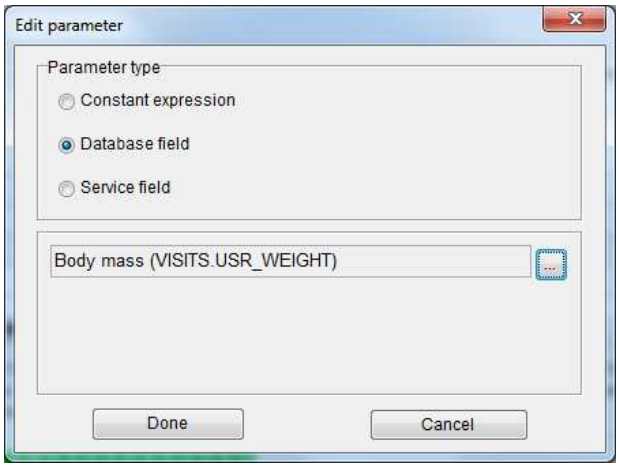
e. Click the **Select database field** button to enter **Body mass**.



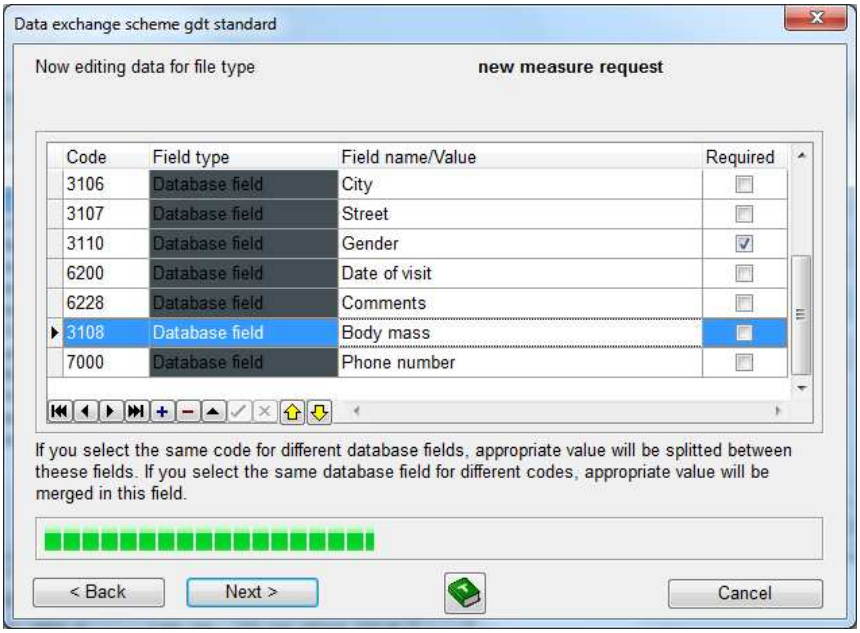
f. The Fields window will appear. Select Visits table and Body mass field.



g. Double-click and the Body mass field will transfer Body mass in the field box.



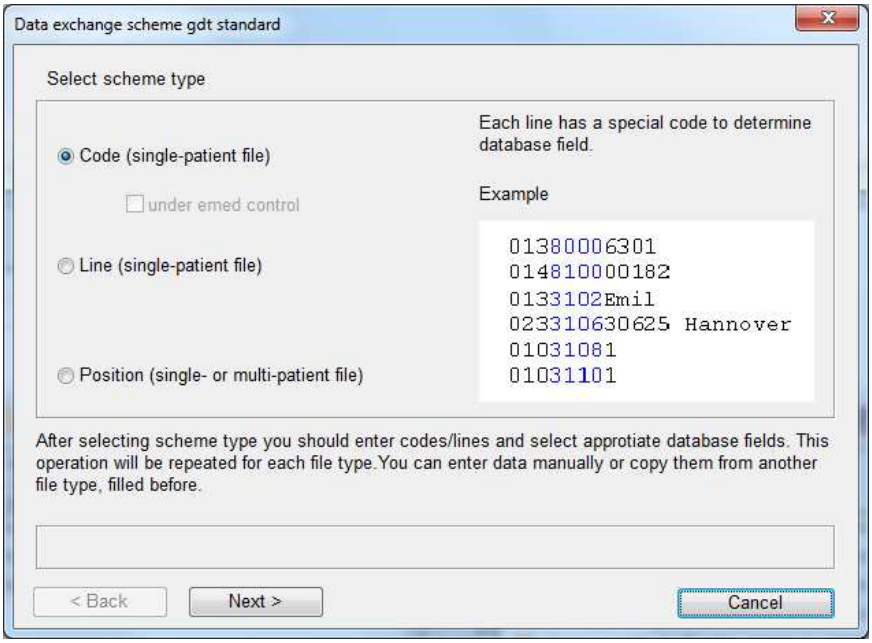
h. Click the Done button to return to the input file table.



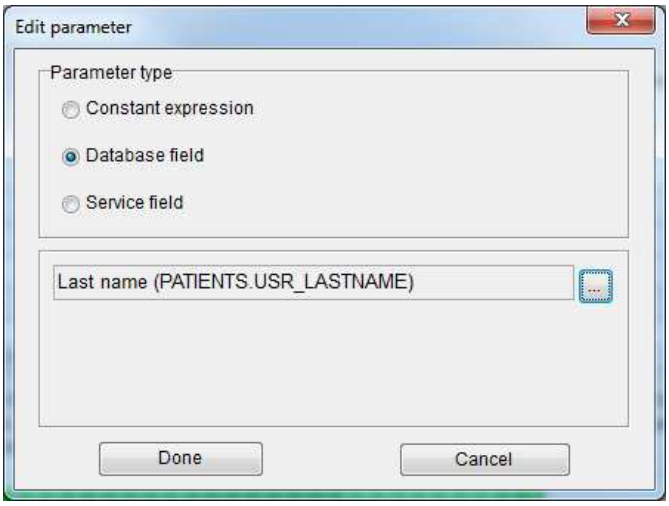
i. Select **Required** checkbox if the presence of this data (Body mass) is required in the input file.

8. Adding a new DES

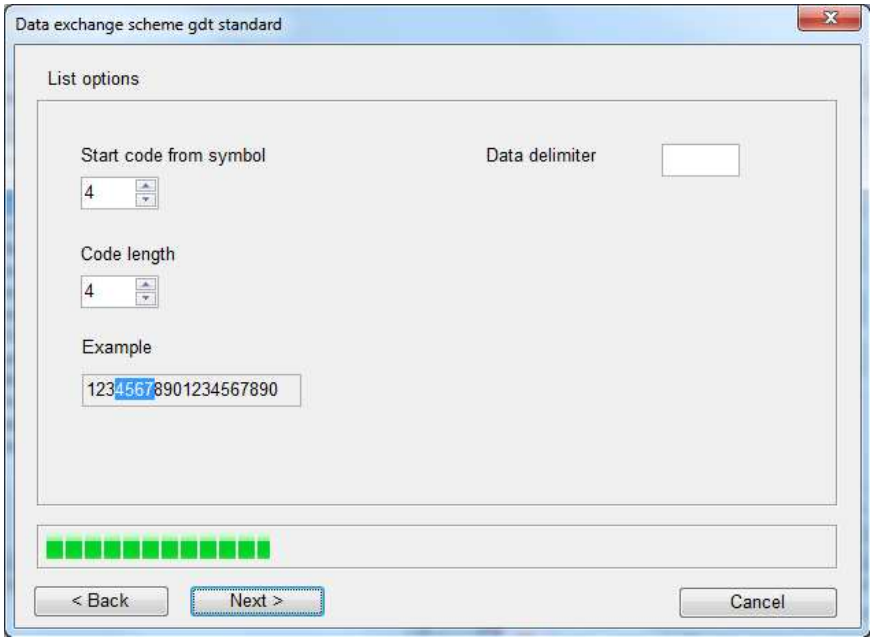
a. Select a DES type: Code, Line or Position.



b. Select the file types, required for the creating DES.

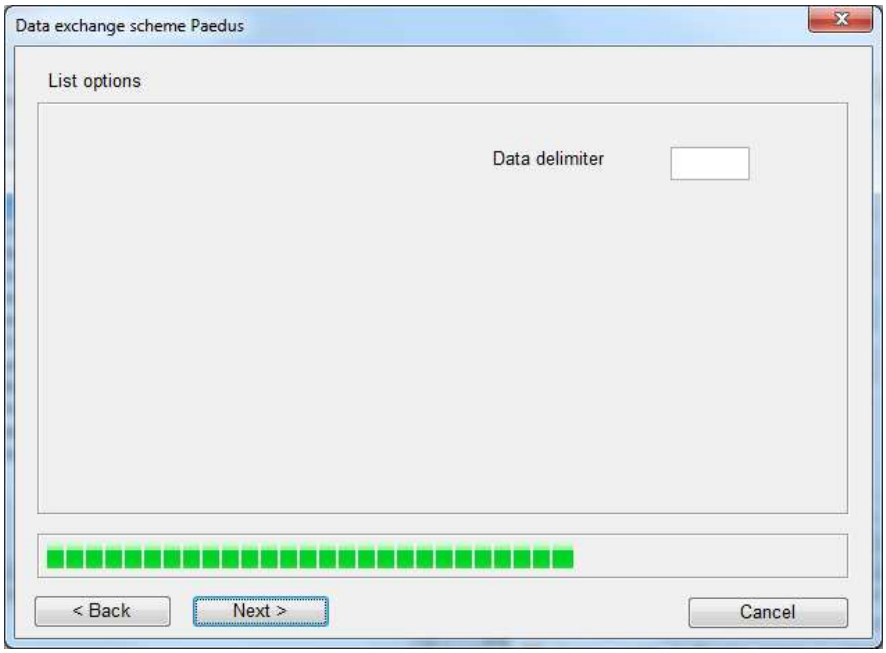


c. Select the options depending on the scheme type



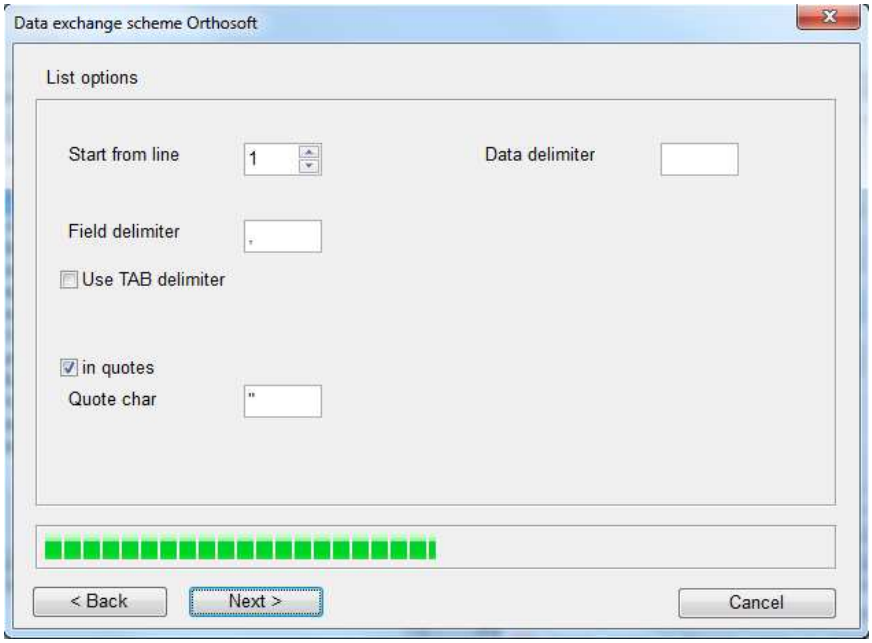
Enter the number of the first symbol position and the code length. Both values are equal to 4 for gdt standard DES, for example. The code position is displayed in the Example text box. Enter the delimiter to split the data in several fields in Data delimiter text box (e.g. ",").

i. Line type of the scheme



Data delimiter information is used for the same purposes as in the code type scheme (is not obligatory).

ii. Position type of the scheme



Enter the line number, from which the data reading will starts, in Start from line box.
If the field names are absent then the line number is equal to 1.
If the field names are present then the line number is equal to 2.

Enter the delimiter, separating the data, in Field delimiter text box or select the Use Tab delimiter check box.

If information is in quotes, then select the In quotes check box and select the Quote char.

Data delimiter information is used for the same purposes as in the code type scheme.

d. Code /data table.
This procedure must be repeated for each selected file type.
It is recommended that the table for input file (New measure request) is created and then copied for the other file types.

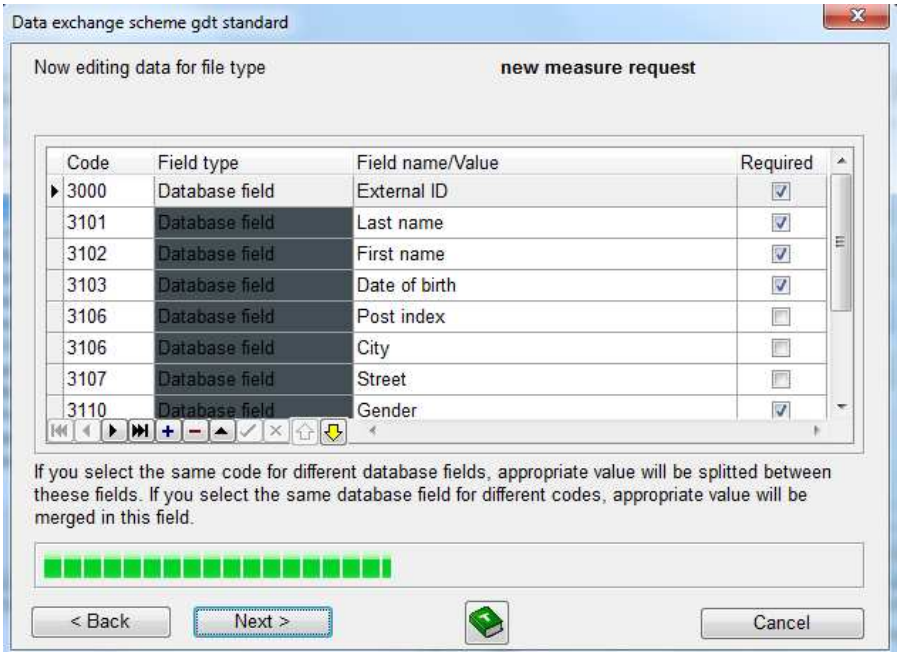
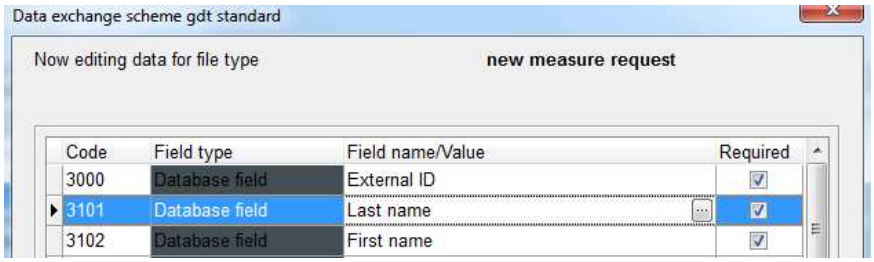


Table columns

- The **code** column contains the code, corresponding to the defined data in the input file
- The **field type** column contains of the following types: Database field or Service field or Constant expression.
- The **Field name/Value** column contains the data in correspondence with the code.
- The **required** column stores the mandatory data indication (absence of this data will stop the procedure of the data exchange).

i. To select the data in the Field name/Value column, click the  in the active box.

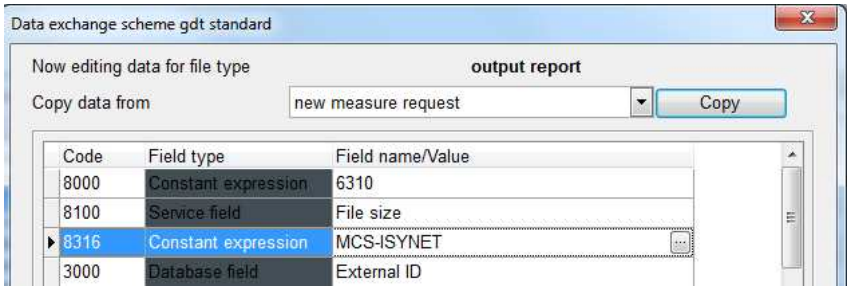


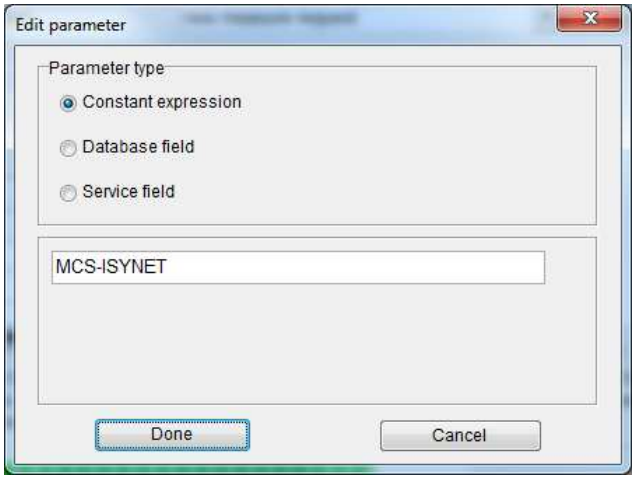
ii. The following window will appear.



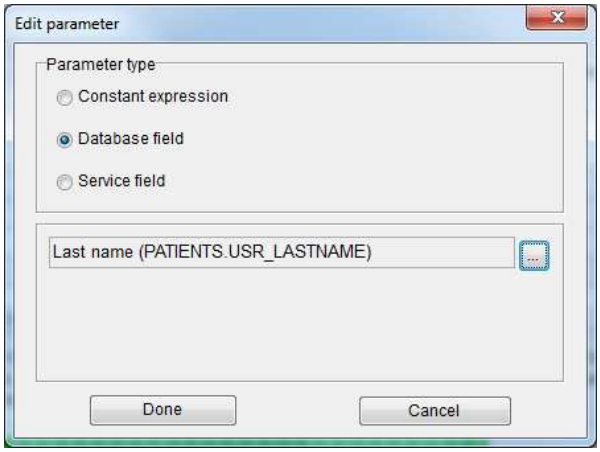
Three types are available.

1. Constant expression type is intended to include any static information. This type is usually used in the output file type (e.g. for the data exchange system name).

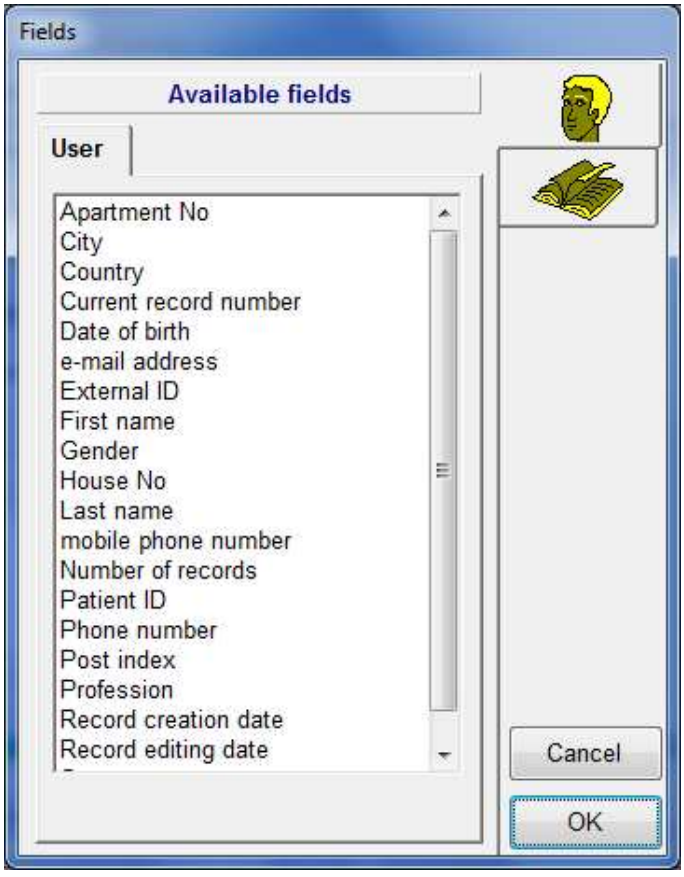




2. Database field type is intended to include the information from the database.

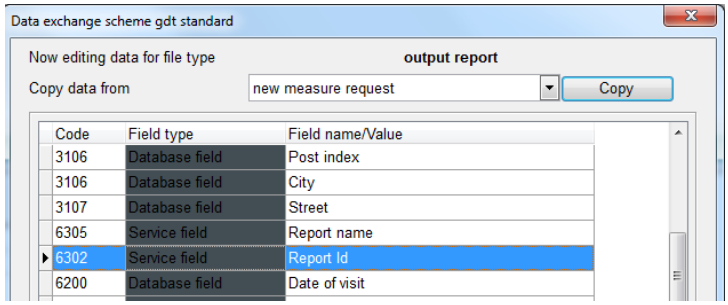


Click the ... button to select the field.
The following window will appear.

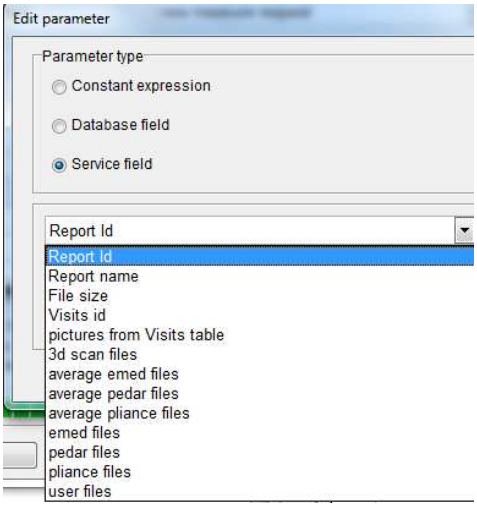


Select the required field and click the OK button or double-click on the selected field.

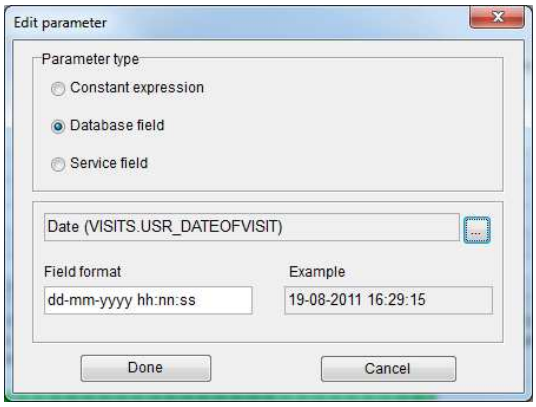
3. Service field type



When the service field type is selected, the following set of the service fields is available.



When the **date** type of the field is selected, additional information is required.

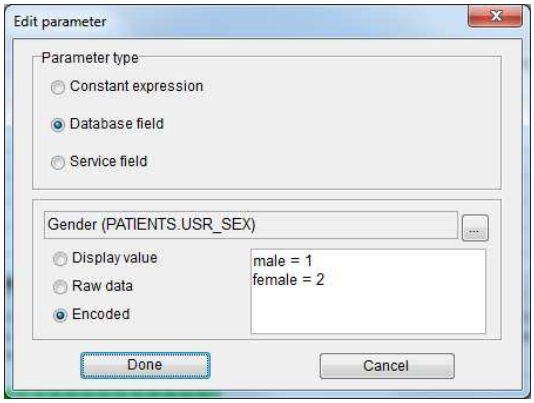


The field format is entered in the Field format box. Use the following letters (case-insensitive): d - for a day; m - for a month; y - for a year.

If time is required, use h - for an hour; n - for minutes; s - for seconds.

The resulting format is displayed in the Example text box.

When the Gradations or Multi gradations type is selected, additional information is also required.



When the Display value is selected, the data will appear with display name (e.g. male or female).

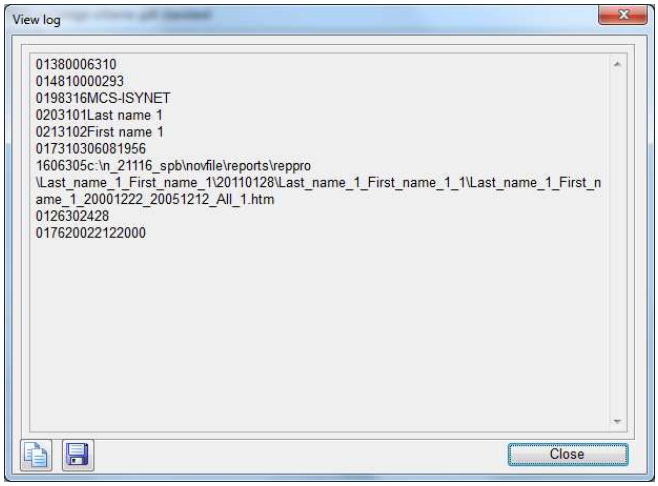
When Raw data is selected, the data will appear as it is stored in the database (e.g. 0 or 1).
NOTE: Use this option carefully.

When Encoded is selected, the codes will appear (e.g. 1 or 2).

Use  and  buttons to move the selected row up and down.

NOTE: If the same code is selected for different database fields, the appropriate value will be split between these fields (e.g. 3106 – Post index, 3106 - City). If the same database field is selected with different codes, the appropriate value will be merged in the field (6228 – Comments, 6229 - Comments).

iii. Use the  button to test the file.

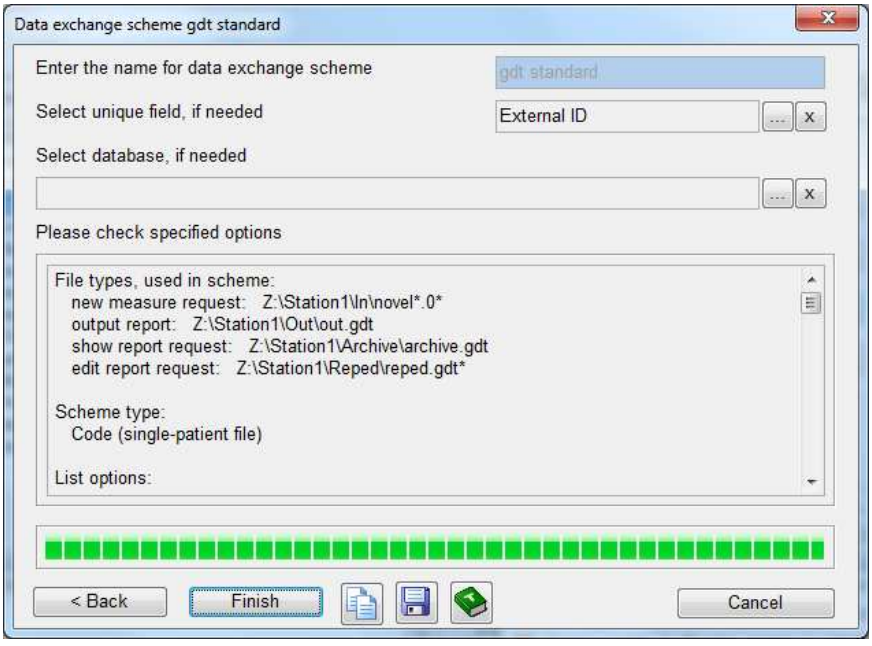


Use the  button to copy the file in the clipboard.

Use the  button to save the file.

NOTE: If the required fields are absent in the input file or corresponding data values are invalid, a warning message will be printed in the log window. This file will be rejected during data exchange.

e. Specify the DES name, the additional options and check the DES.



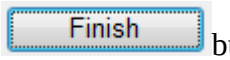
Select the unique DES name in the corresponding text box.
Select the unique field if it is required.
Select the database if it is required.

The  buttons are described above.

Click the Test file  button to test the file.

The results  will appear in the log window.

The Copy and Save as  buttons are available.

f. Click the  button to complete the DES creation.

The examples of the input files for several data exchange schemes are listed below.

1. MCS-ISYNET (gdt standard)

01380006310
014810000302
0268316Clinical Database
01330001234
0173101Mannlich
0173102Timurman
017310301011954
026310683707 Bad Wiessee
0243107Tölzer Allee 50
017620005062006

2. PAEDUS

ro
Falk
Matthias
Berger Str. 1
68523 Weinheim
24.07.1991

m
Herrn

j
106780007
3
0001

100
09547604003

5691560
R122

3. **CosInOS**
Kaufmann;Mike;195426;;;30.10.1965;;

4. **Medistar** (gdt standard)
013800063O1
014810000182
014921801.00
01030007
0183101Testmeier
0133102Emil
017310323111945
020310512982986308
023310630625 Hannover
0203107Hinterhof 3
01031081
01031101

5. **Rothballer**
Anrede;Name;Vorname;Strasse;Ort;Telefon;Geburtstag;Werbekennzeichen;Versicherter;Aktenzeichen;Krankenkasse;KK-Status1;KK-Status2;Eigene-Nr
Herr;Muster ;Maximilian ;Rosenstraße 12 ; Wiesau ;012345/45678 ;31.01.1934;X ;;2159536 ;0000000 ;0;0011;123456789012;1
Herr;Veigl ;Reinhold ;Unterbruck 17 ;95506 Kastl ; ;15.10.1961; ;; ; ; ; ; ;2
Herr;Hösl ;Alfons ;Nelkenweg 2 ;92711 Parkstein ; ;02.02.1950; ;; ; ; ; ;test ;3
Herr;Forster ;Gerhard ;Kirchenstr. 1 ;95703 Plößberg ; ;20.01.1930; ;; ; ;0;000; ;4
Frau;Forster ;Kathja ;Birkenstr. 1 ;92648 Vohenstrauß ; ;27.09.1991; ;; ; ; ; ; ;5

6. **Orthosoft**
16,"Udo Schwarz","","","Daimlerstraße 42","D","69190","Walldorf","4488875","10001","06081958","M"

The example of the output file

MCS-ISYNET (gdt standard)
013800063l0
014810000189
0198316MCS-ISYNET
011300099
0173101Testmann
0143102Peter
017310301011954
026310683707 Bad Wiessee

0243107Tölzer Allee 50
1036305c:\n_13321_mcsisynet\novfile\reports\reppro\Testmann_Peter\20060323\Testmann_Peter_1\index.mht
0126302738
0096228Severe neuropathy
017620023032006

The example of the file to display data
01380006310
014810000189
0198316MCS-ISYNET
011300099
0173101Testmann
0143102Peter
017310301011954
026310683707 Bad Wiessee
0243107Tölzer Allee 50
0126302156
0226228Severe neuropathy